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Thema: Modeling of the 0G state of Large Compliant Thin-Walled Structures with Data from a Smart Jig

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Master thesis

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Topic: Modeling the 0G state of large compliant thin-walled structures with data from a smart jig

Large compliant thin-walled structures (LCTS) made of carbon fiber–reinforced plastic (CFRP) are increasingly used in the automotive and aerospace industries due to their high strength-to-weight ratio. However, the adoption of CFRP-based LCTS introduces new challenges in assembly processes, primarily because of their compliant nature and the associated tendency to deform under gravity.

Current approaches for managing deformation in representative use cases, such as fuselage assembly, rely on expensive assembly jigs to force components into the desired assembly shape or on the use of shimming. Shimming is a widely applied procedure to ensure structural integrity, in which gaps larger than 0.3 mm are compensated using a liquid or solid filler material, referred to as a shim. This process is largely manual and time-consuming, leading to increased production time and additional costs.

Alternative techniques seek to eliminate the need for shimming, including virtual clamping and adaptive tooling approaches that induce targeted deformations such that shimming becomes unnecessary. A key requirement for these methods is knowledge of the gravity-free shape, defined as the theoretical configuration of a structure in the absence of external loads. Under practical conditions, this state can only be approximated numerically.

The objective of this thesis is the development and implementation of algorithms for modeling the gravity-free state of a representative LCTS using optical measurement data obtained from a smart fixture system. The smart fixture enables controlled and quantifiable deformation of the structure, while its geometry is recorded using a stereo-camera-based optical measurement system.

The scope of the thesis focuses on processing the optical measurement data by reconstructing a closed surface mesh from three-dimensional point cloud data. This reconstructed mesh is subsequently registered onto a numerically simulated deformed configuration obtained through finite element analysis under comparable loading conditions. By comparing the measured and simulated geometries, manufacturing-induced deviations are extracted in the form of a displacement field. This field is then transferred onto the nominal geometry to approximate the as-manufactured free-state shape of the component.

This approach enables the approximation of the free-state geometry by removing the influence of external loads, such as gravity, while preserving deviations introduced during manufacturing. The proposed methodology supports more realistic assembly simulations and enables fixtureless inspection

and gap control without reliance on manual shimming. Validation across multiple experimental configurations yields consistent displacement fields, demonstrating the robustness and reliability of the workflow.

The following subtasks are addressed:

- Review of the state of the art in gravity-free shape approximation for thin-walled structures
- Implementation and comparison of surface reconstruction algorithms for three-dimensional point cloud data
- Implementation and comparison of registration techniques for non-rigid structures
- Numerical simulation of gravity- and fixture-induced deformation using finite element modeling
- Estimation of manufacturing deviation fields through comparison of simulated and measured deformations
- Validation of the workflow across multiple actuator configurations
- Documentation of the proposed methodology and automation of the workflow

Prof. Dr.-Ing. Robert H. Schmitt

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Symbol	Unit	Definition
S_{free}^{nom}		Nominal shape
S_{def}^{nom}		Deformed shape
S_{conf}^{meas}		Measured shape
S_{free}^{man}		Gravity-free state shape of a manufactured component
E^{man}	mm	Displacement field
E^{conf}	mm	Deformation field
E^{grav}	mm	Displacement field due to gravitational effect
$S^{meas-disp}$		Measured shape corrected for E^{man}
$S^{nom+grav}$		Nominal shape deformed under gravity only
$S^{corrected}$		Measured shape corrected for E^{man} and E^{grav}
GF		Gravity-free
LCTS		Large Compliant Thin-Walled Structures
CFRP		Carbon Fiber Reinforced Polymer
FE		Finite Element
BCs		Boundary Conditions

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1 Introduction

1.1 Problem Statement

The assembly of large compliant thin-walled structures (LCTS), such as fuselage panels composed of carbon fiber reinforced polymers (CFRP), presents considerable challenges in modern aerospace production. These components, despite having high strength-to-weight ratios, have high elasticity and low plastic deformation, rendering them susceptible to deformations under intrinsic loads such as gravity and external forces such as clamps used during handling and assembly [GUO25]. As a result, the geometric state of these structures during inspection and assembly often differs from their intended design configuration [SABR17].

Traditional assembly processes address these deformations using custom-designed rigid jigs in combination with shimming [SABR17]. While fixtures are used to force the component into the desired geometry during assembly, shimming compensates for the resulting part-to-part gaps when gaps exceed 0.2 mm [GUO25]. Both approaches are costly, as fixtures must be specifically designed and manufactured for each component variant, and shimming involves manual gap measurement and installation [ABEN12]. According to [ESPO22], gap measurement and shim related operations can increase overall project costs by up to 25%.

To overcome these limitations, modern assembly concept known as virtual clamping have been introduced [KRAU23]. This approach is aimed at predicting the deformation during assembly with the use of precise digital models. However, the success of this strategy depends on the approximation of the GF state shape, which represents the part's true design intent. As there are no direct approaches to measure the GF state shape of compliant structures, measured geometries are inevitably influenced by gravity-induced deformation, leading to inaccuracies if this effect is not explicitly removed [LIND20].

This research addresses this limitation by proposing a methodology for approximating the GF state shape of LCTS using deformation measurements obtained under controlled loading configurations. Integrating measurement data with the numerical simulation within a unified workflow allows the proposed approach to approximate the GF state shape while preserving manufacturing deviations. This enables accurate virtual clamping and automated gap control while reducing reliance on custom fixtures and manual shimming.

1.2 Objectives

The primary objective of the research is to approximate the GF state shape of LCTS by integrating measurement data with numerical simulation within a hybrid workflow. As LCTS deform under their own weight during measurement, identifying their unloaded configuration is required for virtual clamping and downstream assembly operations.

The following sub-objectives are pursued in the research:

- Acquire the measured geometry S_{conf}^{meas} of the as-built component using 3D scanning.
- Simulate the deformed geometry S_{def}^{nom} of the nominal model S_{free}^{nom} under identical boundary and loading conditions using finite element analysis (FEA).
- Register the measured geometry S_{conf}^{meas} to the nominal geometry S_{free}^{nom} using point cloud registration approaches to ensure accurate alignment.
- Compute the manufacturing-induced displacement field E^{man} representing deviations between the measured and nominal geometries.
- Map the displacement field E^{man} to the nominal geometry S_{free}^{nom} to approximate the GF state shape of a manufactured component, S_{free}^{man}, compensating for gravitational effects while preserving manufacturing deviations.
- Validate the proposed methodology by evaluating the consistency of the displacement fields obtained from experimental configurations and by assessing the robustness of the proposed workflow using gravity-compensated reconstructions.

Overall, this workflow establishes a framework for gravity-compensated, fixtureless inspection and assembly, enabling the use of digital models that reduce dependence on custom fixtures and manual shimming.

1.3 Thesis Structure

The thesis is organized into seven chapters, structured as follows:

Chapter 2: State of the Art reviews current approaches related to GF state shape approximation and fixtureless inspection of structures. It also introduces fundamental terminology, reviews measurement and registration techniques and discusses methods for evaluating manufacturing deviations.

Chapter 3: Methodology describes the proposed approach for approximating the GF state shape. It outlines the overall workflow, defines the terminology and details the individual methodology steps, including the smart jig setup, surface reconstruction, geometries alignment and displacement field computation.

Chapter 4: Experimentation and Hybrid Pipeline details the experimental setup, data acquisition strategy and design of experiments, followed by the data processing, registration and displacement field computation steps used to ensure repeatability and robustness.

Chapter 5: Numerical Modelling presents the numerical modelling framework employed in this research. It describes the construction of the FE model, including the nominal geometry, material definition, element selection, assembly interactions and simulation conditions. The chapter also addresses the integration of numerical results within the overall workflow.

Chapter 6: Results and Validation reports the results obtained from the proposed methodology and discusses the associated validation procedures. It presents the approximated GF state shape along with the displacement field and workflow validation. The chapter concludes with a discussion of key findings and limitations.

Chapter 7: Conclusion and Future Work concludes the thesis by summarizing the main outcomes and outlining recommendations for future research and potential industrial applications.

2 State of the Art

This chapter builds upon the background established in Chapter 1 and examines the existing approaches relevant to approximation of the GF state shape of a manufactured component. These concepts are introduced to establish the theoretical foundations required to understand how gravity-induced deformations influence measurement, inspection and assembly processes for components. It reviews state of the art methods for estimating the GF state shape of manufactured components, shape reconstruction, 3D measurement techniques and the evaluation of manufacturing deviations. The chapter concludes by evaluating the identified methodologies, highlighting the limitations and defining the research gaps that motivates the development of the proposed approach presented in Chapter 3.

2.1 Fundamental Concepts

Precise terminology is required when analyzing compliant and deformable structures, particularly in contexts where physical experimentation is combined with numerical simulation [THIÉ17]. The use of clearly defined terms is essential for ensuring consistency between measurement-based observations and simulation-based deformations of structural response. The definitions introduced in this section are divided into two fields, structure and fixture, reflecting the two topics that govern the deformation behavior of compliant components in industrial inspection and assembly processes. Together, these definitions provide the basis required to interpret gravity-induced deformation, constraint effects and their interaction in subsequent analyses.

2.1.1 Structural Definitions and Deformation Concepts

The classification of structures as rigid or non-rigid forms the starting point for this research. According to DIN ISO 10579, a non-rigid part is defined as “a part that deforms under its own weight or under small external forces to an extent that, in the free-state, the resulting geometry lies outside the dimensional or geometrical tolerances specified on the drawing” [ISO10]. This definition is useful for lightweight aerospace components, where gravity-induced deformation cannot be neglected during inspection and assembly [KRAU23]. This contrasts with rigid parts, which are defined in ISO and EN standards as “structures whose deformation under gravitational loads is negligible relative to the specified tolerances” [ISO97].

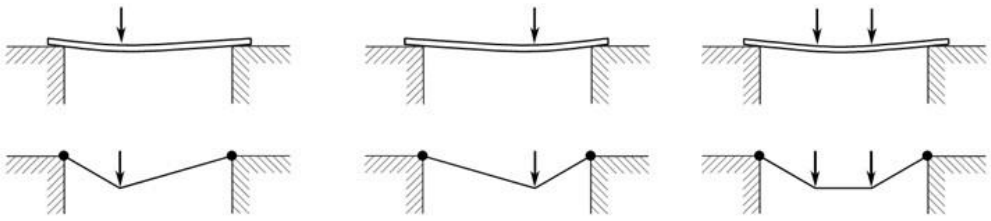


Figure 2-1 Schematic illustration of deformation behavior under applied loading for rigid and non-rigid parts [SOTO24].

As discussed in Section 1.1, compliant and lightweight structures have provided an alternative solution for materials being used in the aerospace industry. Large compliant thin-walled structures (LCTS) are large, deformable carbon fiber reinforced polymer (CFRP) components with weak rigidity and inherent manufacturing errors, which are prone to warping, united deformation and internal damage [GUO25]. As per Guo et al. CFRP refers to a composite material widely used in aerospace applications due to its high brittleness, anisotropic mechanical properties and minimal plastic deformation [GUO25].

Similarly, the shape of the non-rigid LCTS resulting from the forces acting on the structure can be subdivided into a free-state shape and the GF state shape. The free-state shape corresponds to the condition of a part subjected only to the force of gravity as defined in ISO 10579 [ISO10], whereas the GF state shape is described by Claus et al. as the structural configuration corresponding to a state of zero external loading, explicitly excluding gravitational weight [CLAU21]. This distinction is required because gravity induces deformation in LCTS, hence the free-state shape does not represent the intrinsic geometry of the manufactured part but rather a gravity-loaded configuration.

Both these concepts are used to approximate the deviations induced during manufacturing, such manufacturing deviations are form deviations resulting from the intrinsic geometrical difference of an as-built part from its nominal shape, solely due to the manufacturing process variations, independent of deformations caused by gravity, fixturing, or assembly configurations [THIÉ17]. In this context, the nominal shape refers to the theoretical, ideal geometry of a part specified by a Computer-Aided Design (CAD) model, representing the model prior to manufacturing and assembly. On the other hand, as-built parts are the actual manufactured components, including deviations caused by processes such as machining and curing [BABA20].

2.1.2 Fixture Configurations and Virtual Clamping

In addition to structural definitions, it is equally important to identify different fixture configurations to determine how a part is constrained during measurements. A non over constrained fixture refers to a setup where the part is held without excessive constraints, as opposed to *an over constrained fixture* in which the structure is clamped on a measurement fixture that constrains the non-rigid structures in more than six degrees of freedom, to prevent gravity induced deflections and to ensure that the structure’s environment is similar to an assembled situation [CLAU21].

Alternatively, fixtureless inspection is a measurement approach for the dimensional and geometric quality control of compliant mechanical parts, such as sheet metals and skins, that eliminates the need for expensive, specialized physical inspection jigs [BABA20]. While this approach offers increased flexibility and reduced cost, the absence of physical constraints means that the measured geometry of non-rigid components corresponds to a free-state configuration. As a result, gravity-induced deformations are inherently present in the acquired measurement data [SABR16]. Virtual clamping, in contrast, is a numerical technique applied within the digital metrology workflow to compensate for such deformation effects. Rather than

physically constraining the component during measurement, virtual clamping uses FE based numerical models to simulate the effect of mechanical constraints or to remove deformation from the measured shape [TUOM17].

In summary, this section outlines the difference between concepts related to the structure and fixture. The structure-related definitions distinguish between free-state and GF state shapes, whereas fixture-related concepts describe how physical or virtual constraints affect the measured geometry. This integration of measurement data with numerical simulations contributes to the development of a digital twin, a virtual counterpart of the physical component that mirrors and predicts its behavior under varying conditions [KRIT18]. Addressing these distinctions is required for the methodology, which involves hybrid measurement–simulation workflows that approximate the geometry of manufactured components.

2.2 Referencing

This section addresses the processes of measuring, referencing the measured data with respect to the nominal data and digital representation of structures. It outlines the state of the art principles of 3D scanning used to capture the state of components, followed by a discussion of surface reconstruction and registration techniques that relate the measured geometry to the nominal geometry of the structure and concludes with an overview of the current approaches for evaluation of manufacturing deviations [THIÉ17].

LCTS, which tend to deform under their own weight, must be measured using precise scanning equipment to accurately capture their loaded form and associated deviations [SCHM16]. Similarly, it is required to organize and process the acquired data to establish an accurate reference frame to estimate the deviations of the measured structure from the nominal shape. This necessitates a robust approach that integrates advanced scanning technologies with intelligent data-processing and alignment methods capable of handling both local deformations and global displacements [SABR17].

2.2.1 3D Measurement Methods

The increasing level of complexity and flexibility of modern manufacturing processes, particularly in the fields of aerospace and automotive industries, has motivated the development of three-dimensional (3D) measurement technologies [SOOR23]. These techniques capture the surface geometries of structures with high accuracy and consistency, serving as the basis for reverse engineering, fixtureless assembly and digital inspection [SALV10].

The developed solutions can traditionally be categorized into contact and non-contact (optical) approaches [THAL16]. The three fundamental types of 3D measurement devices that are frequently utilized are stereo-camera systems, laser-based or structured-light scanners and tactile measurement probes [SALV10]. Each approach has specific benefits depending on the application, required accuracy, the geometry and the material of the component being measured [THAL16].

a) **Contact-based Approaches**

Tactile or contact measurement systems provide a viable option for surface measurement. These types of systems use a precision probe tip to make physical contact with an object to determine its surface coordinates. Their benefits include repeatability, accuracy and low dependability to external influences such as surface reflectivity or lighting. However, according to [THAL16], three important variables affecting measurement uncertainty in tactile metrology are probe–surface contact, stylus shape and calibration. Hence, these approaches are limited by slow data acquisition speed, the requirement for steady environmental variables and the possibility of surface deformation from probing forces during measurement. Such methods are also less effective for large complex geometries due to longer measurement times and higher operating costs [SALV10].



(a) (b)

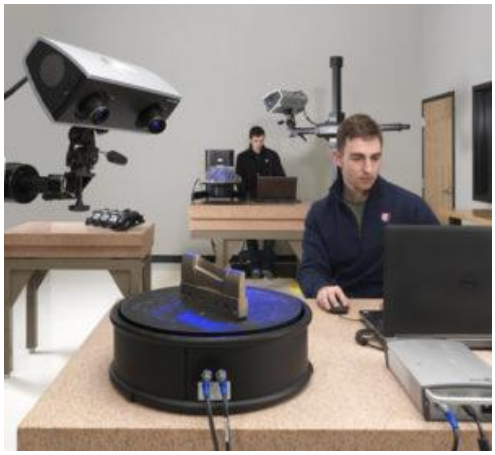
Figure 2-2 (a) Contact based 3D scanning [3DSC25]; (b) non-contact 3D scanning [ALFO20].

b) **Non-contact Approaches**

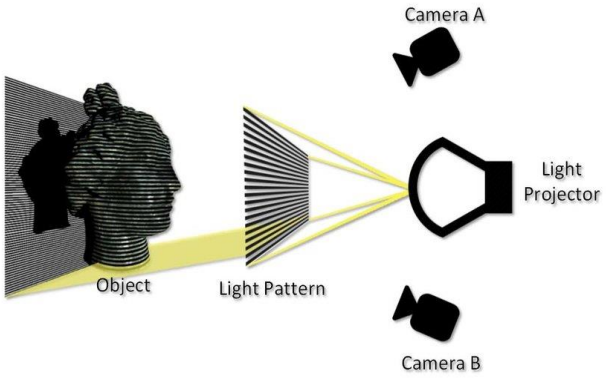
Contactless measuring methods are classified according to their basic working principles: active optical methods, such as structured light and laser scanning, and passive optical methods, such as stereo imaging. Active systems include an illumination source that interacts with the surface, while passive systems rely on ambient light to derive depth information from images. Structured light scanners, depicted in Figure 2-3, beam a known pattern onto an object's surface and evaluate its deformation with cameras to rebuild the 3D surface. Continuous patterns often provide higher resolution and consistency which can we be useful when reconstructing fine surface features [SALV10].

In contrast, laser scanners, such as terrestrial laser scanners (TLSs), operate by producing laser pulses and computing the time-of-flight (TOF) of the reflected signal. Each recorded point is identified by the range and two angle values in spherical coordinates. TLSs have the benefit of capturing millions of points in a very short period, making them ideal for capturing large-scale structures such as aircraft panels or car body parts [NGUY12]. Portable 2D LiDAR sensors, such as those manufactured by Hokuyo, when combined with six degrees of freedom

inertial measurement units (6DOF IMUs), increase scanning flexibility by allowing mobile or hand-held configurations to operate without relying solely on robotic arms or stations [KUMA18].



(a)



(b)

Figure 2-3 (a) Structured light scanner used for non-contact surface measurement [GLAS19]; (b) schematic illustration of the structured light measurement principle [RAYC17]

However, despite their advantages, active methods have certain limitations. Surface reflectivity, fiber patterns and irregular reflectivity can all have an impact on measurement quality. While grayscale structured light systems are more immune to reflections, phase unwrapping in frequency-based systems is still subject to shadows and noise [SARG86]. Partial scan registration poses additional challenges, particularly with repeated patterns such as De Bruijn sequences, which are prone to mistakes caused by obstacles [SALV10]. Furthermore, matching 3D point clouds with 2D video frames in dynamic scanning settings can be computationally demanding [KUMA18].

The third category, stereo-camera systems, provides a balance between flexibility, resolution and automation. It represents a specific implementation of optical triangulation and can be either passive or active, depending on whether an illumination source is used. Passive stereo systems rely primarily on two or more cameras gathering images from various viewpoints and identifying similarities based on surface texture. While useful in certain situations, dependence on surface texture frequently results in sparse and noisy reconstructions when working with texture-free or reflective materials [SALV10]. To address these drawbacks, active stereo vision replaces one of the cameras with a projector, which improves correspondence match by projecting structured light patterns onto the scene.

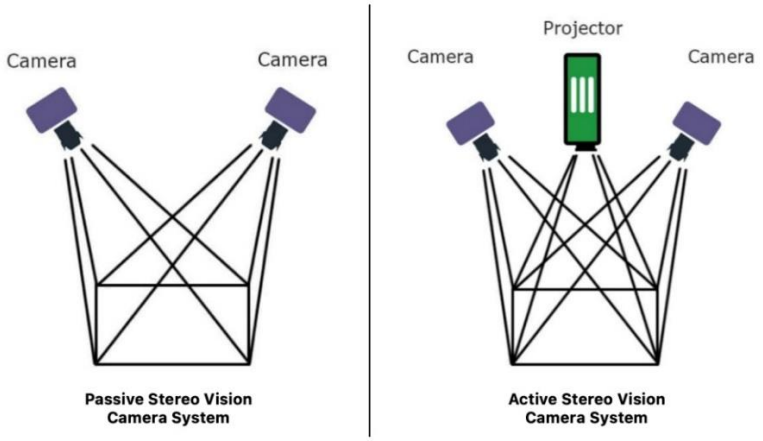


Figure 2-4 Comparison of Stereo Vision Camera Systems. Left: Passive; Right: Active [SHUM24].

Stereo systems are ideal for applications that require measurements of flexible components under variable loads [KUMA18] and are compatible with robotic manipulators and metrology setup making them effective for digital twin frameworks. Industrial systems, such as the Micro-Epsilon surface Control 3D 2500, are lightweight, easy to integrate and can achieve precise depth resolution on the order of 0.05 to 0.1mm within working distances between 0.3 and 1 m [TRÉH25] [EPSI].

Coverage is another important consideration during scanning. Scanners only record data inside their field of view, hence many scan stations are frequently required to accurately record complex geometries. A minimum of 20-30% overlap between consecutive scans is recommended for effective registration, while obstructions or challenges in the scanner's line of sight may induce gaps in the generated point cloud [NGUY12]. Besides these, large-scale inspection environments also require laser trackers to set up a Global Reference System (GRS) that aligns individual acquisitions [SCHM16].

2.2.2 Physical Measurement to Digitalization of Components

The transformation of the structure, available in the form of dense point clouds, into a digital model is an initial step for applications involving range sensing, modelling and simulation, industrial inspection and analysis and reverse engineering [SALV10]. The digitalization concept revolves around the ideology of moving from the physical world to an abstract representation of a component. This digitalization process allows for the creation of a virtual copy, digital shadow, of a physical system, which is suitable for computational manipulation and analysis in virtual environments [SOOR23].

The conversion pipeline comprises of three stages: data acquisition, data processing and surface reconstruction. During data acquisition the component surface is scanned using the 3D measurement techniques discussed in Section 2.2.1, which yields raw point cloud data. Data processing removes noise and outliers, reduces data volume, estimates normal and merges multiple scans into a unified dataset [NGUY12]. Lastly, surface reconstruction then

converts the processed point cloud into a continuous mesh representation suitable for analysis and simulation [BERN99].

The objective of this transformation is to generate a geometric representation that maintains high quality for further computational analysis. Therefore, the resulting representation has to be accurate, have minimal noise effect and be computationally efficiency to be compatible with downstream tasks [QUAN24].

a) Data Processing

Once the raw point cloud is obtained, it can be processed into a structured, usable data set. The raw data gathered from the scanners often suffers from quality issues due to the limitations of sensors and the acquisition environment, such as sparsity, noise and incompleteness resulting in multiple noisy scans of a single configuration of the structure [QUAN24]. Processing the raw data prepares the data to be used for further steps such as surface reconstruction and approximation of the GF shape by improving the geometric accuracy and minimizing the computational load.

Filtering

The first step in this workflow is to remove unwanted points, outliers and reflections, which are very common when dealing with shiny surfaces. This process of filtering removes unwanted points, outliers and reflections, which can distort the underlying geometry. Statistical outlier removal (SOR) and radius outlier removal (ROR) are common techniques: ROR counts the neighbors of each point within a fixed radius and discards points with too few neighbors, while SOR adaptively sets a threshold based on the average point density [SZUT23]. A further refinement for optical scanning systems is background component filtering, which removes illumination inhomogeneities and local surface reflectivity using Fourier transform profilometry to eliminate additive variations [SALV10]. Figure 2-5 indicates the effect of SOR on a scanned surface, highlighting how the underlying geometry is preserved by eliminating noise.

Down-Sampling

Another step involves the resampling of the points in the raw data set to reduce data volume while retaining geometric fidelity and to speed up modeling algorithms. One of the common resampling methods is voxel down-sampling. This method involves dividing the space containing the point cloud into a grid of volumetric cells or voxels because the raw data points are acquired on an irregular grid when obtained from the scanners. Voxel down sampling performs the resampling operation by generating a new grid where all range and amplitude data falling within one cell of the grid is averaged which helps transform the data, often resulting in denser, smoother and more complete datasets [NGUY12].

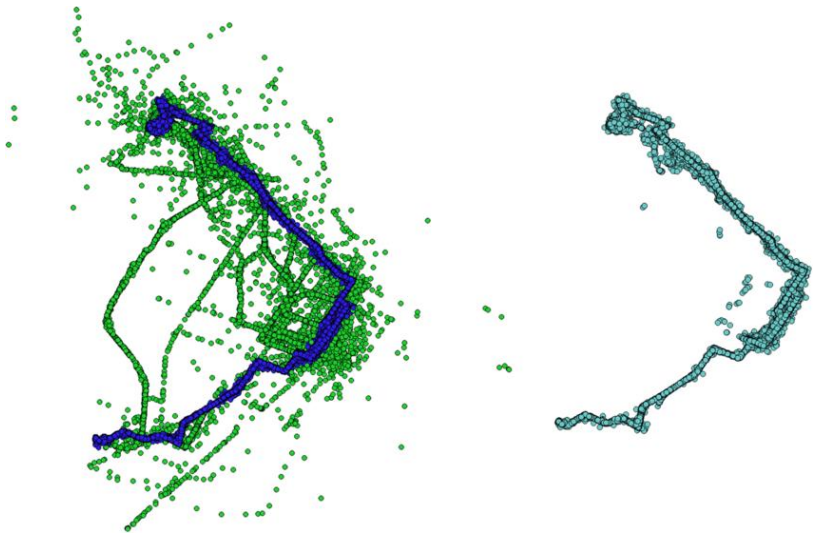


Figure 2-5 A bus line before and after applying the SOR filter [PIRO18].

Normal Estimation

Estimating normal vectors is required as they are essential local geometry properties needed for downstream tasks. At each point, the normal is typically defined as the unit vector perpendicular to a local tangent plane approximated from the surrounding neighborhood [ZHOU22]. One approach computes normals by fitting a plane via principal component analysis (PCA) to the point's k-nearest neighbors and selecting the eigenvector associated with the smallest eigenvalue. Adaptive statistical methods improve on PCA by varying the neighborhood size based on curvature, noise and sampling density, helping to preserve sharp edges and reduce outlier influence [MURA18]. More recent learning-based techniques train neural networks to predict normal from local point neighborhoods, allowing the models to capture complex patterns and improve robustness to noise [LU20].

Merging Point Clouds

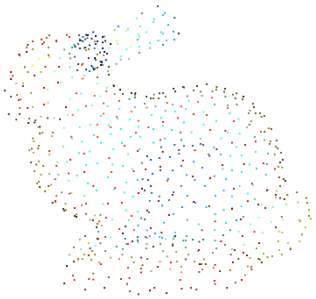
Data processing also involves merging the individual scans acquired from different viewpoints to generate a unified, complete 3D representation. This procedure is required not only to address occlusions and blind spots, but also because the measurement volume and achievable uncertainty of many optical systems necessitate multiple acquisitions to fully capture large or complex structures. Once stitched, the outcome is a unique point cloud that captures the scanned real-world scene, forming a complete 3D model of the structure. There are two stitching techniques outlined in the literature: coarse and fine stitching [THOM20].

b) Surface Reconstruction

After processing the data, the next step is to convert it into a 3D model which can then be used for shape comparison and analysis. This transformation of the processed and stitched point cloud to a continuous surface model is achieved through surface reconstruction. Surface reconstruction determines an approximate continuous surface from an unorganized set of points [HUGU92]. Three widely used techniques are Alpha Shape Reconstruction, the Ball Pivoting Algorithm and Poisson Surface Reconstruction [MICH06].

Alpha Shape Reconstruction

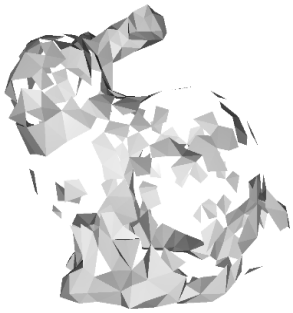
The alpha shape method derives a triangulated boundary from the Delaunay triangulation of the point set, which is a geometric construction that partitions space into tetrahedra such that no input point lies inside any tetrahedron, and is controlled by a parameter α. The core concepts defining three-dimensional alpha shapes were developed by Edelsbrunner and Mücke in [EDEL83]. These schemes typically generate a triangle mesh that interpolates all or most of the input points. Smaller α values lead to coarse surfaces approximating the convex hull, as shown in Figure 2-6(b), whereas larger α values capture fine details, as shown in Figure 2-6(c), but increase computational cost and sensitivity to noise. The method's flexibility makes it useful for irregular datasets, although it can be memory intensive and noisy data often causes additional smoothing steps.



(a)



(b)



(c)

Figure 2-6 (a) Raw point cloud; (b) alpha shape reconstructed surface ($\alpha = 0.030$*);* (c) alpha shape reconstructed surface ($\alpha = 0.010$) [OPEN25b].

Ball Pivoting Algorithm (BPA)

The BPA is an advancing-front, region growing method that incrementally constructs an interpolating triangulation. The core principle is that three sample points form a new triangle if a ball of a specified radius (ρ) touches them without containing any other point, and the algorithm proceeds by pivoting this ball around the existing mesh edges. The radius ρ should be set according to the point spacing; for dense, uniformly sampled data, the resulting mesh

is homeomorphic to the original surface, Figure 2-7. Due to its local nature, BPA operates with linear time complexity and modest memory usage, and it can handle uneven sampling by performing multiple passes with different radii [BERN99].



(a) (b)

Figure 2-7 (a) Raw point cloud; (b) BPA reconstructed surface ($\rho = [0.005, 0.01, 0.02, 0.04]$) [OPEN25b].

Poisson Surface Reconstruction

Poisson Surface Reconstruction is an implicit method that conceptualizes surface reconstruction from oriented points as solving a spatial Poisson equation. It computes a three-dimensional indicator function whose gradient field approximates the oriented normal of the point cloud; the surface is extracted as the zero set of this indicator function. As it considers all data globally rather than relying on local heuristics, the method is resilient to noise and produces smooth, watertight surfaces as indicated in Figure 2-8. Its hierarchical octree-based implementation yields efficient solutions even for large and high-resolution datasets [MICH06].



(a) (b)

Figure 2-8 (a) Raw point cloud (796825 points); (b) Poisson surface reconstructed mesh (563112 points, 1126072 triangles) [OPEN25b].

In conclusion, acquiring geometry, cleaning and merging point clouds, and reconstructing surfaces are all necessary steps in converting actual data into digital models. Every processing and reconstruction technique has trade-offs between computing cost, robustness to noise and detail retention. The generated mesh is used as the foundation for further research including registration against nominal models and for the approximation of GF state shapes.

2.2.3 Registration Techniques

Registration is the process of aligning a measured, processed 3D model of a point cloud with its nominal reference model so that they share a common coordinate system [KUMA18]. This procedure is mathematically defined using the translation and rotation of the design coordinate system (DCS) with respect to the measurement coordinate system (MCS). The registration process involves searching for the correspondence relationship between the measured and nominal surfaces and determining an optimal transformation matrix between the DCS and MCS [SABR16]. Registration is fundamental to quality inspection, enabling both comparison of measured geometry against CAD models and mapping of deviation fields for analysis and corrective actions [RAVI12].

Registration approaches can be broadly classified according to the expected deformation and the goal of the alignment. Coarse registration establishes an approximate pose when two datasets are far apart [RAVI12], whereas, rigid registration estimates rotation and translation under the assumption that the object does not deform [MONJ23] and non-rigid registration accommodates shape changes due to compliance or residual stress [SABR17].

a) Coarse Alignment

Coarse registration serves as the initial stage in the point cloud registration process. This step establishes an approximate alignment (pose) between the two models without seeking to optimize minute local differences. The primary role of coarse registration is to produce a good initial guess for subsequent fine-alignment methods, like the Iterative Closest Point (ICP) algorithm or its variants [YUE22]. The importance of coarse registration lies in its ability to reduce the search space and improve the robustness of later stages [RAVI12].

Simple (Manual) Coarse Alignment

Manual registration relies on an operator to select a few corresponding features on the measured and nominal surfaces, such as fixation holes, target datums, or contact surfaces. These points define an initial transformation matrix that positions the datasets close enough for fine alignment. Manual selection remains a dependable strategy for smooth or repetitive surfaces where automatic feature detectors may fail [SABR17]. Manual point selection can be formalized into two frameworks: the Generalized Numerical Inspection Fixture (GNIF) which locates constraint areas by leveraging geometric properties such as geodesic distance (Figure 2-9) [RADV14], and the As-is-Where-is Inspection (AIWIN) procedure which uses physical fiducials, typically calibrated spherical gauges, attached to the component [RAVI12].

Automated Course Alignment:

Automated coarse registration approach typically relies on feature-based approaches, which establish approximate alignment by identifying and matching salient features, key points, between the models. The fundamental pipeline for feature-based registration includes key point detection, feature description, feature matching, correspondence screening and calculating the transformation parameters [YUE22].

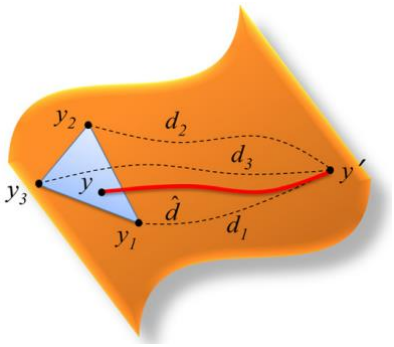


Figure 2-9 Geodesic distance interpolation for GNIF [RADV12].

Key points capture local geometric structures invariant to rotation and translation. Descriptors such as Point-Pair Features (PPF) or Local Point-Pair Features (LPPF) and Fast Point Feature Histograms (FPFH) encode the spatial relationship of oriented points. However, these methods include some assumptions and limitations. Feature-based methods presume distinctive geometric features exist on the surface, such as corners or high-curvature areas [YUE22]. Additionally, highly descriptive descriptors can lead to significant computational cost, limiting their practical usage for large point clouds [LU25].

b) Fine Alignment

Fine registration is usually performed after coarse registration, dedicated to achieving the precise alignment between models. This phase aims to iteratively refine the approximate alignment to minimize the spatial differences between the source and target models as accurately as possible. Fine registration techniques are broadly categorized into rigid and non-rigid methods, depending on whether the objects being aligned maintain their original shape or deform during the process [THOM20].

Rigid Registration

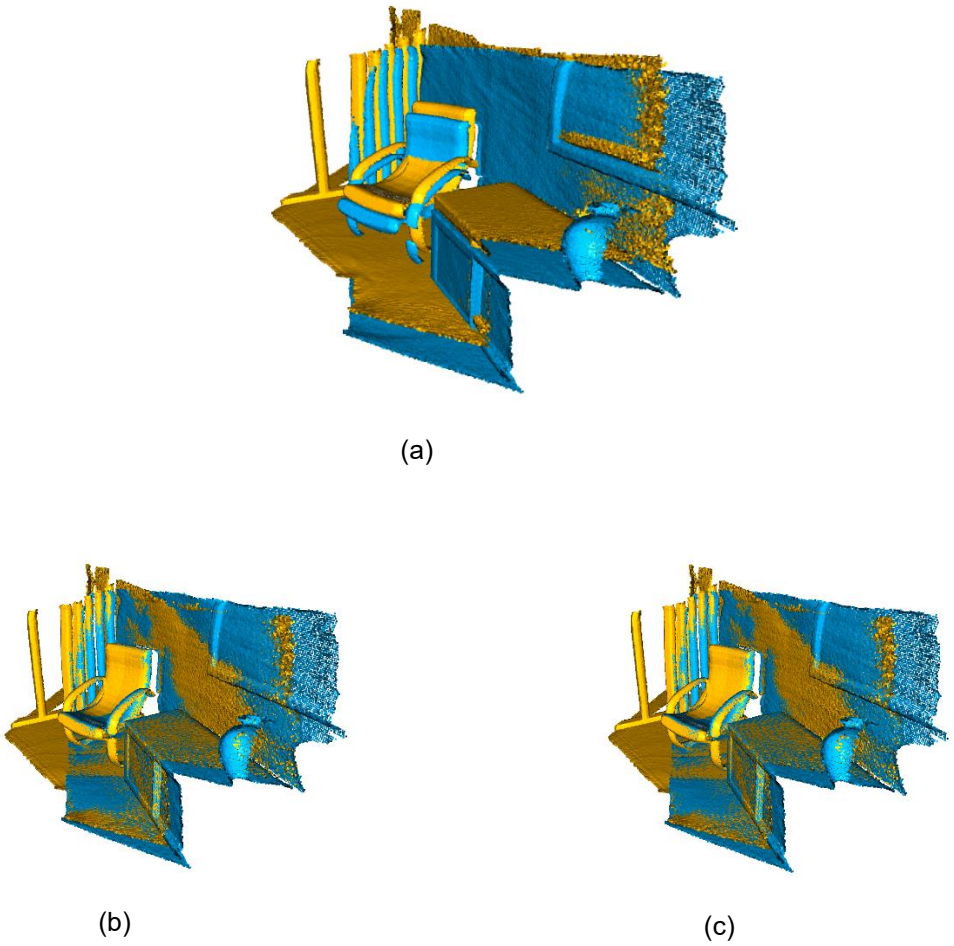
Rigid registration refers to techniques where the measured shape is assumed to undergo only rigid transformations, specifically rotation (R) and translation (T), relative to the nominal model. This type of transformation ensures that the distance between any two points on the object remains unchanged [MONJ23]. Rigid registration is then further classified into Iterative Closest Point (ICP), Robust Point Matching (RPM) and 4-Points Congruent Sets (4PCS).

The Iterative Closest Point (ICP) algorithm is used as one of the approaches for rigid point set registration due to its simplicity and relatively low computational complexity [LU25]. ICP begins by taking an initial guess for the transformation (often provided by coarse registration). The algorithm then iteratively calculates the optimal rigid body transformation matrix (rotation (R) and translation (T)) that minimizes a distance error metric between the source model (obtained from 2.2.2) and the nominal model [YUE22]. ICP consists of the following workflow:

- Determine Correspondence: Pairs of corresponding points are identified between the models, typically based on a closest distance criterion [THAL16].
- Estimate Transformation: A rigid transformation is computed that minimizes the sum of squared distances between these established correspondences [MYRO10].

- Apply Transformation: The calculated transformation is applied to align the input point cloud with the target point cloud [THAL16].

However, ICP does have some limitations such as the initial pose requirement, speed and sensitivity to outliers and missing data [LU25]. Moreover, the ICP algorithm is prone to converging incorrectly to a suboptimal local minimum if the initial transformation is poor. To address these shortcomings, various variants have been developed such as Point-to-Point vs. Point-to-Plane, Figure 2-10, according to [THAL16] and Probabilistic Methods according to [MYRO10].



(a)

(b) (c)

Figure 2-10 (a) Coarse alignment of source point cloud (yellow) with the target point cloud (cyan); (b) point-to-point ICP registration; (c) point-to-plane ICP registration [OPEN23]

Other rigid registration algorithms include Robust Point Matching (RPM) and 4-Points Congruent Sets (4PCS). RPM is a probabilistic method that alternates between correspondence estimation and transformation calculation using soft assignment, allowing correspondences to be defined probabilistically rather than deterministically [MYRO10]. In contrast, the 4PCS algorithm is a global registration technique inspired by RANSAC, in which sets of four congruent points are used to estimate the rigid transformation. While effective for complex registration scenarios, 4PCS can be computationally expensive [YUE22].

Non-rigid Registration Algorithms

The shift from rigid to non-rigid registration is necessary when dealing with structures that change shape internally, meaning the distances between points within the object do not remain constant during transformation [MONJ23]. Non-rigid registration methods model complex deformations beyond simple rotation and translation. Non-rigid registration can be classified into Multi-Dimensional Scaling (MDS), Coherent Point Drift (CPD), Finite Element–Based Non-Rigid Registration (FENR) and Generalized MDS (GMDS).

MDS is fundamentally based on the concept that isometric deformation does not alter the intrinsic geometry of the surface. The working mechanism relies on preserving pairwise geodesic distances (the shortest path along the surface) between points. By maintaining these intrinsic distances, MDS transforms the challenging non-rigid surface matching problem into a simpler rigid similarity problem within the embedding space [RADV12]. The CPD algorithm treats one point set as centroids of a Gaussian mixture and aligns them to the other by maximizing likelihood, enforcing motion coherence through regularization, Figure 2-11. This algorithm is particularly robust to noise, outliers and missing points [MYRO10].



(a) (b)

Figure 2-11 (a) Initialization; (b) non-rigid CPD registration of 2D fish point sets [MYRO10].

Besides these, another algorithm, the FENR is a physics-driven approach typically employed for the dimensional inspection of compliant parts [CLAU21]. It uses FE simulations to deform the nominal geometry toward the measured shape by applying required BCs [SABR17]. The GMDS algorithm is often used as a pre-step for FENR to find correspondences that preserve geodesic distances between the measured surface and the nominal surface [RADV12]. These correspondences are then used to define the BCs for FENR [SABR17]. Both FENR and GMDS are effective alignment approaches, but they are generally characterized as computationally intensive [RADV14].

In summary, registration methods offer a range of trade-offs between computational effort, robustness to noise or missing data and the ability to handle rigid or deformable shapes. Some techniques focus on simplicity and efficiency, while others incorporate more complex modelling to manage deformations and outliers, often at higher computational expense. The most appropriate choice depends on factors such as the geometric characteristics of the surfaces, the presence of distinctive features, the degree of expected deformation and the required accuracy. The registration approach aligns the measured and nominal models in the same

coordinate system. This common alignment lays the groundwork for evaluating geometric deviations, covered in the following section.

2.2.4 Evaluating Manufacturing Deviations

After registration, it is required to evaluate how a manufactured measured structure's geometry differs from its nominal design [SCIM22]. This evaluation revolves around characterizing and modelling these manufacturing (geometric) deviations and is useful for quality control and process understanding. Deviations are defined as the measurable differences between the produced geometry and the nominal shape. They manifest in metrics such as dimensional, form, positional and orientation errors, and can be quantified by computing the shortest distances from each point in the scanned dataset to the nominal surface.

Distinguishing systematic from random deviations helps decide on appropriate modelling strategies: systematic deviations arise from consistent process errors and can often be modelled or compensated, whereas random deviations stem from unpredictable fluctuations such as tool wear or environmental variations. In advanced manufacturing such as additive processes, the ability to model and monitor these deviations is essential for quality assurance, statistical process control and diagnostic analysis [ZHU17].

a) Manufacturing Deviations

The evaluation of manufacturing deviations serves several purposes. Controlling deviations is central to manufacturing accuracy and quality assurance: dimensional and form errors, along with surface roughness, remain key challenges, especially in additive manufacturing. Effective geometric variation management ensures that the final product meets functional requirements despite inherent process variability. Deviation analysis supports process control and statistical quality monitoring as it characterizes and monitors the natural variability of complex shapes [ZHU17].

Systematic Deviations

Systematic deviations are deterministic, repeatable and predictable. They typically stem from characteristic errors of the manufacturing process, such as machine dynamics, clamping errors or residual stress. Due of their consistent nature, systematic deviations can often be modelled and compensated. Modelling approaches fit mathematical descriptions, commonly quadric (second-order) surfaces, to the deviation field [SCHL14]. These effects may be integrated into geometric models or subtracted from the nominal shape as part of compensation strategies [ZHU17].

Random Deviations

Random deviations arise from unpredictable fluctuations in production, such as tool wear, material inhomogeneities, or environmental conditions like temperature or humidity. They cannot be modelled with a single deterministic function and instead require statistical approaches. Figure 2-12 propose methods such as the 1D-Gaussian and multi-Gaussian models or, more effectively, random field theory [SCHL14]. In additive manufacturing (AM),

statistical shape analysis decomposes deviations into systematic and random components. The main modes of random variation are derived from experimental data to predict random shrinkage [ZHU17].

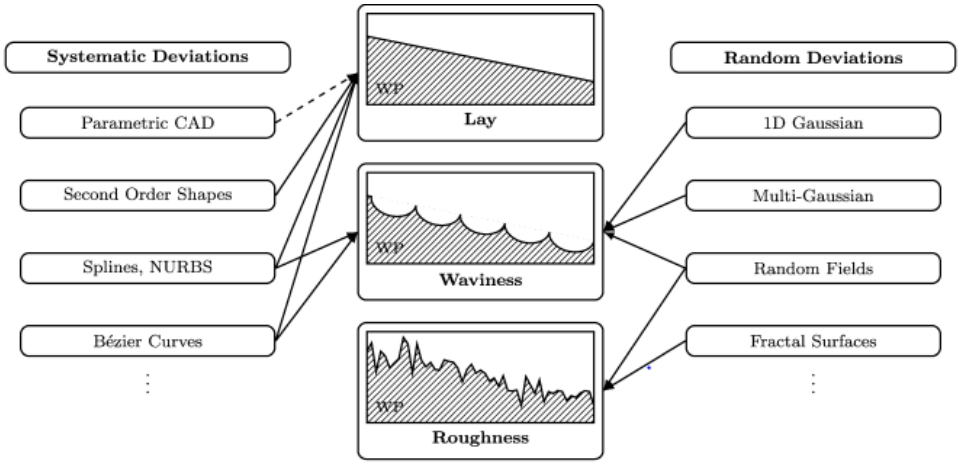


Figure 2-12 Overview over methods for the modelling of geometric imperfections [SCHL14].

Spatial Pattern and Characteristic Form

Beyond this classification, deviations can also be distinguished by their spatial pattern and characteristic form. Global deviations affect the entire component, warpage or overall shrinkage, and are generally smooth, continuous models like quadric surfaces or low-order splines describe them well. Local deviations occur in restricted areas, for example, missing struts or excess material around specific features and can introduce structural weakness. Characteristic form deviations relate to surface texture and are classified into lay, waviness and roughness. Understanding these spatial patterns supports the selection of appropriate modelling and interpolation strategies for geometric deviation analysis [SCHL14].

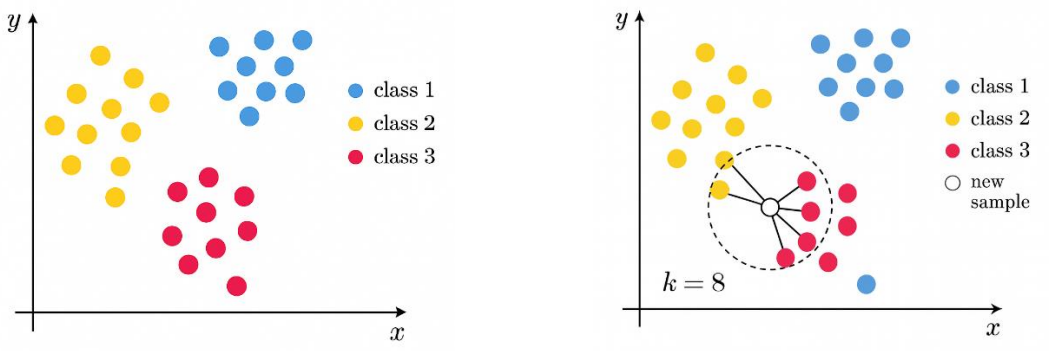
b) Mapping Techniques

Establishing correspondence between the measured geometry with the nominal surface enables the subsequent computation of deviation or displacement fields. This mapping process handles discrepancies arising from different sampling densities or mesh tessellations between the measured and nominal models [SCIM22]. Several key techniques and strategies are employed in practice to achieve this correspondence.

Nearest-Neighbor (NN) and k-Nearest-Neighbor (k-NN) Mapping

The simplest approach assigns each measured point to the nearest point in the nominal dataset using a distance metric typically Euclidean distance. In large scanning datasets, which often contain millions of points, brute-force search is infeasible. Efficient implementations therefore employ k-d trees, which hierarchically partition the nominal point cloud and enable nearest-neighbor queries with reduced computational complexity, lowering $O(N^2)$ comparisons to roughly $O(N\ log\ N)$ [PINK20].

The k-nearest-neighbor (k-NN) variant extends this by finding multiple neighboring nominal points. In mapping, the barycenter of the k neighbors can be used as the corresponding point, or the neighbors may be used later for interpolation [TREM19]. Figure 2-13 indicates the principle of k-NN correspondence, in which a sample point is associated based on the proximity of multiple neighboring samples. While NN/k-NN mapping is straightforward and fast, but it is sensitive to noise as noisy points may map incorrectly if the nearest nominal point is not the true correspondence.



(a) (b)

Figure 2-13 (a) Initial dataset consisting of class 1, 2 and 3; (b) k-NN used to categorize class of the new sample ($k = 8$) [DANK24].

Barycentric Mapping

Barycentric mapping exploits the structure of a mesh. For each measured point, the algorithm first determines which triangle (or polygon) on the nominal mesh is closest or encloses the projected point. It then expresses the point's position as a set of barycentric coordinates relative to the vertices of that triangle [RUST10]. These coordinates are weights that sum to one and are non-negative and they indicate how far the point is from each vertex. As barycentric coordinates are linear interpolants, they provide a continuous mapping where small movements of the point result in small changes in the weights. To transfer this correspondence to another mesh or to refine it, the same weights are applied to the corresponding vertices on the target mesh [DODI23].

Figure 2-14 illustrates this principle. A point q defined in a parametric domain is mapped onto a curved surface by expressing its position as barycentric weights with respect to the vertices v_1–v_4 of the enclosing surface element. Applying these weights reconstructs the corresponding point p on the 3D surface, demonstrating how barycentric coordinates enable consistent correspondence between planar parameterization and curved geometry [RUST10].

Advanced Mapping Strategies and Context

For complex shapes where explicit correspondence is difficult to establish, comparison metrics such as the Hausdorff distance are used. Instead of storing point-to-point correspondences, this approach computes bi-directional deviation maps that record distances between measured and nominal surfaces [SCIM22]. These maps enable the identification of defects such as

excess or missing material and are particularly effective for lattice structures or geometries with differing topology, where traditional correspondence methods may fail.

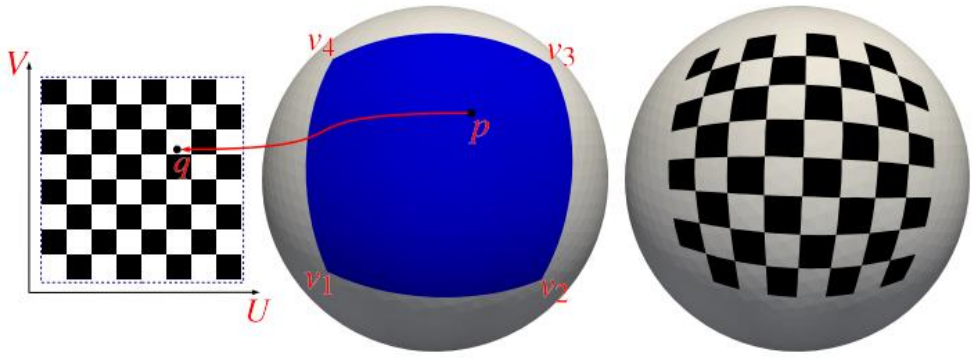


Figure 2-14 Barycentric mapping used to map point $\boldsymbol{q}$ onto point $\boldsymbol{p}$ [RUST10].

c) Interpolation

After mapping the measured geometry to the nominal surface and determining discrete deviation values, generating a continuous deviation field typically involves a spatial interpolation technique [RUST10]. Spatial interpolation is the process of estimating approximate values for locations where measurements are unavailable, based on known values measured at other sites [EMME20]. This method seeks a function that best represents the entire surface and allows for prediction at unsampled points [ZHEN18].

Inverse Distance Weighting (IDW)

IDW is an intuitive method that estimates a value at an unsampled location by averaging nearby measurements, with weights inversely proportional to distance. To perform IDW, one selects all measured points within a radius or chooses the k nearest neighbours. For each neighbour points a weight, $w_{i,j}$, is computed as:

$$w_{i,j} = \frac{d_{i,j}^{-\alpha}}{\sum_{k=1}^{n} d_{k,j}^{-\alpha}} \tag{2.1}$$

where $d_{i,j}$ is the distance to the interpolation point and α is a user-chosen exponent, often 2.

The interpolated value, $\hat{y}_j$, is then the weighted sum of the neighbors' deviation values:

$$\hat{y}_j^{IDW} = \sum_{i=1}^{n} w_{i,j} y_i \tag{2.2}$$

according to [EMME20].

IDW is deterministic and reproduces exact values at sample locations as shown in Figure 2-15. Since weights decrease rapidly with distance, points farther away have little influence.

However, IDW cannot predict values beyond the range of the data and it is sensitive to clustering as dense clusters exert disproportionate influence, potentially distorting the interpolated field [EMME20]. Adjusted versions of IDW introduce distance constraints or adapt the exponent α to the local point distribution, improving performance in unevenly sampled data.

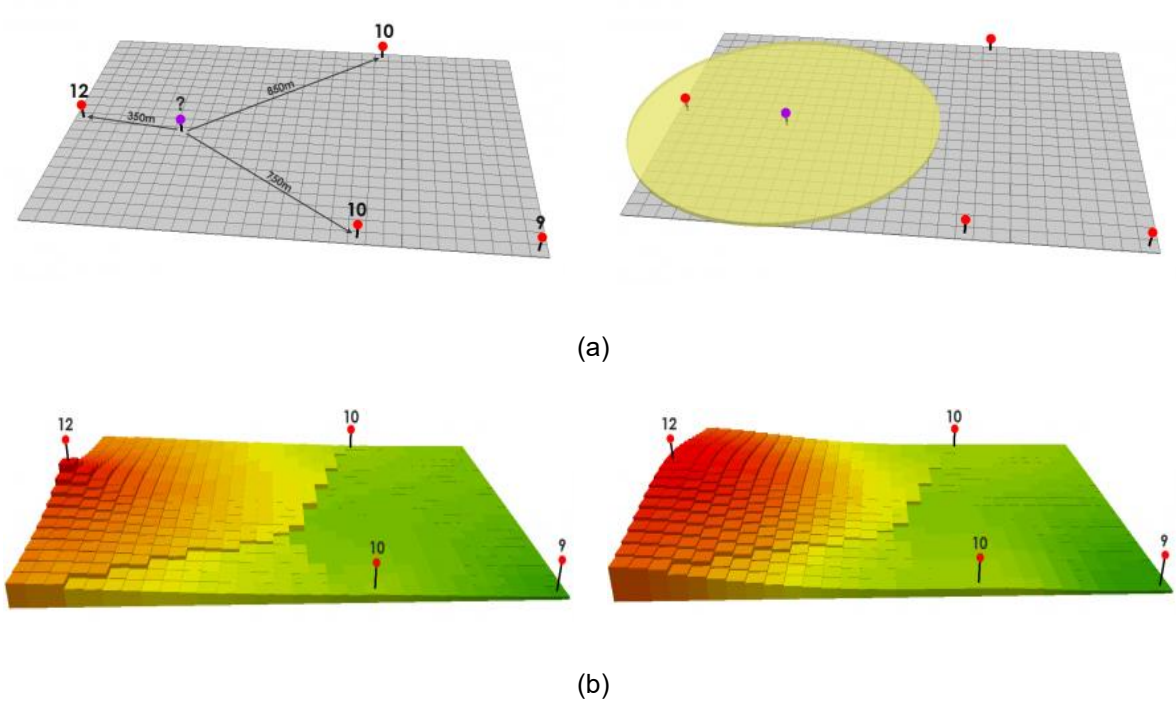


Figure 2-15 (a) IDW interpolation to identify the closest lying points to the purple point; (b) interpolation of height to generate an accurate heatmap [GISG16].

Radial Basis Function (RBF)

RBF methods model the interpolated value as a sum of radially symmetric basis functions centred at each data point [55]. A common kernel is the **Gaussian**, but other functions are also used. The interpolation function has the form:

$$f(x) = \sum_{i=1}^{n} c_i \phi(\|x - x_i\|) \tag{2.3}$$

where x_i are the data point positions, c_i are weights and ϕ is a radial basis function. To determine the weights, a linear system is solved such that $f(x_i) = y_i$ for all data points i (where y_i are the observed deviations) [ZOU13]. Once the weights are obtained, the interpolant is evaluated at any location by computing the basis functions' values at that location and summing with the weights. As each basis function affects the entire domain, RBF interpolation yields smooth, global surfaces. The main disadvantages of RBF are computational because solving the linear system has $O(N^3)$ complexity and must be repeated when the point set changes [ZOU13].

Spline-Based Interpolation

Spline-based methods fit piecewise polynomials across a domain. In two dimensions, bicubic splines or Bézier surfaces are common; on surfaces, spherical bivariate splines are used to interpolate data on the sphere [TREM19]. To apply splines, the domain is partitioned (e.g., into triangles or spherical segments), and polynomials are fitted subject to continuity conditions at the boundaries. Spline interpolation yields smooth, differentiable surfaces and captures global trends well. However, constructing splines on irregular domains can be challenging as selecting knot positions and controlling smoothness require expertise and the method may become unstable if data points are sparse or contain outliers [ZHEN18].

The combination of manufacturing deviation calculation, mapping and interpolation allows for the creation of a continuous representation of the deviations in form of a scalar field [ZHU17]. The accuracy of this process depends on factors such as point cloud density, nominal mesh quality and outlier treatment. The nominal surface is often approximated by a triangular mesh, which can introduce chordal errors, while interpolated surfaces are sensitive to noise and isolated anomalies [ZHEN18]. Computational cost is also a key consideration, as global methods such as RBF or splines typically involve higher computational complexity than local approaches [BUCE24].

In summary, Section 2.2 presents the state of the art related to the measurement, surface registration and deviation evaluation of manufactured structures. These concepts provide the theoretical and methodological background required for analyzing manufacturing-induced deviations and for interpreting deformation behavior in subsequent sections. Building on this, the following section focuses on approaches for approximating the GF state shape without jigs.

2.3 Jig-less Approximation of GF State

Accurately approximating the GF state shape has been an integral area of research because compliant components measured without a physical jig are susceptible to geometry changes caused by self-weight and clamping-induced effects. This section provides an overview of the state of the art in GF state shape reconstruction, beginning with fixtureless inspection principles and the development of jig-less assembly concepts, followed by deformation modeling and concluding by summarizing the existing approximation techniques.

2.3.1 Jig-less Assembly

Jig-less or fixtureless assembly refers to manufacturing and inspection processes that do not rely on traditional physical fixtures to position or constrain parts [SCHL16]. The main goal of fixtureless dimensional metrology is to remove the need for dedicated tooling while maintaining measurement accuracy. This approach is especially required for parts that naturally deform when unsupported, such as thin sheets or skins used in transportation industries [ABEN12]. In these methods, digital measurements and computational referencing replace mechanical constraints. The component surface is measured in a constrained position using simple support systems, and then numerical algorithms realign or virtually fix it for analysis [SABR17],

[TUOM17] . Figure 2-16 represents a support system used to hold an aerospace panel without applying any additional load.

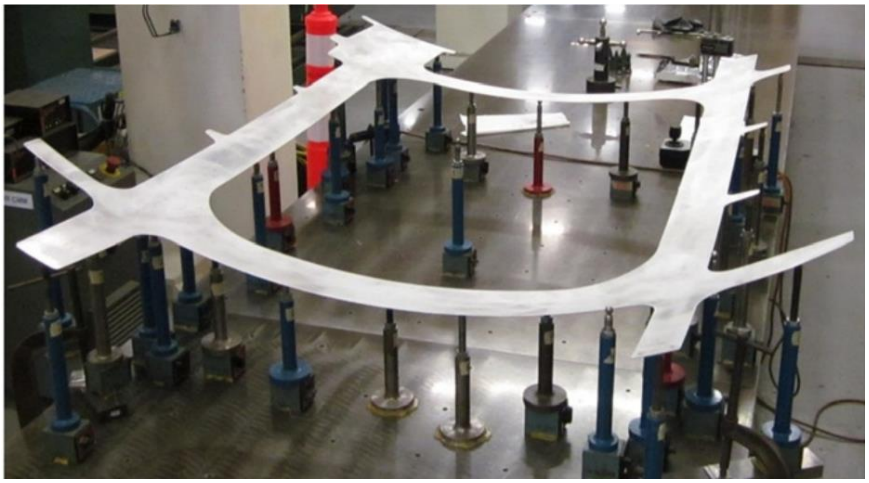


Figure 2-16 A regular aerospace panel constrained by fixing jigs in a simple support system [KARG17].

Industries have increasingly adopted jig-less assembly to reduce tooling cost, increase flexibility and shorten setup times [SCHL16]. Conventional jigs are typically large, heavy and expensive and they must be custom designed for each component. Their fabrication and maintenance requires significant time and resources. In some aerospace applications, fixture setup alone can take several dozen hours [SATT16]. The transition from rigid tooling toward flexible and fully digital assembly concepts is commonly structured into three stages: traditional rigid fixtures, reconfigurable or flexible fixtures and fully digital systems [KRIT18]. These approaches and their characteristics are summarized in Table 2-1, which highlights the shift from hardware-intensive setups to solutions relying predominantly on non-contact metrology and computational referencing. In practice, even digital systems often employ minimal physical clamping during measurement to prevent rigid-body motion [ABEN12].

Within this context, virtual clamping provides an alternative, as it numerically reproduces the behavior of physical fixtures by applying FE models to the CAD geometry. By imposing simulated constraints and computing the resulting elastic deformation, both free-state and virtually clamped configurations can be obtained from a single measurement [TUOM17]. This digital replacement of mechanical jigs enables flexible inspection and adaptive tooling concepts, but its accuracy is influenced by the definition of the GF state shape. Recent research emphasizes the need to first estimate the GF state shape from measurement data before applying virtual constraints [ABEN12], [TUOM17].

Table 2-1 Comparison of traditional, flexible and digital assembly approaches [KARG17].

Aspect	Traditional/Rigid Fixtures	Flexible Fixtures	Digital Systems
Primary concept	Dedicated mechanical jigs used to fix geometry and position of each part.	Modular or flexible structures with adjustable support for multiple part types.	Physical fixtures replaced by digital constraints and simulation models.
Setup time	Long setup time; manual alignment required.	Moderate; reconfiguration effort still significant.	Short; geometry alignment achieved digitally using scanning and registration.
Flexibility	Very low; one fixture per part design.	Medium; limited range of adaptable fixtures.	Very high; digital referencing supports any geometry.
Accuracy	High accuracy under controlled conditions.	Slightly reduced accuracy due to mechanical flexibility.	Accuracy dependent on boundary-condition modeling and measurement.
Compensation methods	Manual shimming or mechanical adjustment.	Semi-automatic adjustment or mechanical tuning.	Virtual clamping, virtual shimming or adaptive tooling.
Industrial examples	Automotive body-in-white lines, legacy aerospace tooling.	Modular fixtures in aircraft subassembly cells.	Digital twins, robotic fixtureless assembly, adaptive tooling systems.
Evaluation	Reliable but inflexible and costly.	Transitional solution towards digital manufacturing.	Core of Industry 4.0; high flexibility and data integration.

Building on these advancements, adaptive tooling has become a useful tool for fully digital and reconfigurable assembly. Instead of relying on rigid jigs or manual shimming, adaptive fixtures employ actuated supports capable of adjusting position based on feedback from measurement or simulation [SCHL16]. The GF state shape, once reconstructed from measured data, provides the reference shape for configuring these adaptive tooling systems. By combining feedback control with digital simulation, adaptive tooling, an example shown in Figure 2-17, allows shape correction and compensation during assembly without requiring physical jigs [ESPO22].

Alternatively, shimming remains a widely applied compensation method, particularly in aerospace assembly. In conventional workflows, part-to-part gaps are measured, bespoke shims are fabricated and inserted in the respective gap. Figure 2-18(a) illustrates manual gap measurement using a feeler gauge, while Figure 2-18(b) shows a gap compensated by a fabricated shim [ESPO22]. In contrast, virtual shimming embeds this procedure within a computational framework by performing numerical gap analysis and generating 3D shim geometries, typically for additive manufacturing [ESPO22].

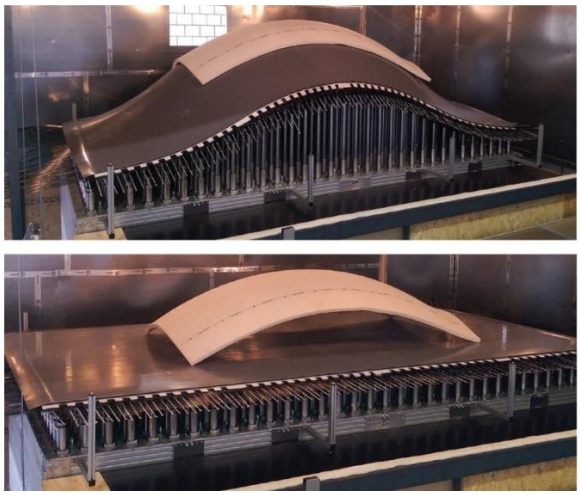
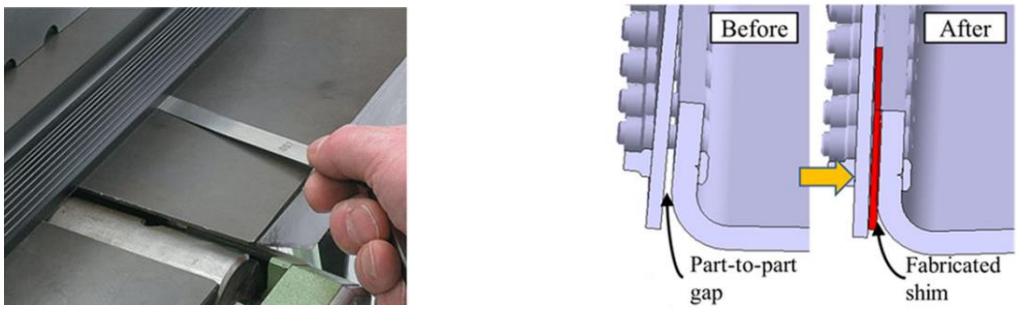


Figure 2-17 Adaptive tooling being used to deform a thermoplastic into desired shape [GARD17].

There are several industrial advantages to the reviewed integrated systems. They reduce the number of physical tools, shorten inspection and setup times and enhance process flexibility and accuracy [SABR17], [TUOM17], [GLOR18]. By coupling the GF state shape with virtual clamping and adaptive tooling, the design and optimization of assembly processes can be achieved virtually before physical production begins [CLAU21]. However, challenges remain with respect to deformation modeling, BC accuracy and measurement reliability. Therefore, resolving these problems is useful to obtaining high-precision outcomes in assembly settings [SABR17], [KARG17]. For this reason, the following section reviews state of the art for modeling deformation behavior of structures.



(a) (b)

Figure 2-18 (a) Manual feeler gauge to record gap measurement; (b) Visualization of a gap filled with a fabricated shim [ESPO22].

2.3.2 Modeling Deformation Behavior of Structures

The prediction of deformation behavior is a requirement across structural analysis, digital metrology and simulation-based workflows. Deformation models allow the structural response of a component to be analyzed under different conditions. These models are used in processes where the free-state geometry must be computationally related to a reference nominal form, defined in Section 2.1 [SABR16]. They also provide the basis for estimating unconstrained state shapes, including GF configurations [KARI20].

a) Types of Boundary Conditions

A deformation model requires a definition of boundary conditions (BCs), which determines how a virtual structure interacts with its surroundings and thus influences the global deformation pattern. The main BCs used in structural analysis are displacement (Dirichlet), traction (Neumann), mixed and contact/interface conditions [MITT23]. BCs identification ensures that equilibrium is fulfilled at the model boundaries as incomplete conditions can significantly distort the predicted displacement field [KARI20]. According to [KARG17] an accurate BC definition is often more critical to predictive accuracy than high-resolution meshing or advanced material modelling.

The mathematical representation of BCs follows from elasticity theory. Displacement BCs enforce specified positions or displacements:

$$x_i' = \bar{x}_i' \tag{2.4}$$

$$u_i' = \bar{u}_i' + \bar{g}_i' \tag{2.5}$$

where $\bar{g}_i'$ denotes any initial geometric gap on the boundary. Traction conditions specify the surface forces through:

$$t_i' = \bar{t}_i' \tag{2.6}$$

$$t_i = \sigma_{ji} n_j \tag{2.7}$$

where t_i is the traction vector, σ_{ji} the Cauchy stress and n_j is the outward normal [ZIEN14]. These equations provide the mathematical interface between the structural domain and its external environment.

Contact and interface BCs are used in modeling where involved surfaces are separable and can slide or interact through partial supports. The state of the art indicates that contact in assembly simulations is provided using node-to-surface search algorithms and gap evaluation functions [SABR16]. These methods enable the modeling of frictional effects and local load transfer during aligning or assembling structures. Interface BCs are also used for multi-material

modelling, where strong bonding is often assumed to simplify the problem and avoid explicit modelling of interfacial traction–separation laws [KRAU23].

b) Influence of Material

Material and geometric properties strongly influence a structure's deformation characteristics. Studies indicate that isotropic and layered materials, shown in Figure 2-19, exhibit distinct stiffness distributions, which affect bending, membrane behavior and load-transfer mechanisms near boundaries [KRAU23]. In addition, geometric attributes such as curvature, sectional stiffness or the presence of stiffeners can amplify sensitivity to the boundary conditions applied. Hence, modelling approaches incorporate material and geometric effects consistently to achieve physically accurate results [DENG25].

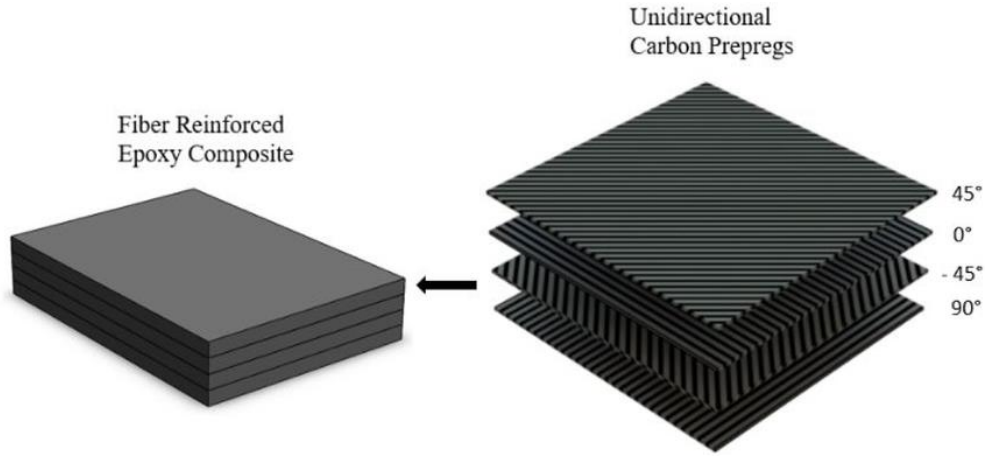


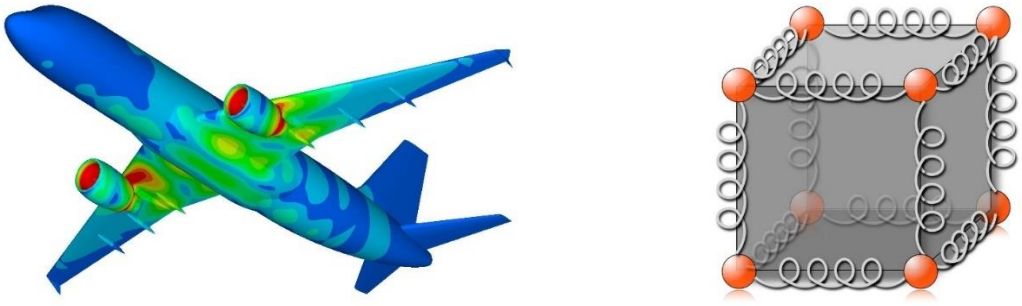
Figure 2-19 Schematic of a fiber-reinforced epoxy composite laminate built from stacked unidirectional carbon plies (left) with fiber orientations at 45°, 0°, −45° and 90° (right) [İPLI23].

c) Numerical Techniques

Amongst the techniques for predicting structural deformation finite element methods (FEM) remains the principal approach [MITT23]. Linear static FEM is commonly used for modeling involving small deformations and linear elastic behavior, particularly within fixtureless inspection and assembly simulation workflows [KARG17]. When rotations or displacements become large relative to the structural dimensions, nonlinear FEM approach is required to account for configuration-dependent stiffness [ZIEN14]. FEM, shown in Figure 2-20(a), is regarded as the reference method due to its generality, ability to incorporate complex geometries and suitability for detailed stress and displacement analysis.

Several approaches are used as an alternative to FEM. Mass–spring modeling, Figure 2-20(b), discretizes a structure using nodes and springs and is useful for the approximation of the deformation patterns with lower computational cost. However, this method is applicable for certain classes of components [MSS09]. Inverse finite element methods (iFEM) reconstruct deformation field from strain or displacement measurements, offering a means of rapid shape sensing without explicit load information [OBOE21]. More recently, physics-informed neural networks (PINNs) have emerged as mesh-free alternatives that embed elasticity equations within the training loss function, demonstrating accurate predictions for structures while avoiding certain FEM-related numerical challenges [BAST23].

These methods are useful in predicting the behavior of clamped or assembled shapes, assessing distortion after assembly and load application and identifying unconstrained or reference geometries [SABR16]. However, the methods face limitations related to boundary uncertainty, material behavior and computational cost. [MOGA24] states that inaccuracies in BCs represent one of the most significant sources of modelling error. Linear models may be inaccurate when materials exhibit anisotropy or nonlinear responses, whereas nonlinear FEM increases computational effort [BAST23]. Reduced-order models such as mass-spring can improve efficiency but may reduce predictive quality under complex loading or support conditions [JARR11].



(a) (b)

Figure 2-20 (a) FE simulation showing the spatial distribution of deformation across a complex structure [BATC14]; (b) a reduced-order mass–spring model, where nodal masses are connected through spring elements [JARR11].

In summary, modeling the deformation behavior of structures provides the basis for understanding structural response under various loads and supports. Through the appropriate selection of BCs, material models and numerical techniques, these methods support virtual assembly, shape prediction and inspection workflows [SABR16]. The following subsection builds on this by reviewing the state of the art techniques for reconstructing the GF state shape of a structure.

2.3.3 Methods for Approximating the GF State Shape

The GF state shape is defined in Section 2.1 as the theoretical geometry of a component in the absence of external loads, including its own weight. This definition is different from the free-state shape, which represents the measured configuration of a part subjected to gravity and residual stresses during inspection [CLAU21]. The distinction is shown in Figure 2-21, which shows how the free-state geometry, under gravitational loading, deviates from the GF state configuration.

As flexible and thin-walled structures deform under self-weight, the GF state shape cannot be directly captured without the introduction of additional distortions which are induced using clamping [ABEN12]. Hence, the approximation approach is required to estimate a geometry that reflects manufacturing deviations rather than gravitational effect. The reconstructed geometry is then used for dimensional inspection and digital assembly processes. The shape’s

accuracy is influenced by factors including material representation, BC definition and measurement quality [THIÉ17].

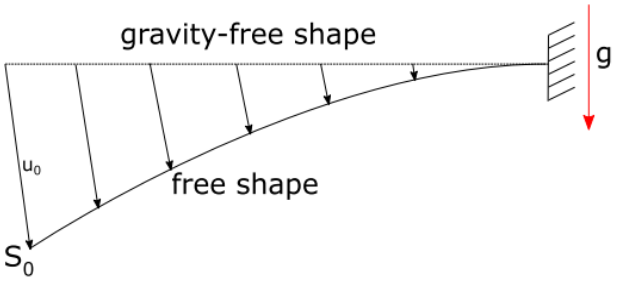


Figure 2-21 Conceptual difference between the free-state shape and the GF state shape, representing the configurations of the same part [CLAU21].

a) Approximation Techniques

A variety of approximation techniques have been proposed in the state of the art. Linear FEM-based gravity unloading is the common approach and assumes small deformations and linear material behavior [SABR17]. Nonlinear FE formulations have also been introduced to account for large deflections or complex geometries, particularly in cases where linear assumptions fail to approximate the physical behavior [ZIEN14]. Reduced-order approaches based on modal representations provide a simplified reconstruction of gravity-induced deformation and can reduce computational effort [ABEN12]. Across these approaches, approximating the GF state shape involves predicting and removing the deflection field from the measured geometry.

Several hybrid methods integrate measurements and simulation to increase robustness. These methods combine scanned data with an FE model that enforces physical consistency, allowing the GF state shape to satisfy both the observed geometry and the structural response of the component [THIÉ17]. Multi-posture scanning, where the part is measured in two or more orientations, provides additional information to isolate gravity effects by exploiting differences in deformation patterns. Other approaches rely on partial data fusion strategies to relate measured points to the expected unloaded configuration. These hybrid systems have demonstrated improvements in accuracy, particularly when material or BC parameters are difficult to determine precisely [CLAU21].

On the other hand, inverse and data-driven reconstruction methods provide an alternate approach. Inverse FEM (iFEM) formulations compute the GF state shape directly from the observed deformed configuration by solving the inverse equilibrium problem [CLAU21]. These methods reduce reliance on iterative forward simulations and can achieve faster convergence, however they require specialized solver implementations. Data-driven methods, including regression and learning-based deformation compensation, have been proposed as alternatives that use datasets to learn the mapping between free-state and GF state shapes. Their performance depends on the training data and the ability to remain physically consistent [ABEN12].

b) Validation Approach

Validation for the approximation methods is indirect because the GF state shape cannot be measured physically. A common approach is to compare the GF state shape with a fixture-supported or assembled measurement that acts as a surrogate ground truth [THIÉ17]. The reconstructed GF shape is virtually reloaded using the same fixture constraints, and its simulated deformation is compared with the same measured reference configuration to analyze accuracy. Simulation-based verification is also common, where synthetic deformed shapes generated from FE models are used to evaluate whether an approximation technique can correctly recover the known GF state shape under controlled conditions [ABEN12].

Another approach involves multi-posture validation, where the part is measured in two or more configurations and the GF state shape obtained from one configuration is used to predict the part deformation in another configuration. Consistency between the predicted and observed shapes indicates that the reconstructed GF state shape is physically meaningful [CLAU21]. The workflow is shown in Figure 2-22, where the GF state shape computed from situation 1 is used to predict the deformation of situation 2, and agreement between the actual and predicted shapes provide the required validation. These approaches provide evidence of reconstruction robustness and help quantify the sensitivity of GF state shape approximation methods.

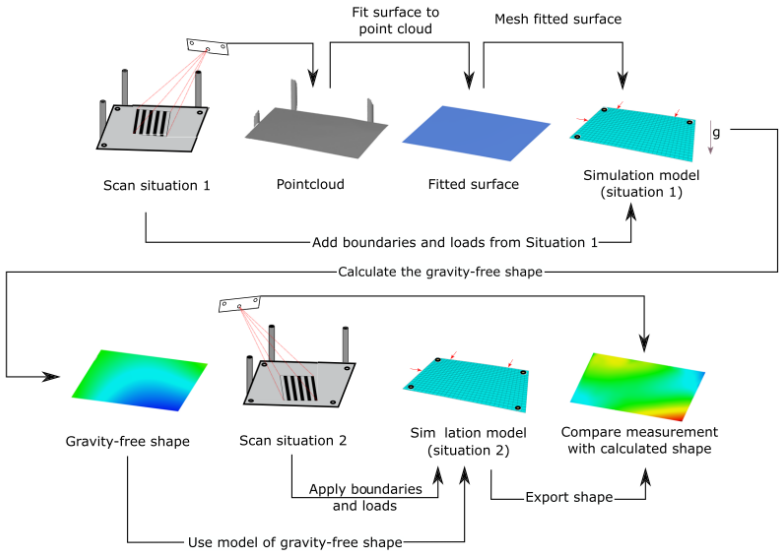


Figure 2-22 Overview of the validation approach adopted by [CLAU21].

In conclusion, GF state shape reconstruction enables the separation of manufacturing deviations from gravity-induced deformation and allows for accurate evaluation of nonrigid components. Current methods range from linear FEM unloading to inverse and hybrid approaches, each offering different balances between computational effort, accuracy and required inputs. Although challenges remain in material modelling and BC identification, these methods have demonstrated usefulness in flexible part inspection and virtual assembly workflows [SABR17]. The following section examines the advantages and limitations of the state of the art and provides an evaluation of their applicability within the broader context of flexible-part inspection.

2.4 Comparison of Related Work

The GF state shape approximation methods discussed in Section 2.3.3 demonstrate progress toward fixtureless inspection of non-rigid components. However, these methods differ in terms of modelling assumptions, treatment of measurement data and validation strategies. To clarify these differences and to identify the challenges, Table 2-2 provides an overview of the state of the art approaches. The comparison focuses on methodological characteristics and on the research gaps that limit the applicability of existing approaches.

Table 2-2 indicates that most existing methods rely on forward or inverse structural modelling to remove gravity-induced deformation from measured free-state geometry. Linear and nonlinear FEM-based unloading approaches are used due to their physical foundation and compatibility with standard simulation tools [SABR16], [ZIEN14]. Reduced-order and modal techniques offer computational efficiency but are generally limited to capturing global deformation behavior [ABEN12]. Inverse and hybrid approaches attempt to improve robustness by directly incorporating measurement data into the reconstruction process, often through multi-posture measurements or constraint-based inference [SABR17].

Despite their advantages, common research gaps are identified in the comparison. A primary limitation across the approaches is that the GF state shape is implicitly referenced to the nominal geometry, rather than representing the actual manufactured component in its unloaded state [CLAU21]. Manufacturing deviations are frequently treated as secondary effects or absorbed into deformation fields, rather than being explicitly preserved and propagated through the reconstruction process [SABR17]. As a result, the approximated GF state shape often reflects a corrected nominal model instead of the true unloaded configuration of the manufactured component.

Hybrid measurement–simulation approaches reduce reliance on purely numerical assumptions by combining measured geometry with structural modelling, hence offering a more consistent representation of the physical behavior of compliant structures. These methods are suited to problems where manufacturing deviations cannot be neglected. However, existing hybrid pipelines lack an integrated strategy that combines surface registration, manufacturing deviation evaluation and gravity compensation within a single coherent workflow [CLAU21], [THIÉ17]. Moreover, validation is often limited to controlled laboratory cases or simplified geometries and their applicability to large, geometrically complex industrial structures remain insufficiently demonstrated.

In conclusion, the comparison indicates that although state of the art GF state shape approximation methods enables fixtureless inspection and virtual assembly analysis, they remain constrained by workflows and implicit assumptions about reference geometry. In particular, the absence of methods that explicitly reconstruct the GF state shape of a manufactured component represents a critical research gap. In addition, the reported studies validate their methodology using moderately sized test parts, leaving uncertainty regarding scalability to large, geometrically complex structures being used in industrial assembly contexts. Based on this assessment, hybrid approaches were selected as the most suitable

foundation. This approach motivates the methodology presented in Chapter 3, which addresses the gaps by proposing an integrated and hybrid workflow for estimating the GF state shape of a manufactured LCTS from measured and simulated data.

Table 2-2 Comparison of GF state shape approximation approaches.

Approach	Core Principle	Validation Strategy	Research Gap
Linear FEM unloading [SABR16], [SABR17]	Identified deformation is simulated and subtracted from free-state measurements	Synthetic data and fixture-based comparison	GF state shape remains nominal-based and manufacturing deviations are not explicitly preserved
Nonlinear FEM unloading [ZIEN14]	Accounts for large displacements and geometric nonlinearity	Multi-posture and physical experimentation	High computational modelling requirement, strong dependence on material and BC accuracy and limited manufacturing deviation decoupling
Reduced-order methods [ABEN12]	Deformation approximated using limited basis functions	Numerical validation	Inability to capture local deviations and unsuitable for reconstructing GF state shape of a manufactured component.
Inverse FEM methods [CLAU21]	Unloaded shape computed by solving inverse equilibrium	Numerical and partial experimental validation	Limited validation on large, geometrically complex industrial structures and difficulty preserving deviation fields
Hybrid measurement–simulation [CLAU21], [THIÉ17]	Combines scan data with structural models and constraints	Fixture-based and multi-posture validation	Lack of integrated scan-to-GF workflow and approximated GF state shape does not preserve manufacturing deviations

3 Methodology

The following chapter outlines the methodology developed to approximate the GF state shape of a large compliant thin-walled structure (LCTS). It provides an organized description of the complete process, beginning with data acquisition and concluding with the approximation of the GF state shape of the manufactured component. This chapter explains the conceptual workflow and the motivation behind each methodological step, rather than focusing on the experimental conditions or numerical parameters, which are covered in the following chapters. The methodology is designed to address the limitations identified in the state of the art review in Section 2.4 by integrating measurement data and numerical simulation within a unified and hybrid workflow. The proposed approach is structured around three main approaches: the experimental procedure used to generate the measurement data, the numerical simulation that provides the reference behavior of the part and the automated pipeline which adopts methods that enable the approximation of the GF state shape of a manufactured component.

3.1 General Workflow

Section 3.1 introduces the workflow, providing a detailed description of the methodology and a brief description of each step's function within the workflow. The section also provides a summary of the terms and abbreviations used throughout the methodology. The subsequent sections go into further detail about each step, starting with the scanning conditions and fixture setup and moving on to alignment of the scanned component to the ideal deformed state of the component, displacement field calculation and GF state shape approximation of a manufactured component.

3.1.1 Terminology

Throughout the workflow, several terms and abbreviations are used to describe the geometric configurations and deformation states involved in the methodology. These notations distinguish between the measured, nominal, and simulated representations of the component, as well as the distance fields derived from their comparison. The definitions, which are summarized in Table 3.1, provide a reference for interpreting the relations between the measured and the simulated data.

The nominal shape, denoted as S_{free}^{nom}, represents the ideal GF state shape of the component in an unloaded condition and without any form deviations. When the same model is subjected to external loads such as gravity or clamping forces in the simulation, it deforms into the shape, S_{def}^{nom}. The measured configuration of the as-built component, obtained from 3D scanning under identical load conditions, is referred to as S_{conf}^{meas} [THIÉ17]. The objective of the methodology is to approximate the theoretical GF state shape of the manufactured part, S_{free}^{man}, which includes manufacturing deviations but excludes any external forces.

Table 3-1 Summary of abbreviations and definitions used to describe the geometric shapes and distance fields.

Abbreviation	Definition	Description
S_{free}^{nom}	Nominal shape	Represents the ideal GF shape as it is only subjected to the force of gravity [ISO10].
S_{def}^{nom}	Deformed shape	Simulated S_{free}^{nom} including the effect of external forces.
S_{conf}^{meas}	Measured shape	Measured shape of the as-built component
S_{free}^{man}	Gravity-free shape of a manufactured component	Includes manufacturing deviations but excludes any external forces.
E^{man}	Displacement field	Displacements due to manufacturing deviations
E^{conf}	Deformation field	Deformations due to external forces of gravity and clamping setup.

To describe the geometric relationship between these configurations, two distance fields are introduced. The first, E^{conf}, defines the deformation field caused by external effects such as gravity and clamping setup interaction, while the second, E^{man}, represents the displacement field induced by manufacturing deviations. Manufacturing deviations are defined as form errors arising from geometric inaccuracies that are either inherent in the material or introduced during the manufacturing process of the component. These two deformation fields are required for estimating the GF state shape of the manufactured component.

3.1.2 Workflow Overview

The overall workflow developed to approximate the GF state shape of the component is illustrated in Figure 3.1. It consists of two branches that combine into a unified process of alignment, deviation mapping and reconstruction. The left branch, Process A, represents the experimental pathway, where the geometry of the component is captured on the smart fixture and converted into a continuous digital model. The right branch, Process B, describes the numerical pathway, in which the nominal model is simulated under ideally boundary and loading conditions to reproduce the deformation caused by gravity and fixture interaction.

In Process A, the scanned point cloud of the large compliant thin-walled structure (LCTS) is transformed into a closed surface through an alpha-shape reconstruction algorithm. This yields the measured configuration, S_{conf}^{meas}. On the other hand, Process B begins with the nominal GF state model, S_{free}^{nom} , which is subjected to the same external conditions in a finite element simulation, resulting in the deformed configuration S_{def}^{nom}. These two datasets form the input files of the subsequent alignment steps.

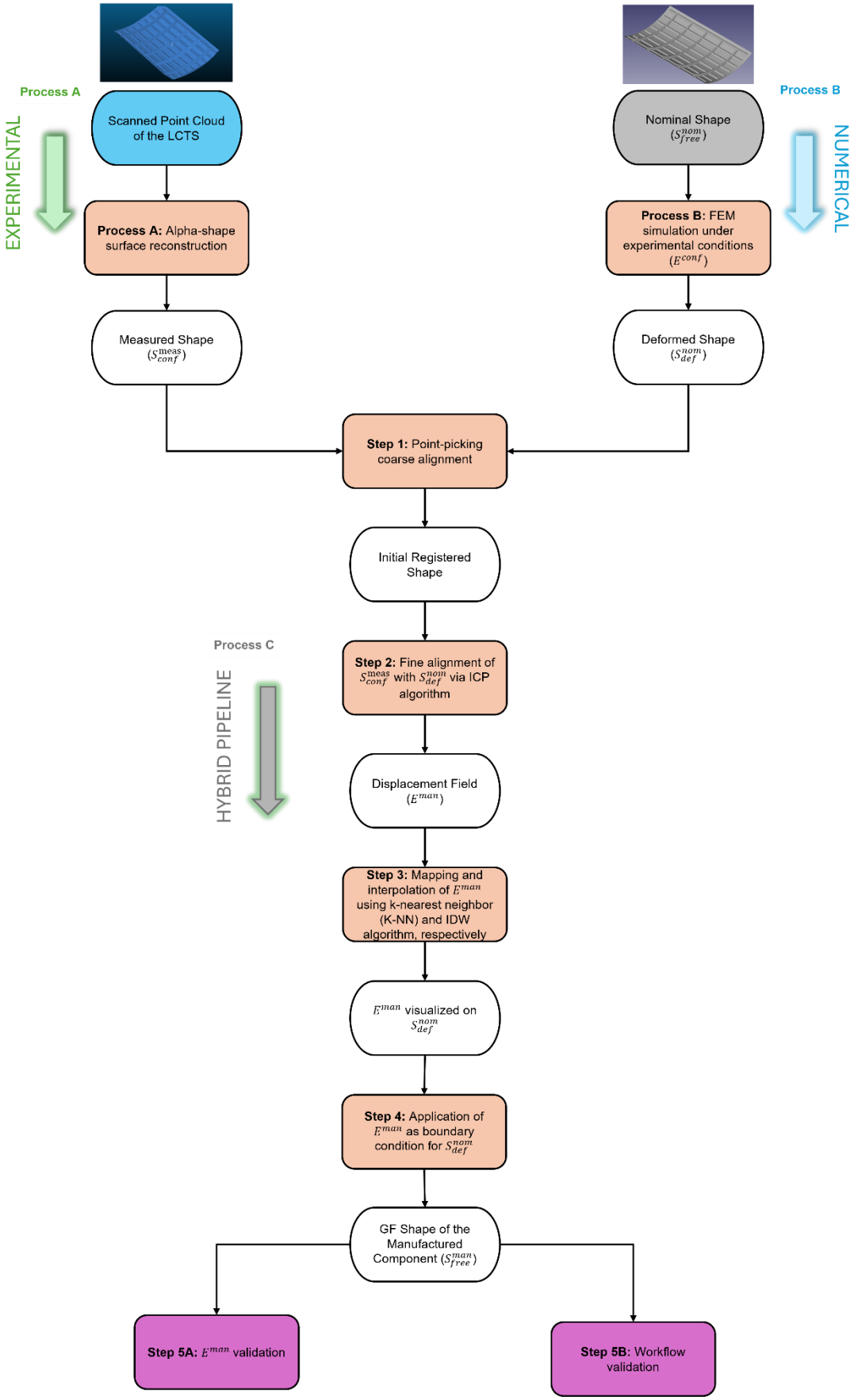


Figure 3-1 Workflow for the approximation of the GF shape, illustrating the data flow between the experimental and numerical domains.

The registration phase starts with a manual coarse alignment based on the point-picking method, where characteristic features of both geometries are manually selected to establish an initial registration. This initial fit is refined using an Iterative Closest Point (ICP) algorithm, which further minimizes spatial deviations between the measured and simulated surfaces. The outcome is a common coordinate frame that enables the calculation of the manufacturing deviation field, E^{man}.

Once the alignment is completed, the displacement field E^{man} is defined on the nodes of the measured configuration S_{conf}^{meas}. To ensure proper visualization and further numerical use, this field is mapped onto the nominal shape, S_{free}^{nom}, after which the missing values are interpolated. This step is essential for compensating differences in sampling density and mesh discretization between the measured and nominal datasets [SCIM22]. The mapping is performed using a k-nearest-neighbour (KNN) algorithm combined with inverse distance weighting (IDW) interpolation, allowing the displacements from the measured model to be accurately projected onto the nodes of the mesh of S_{free}^{nom}. The mapped displacement field, E^{man}, is then applied as a boundary condition to the nominal model, S_{free}^{nom}, effectively including the manufacturing deviations into an ideal GF shape, nominal shape. The resulting configuration represents the reconstructed GF state shape of the manufactured component, S_{free}^{man}.

Finally, the thesis includes two validation steps that verify the overall approach. The first assesses the accuracy of the computed displacement field, while the second verifies the reliability of the entire workflow through comparative simulations and consistency checks. Together, these stages ensure that the estimated shape, S_{free}^{man}, accurately reflects the manufactured component under ideal conditions. Building on this framework, the next section outlines the jig configuration and scanning conditions that form the basis for the experimental data used throughout the workflow.

3.2 Smart Jig Setup

The acquisition of the components' geometry requires a controlled setup capable of producing representative boundary conditions and maintaining consistent orientation relative to gravity. To achieve this, a dedicated test cell is used to capture both geometric and physical data of a LCTS made of CFRP. The configuration enables the measurement of this panel's deformation under its own weight as well as under additional controlled loads. This smart jig combines optical scanning, coordinate tracking and actuation systems in a single environment to ensure that the data collected under different loading states are directly comparable and spatially consistent. The schematic representation of the setup, showing the clamping configuration, gravity direction and scanning layout, is provided in Figure 3-2.

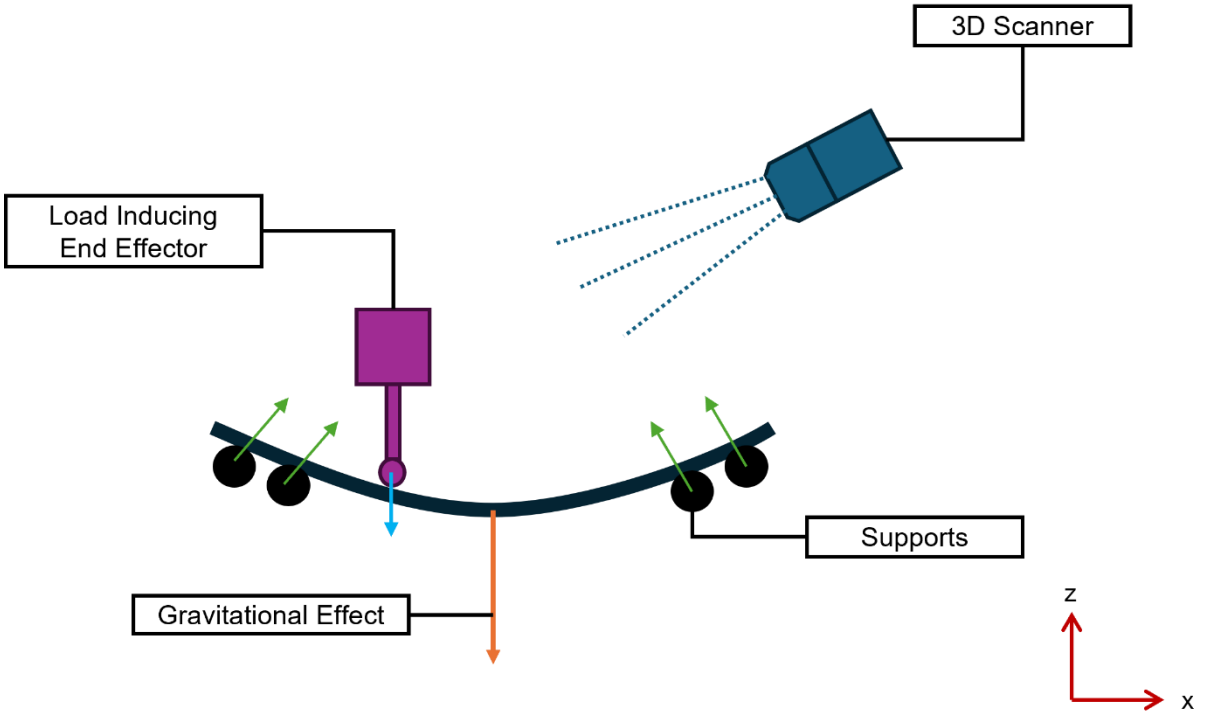


Figure 3-2 Conceptual schematic of the clamping and scanning setup showing the interaction between the load-inducing end effector, 3D scanner and supports used to measure the panel's deformation under gravity and external loads.

The panel investigated in this research is a non-rigid structure that deforms under its own weight. It is reinforced with stringers and frames and represents approximately one-sixth of the fuselage circumferential joint. In the full aircraft assembly, this panel forms part of the outer skin between neighboring frames, capturing the characteristic geometry and structural behavior of a typical fuselage subsection, as shown in Figure 3-3. It is supported on a test bed that follows a fixtureless clamping approach, in which the component is held only at essential reference points to avoid over-constraining the structure. Four linear actuators equipped with vacuum suction cups clamp the panel in position while allowing intrinsic deformation under gravitational loading. Each actuator integrates a force-torque sensor and an inertial measurement unit to record the magnitude and orientation of the forces applied to the panel. These reaction forces define the boundary conditions later reproduced in finite element simulations. Spherically mounted retroreflectors (SMRs) located at the base of the actuators allow for spatial referencing within a global coordinate system established by a laser tracker, ensuring that the mechanical and geometric data are recorded within a unified reference frame.



Figure 3-3 A typical fuselage section consisting of LCTS panels reinforced by stringers and frames [HEXA25].

The clamping configuration is used as part of a smart-jig setup to observe the panel's elastic response, rather than to replicate assembly conditions. The suction cups provide holding forces with minimal friction, while the actuators maintain quasi-static equilibrium during measurement. This design allows the panel to deflect under its own weight and the influence of the clamp, enabling accurate measurement of its elastic behavior. The assumption of rigid and frictionless support ensures that any deformation observed arises solely from the panel's self-weight or the clamping loads. The defined reference system and the recorded actuator forces therefore provide a consistent framework for comparing experimental and simulated data.

Gravity is a relevant factor for large, thin-walled aerospace panels, as their low bending stiffness makes them susceptible to sagging under self-weight. This behavior has also been reported in studies such as [CLAU21], which highlight that gravity-induced shape changes can influence measurement accuracy. In the measurement setup, the panel is positioned so that gravity acts normal to its curved surface, in the negative z-axis. The same gravitational orientation, in the negative z-axis, is reproduced in the finite element simulations to maintain consistency between the measured and simulated configurations. The laser tracker defines the global coordinate system, which is shared by all sensors and robots, ensuring that all measurements, whether of geometry, load, or displacement, are recorded in the same spatial reference frame.

The surface geometry of the panel is captured by a structured-light scanning system mounted on a robotic arm placed on a linear axle. The scanning robot provides flexibility in positioning and ensures full coverage of the large panel surface through sequential measurements from multiple viewpoints. Sequential acquisition is required because the scanner operates as a structured-light system with a fixed field of view (approximately $570 \times 460mm$ at the $600mm$ focus working distance). Due to this limited coverage per capture, the panel must be scanned in overlapping segments from different robot poses, which are later merged into a complete

reconstruction. The scanner's end-effector carries both the optical sensor and a 6D tracking target, allowing the laser tracker to determine its pose in the global reference system. Because the field of view of the scanner is smaller than the entire panel, several overlapping scans are taken and later merged to generate a continuous surface representation. The scanning process follows a non-contact principle to prevent additional deformation and the measurement resolution is selected to capture local shape deviations while maintaining efficient acquisition times. Surface reflections on the CFRP skin are reduced by applying a thin antireflective coating, whose thickness remains negligible relative to the scanner's spatial resolution.

In addition to self-weight measurements, the test cell allows the investigation of the panel under external loads. For this purpose, a secondary robot is integrated into the setup to apply controlled, pointwise forces to the structure. The robot operates with a force-torque sensor mounted on its end-effector and can impose precisely defined load magnitudes and directions. By measuring the geometry under several load cases, the panel's deformation response can be observed and compared against simulation results, forming part of the validation strategy for the reconstructed GF state shape. The applied loads are quasi-static and the robot's compliance is considered negligible compared to that of the panel.

The scanning operations are conducted under stable environmental conditions, with the jig placed on a rigid, vibration-damped surface to minimize thermal drift and external disturbances during acquisition. Each configuration, both self-weight and loaded, is measured following identical scanning procedures to ensure consistency across datasets. The resulting point clouds represent the measured configuration S_{conf}^{meas} of the panel under specific load states. These datasets, together with the recorded force and positional information from the test cell, serve as input for the registration and deviation analysis described in Section 3.3. The smart jig system developed for this setup is designed to be universally adaptable. Its actuators can be repositioned through manually operated lever mechanisms, each providing a stroke of approximately 30 cm, which allows the fixture to be configured for various geometries, including large compliant thin-walled structures.

3.3 Reconstruction and Registration

The alignment of the measured and simulated geometries is a prerequisite for a single coordinate frame and for quantifying the displacement field associated with manufacturing deviations. The scanning procedure described in 3.2 yields a dense point cloud of the panel in its loaded configuration, whereas the finite element simulation produces a discretized surface representing the panel under similar conditions. These two geometries originate from different coordinate systems and exhibit different sampling characteristics. A specific reconstruction and registration procedure is therefore required to establish a consistent relationship between them. This section introduces the surface reconstruction approach, evaluates alignment techniques and describes the alignment procedure used to relate the measured configuration S_{conf}^{meas}, to the deformed configuration, S_{def}^{nom}.

3.3.1 Surface Reconstruction from Measured Data

Although the raw point cloud from the optical scanner provides a dense sampling of the measured surface, it typically contains noise, outliers and local irregularities that make it unsuitable for direct registration without prior filtering and preprocessing. Differences in point density and the absence of neighbourhood information can lead to unstable correspondence matching and poor convergence during alignment. To overcome these limitations, the point cloud is converted into a continuous surface using an alpha-shape reconstruction technique [EDEL83]. The alpha-shape method constructs a surface as a subset of the Delaunay triangulation, a geometric structure that connects the points in a way that avoids overly sharp or skinny triangles. Based on this representation, the alpha parameter, α, controls how coarse or detailed the reconstructed surface becomes. For large values of alpha, the reconstructed shape resembles the convex hull, while smaller values reveal finer structural characteristics. This behavior allows the method to adapt naturally to the curvature and boundary behavior of the panel.

In this context, the alpha-shape reconstruction is used to obtain a stable and watertight mesh that preserves the global form of the measured geometry. This method was selected primarily because it is computationally efficient and, for the given data, produced more reliable and consistent reconstructions than alternative surface-reconstruction techniques. The resulting surface is sufficiently smooth to support the identification of geometric features used in the initial alignment and is also detailed enough to capture the shape variations relevant for manufacturing deviation analysis. Although the exact reconstruction parameters are presented later in Chapter 4, the conceptual role of this step is to transform the unstructured point cloud into a surface representation compatible with the simulated model. This mesh becomes the basis for all registration operations that follow.

3.3.2 Geometry Alignment

Upon completion of the surface reconstruction described in 3.3.1, the measured geometry, S_{conf}^{meas}, must be registered with the deformed shape, S_{def}^{nom}, to ensure that the observed differences represent physical deviations rather than inconsistencies in orientation or scale. Several registration approaches are considered for this purpose. Coherent Point Drift (CPD) is a probabilistic method that models the alignment as a density estimation problem and offers considerable flexibility for non-rigid shapes. The Spectral Decomposition Technique (SDT) embeds surfaces into a spectral domain and constructs correspondences using eigenvalue-based descriptions of geometry. Both CPD and SDT are well suited for applications where shapes differ significantly or where non-rigid behavior must be modelled explicitly [MYRO10].

However, the use of non-rigid registration introduces the risk of distorting the geometry of the panel, since such methods permit local deformations that may not reflect physical behavior. These techniques also require substantial computational effort and are more commonly applied to biomedical or animated surfaces rather than mechanical components. Maintaining the original geometry is required because one of the main objectives of the current work is to

align two configurations of the same structure under various loading conditions. For these reasons, a rigid registration approach is implemented.

In contrast, the panel considered in this work, although compliant, is reinforced through stringers and frames that stabilise its geometry, Figure 3-3. These stiffening features restrict large-scale rigid-body motions, which allows the deformation behavior under applied loads to be interpreted in a way that is consistent across measurement and simulation. In this configuration, the underlying assumption of linear superposition of loads can reasonably be applied. The measured configuration, S_{conf}^{meas}, and the simulated configuration, S_{def}^{nom}, therefore are assumed to differ only by a rigid transformation, since each dataset already incorporates the elastic deformation generated by its respective loading conditions. As a result, rigid registration is sufficient and preferable. Table 3-2 shows a qualitative comparison of the alignment algorithms, supported by Harvey Balls.

Table 3-2 Harvey Ball assessment of CPD, SDT and point-picking + ICP for the registration stage of the workflow.

Criterion	CPD	SDT	Point-Picking + Rigid ICP
Preservation of geometry	◔	◔	◕
Suitability for LCTS	◕	◕	◑
Robustness to noise	◕	◑	◑
Sensitivity to initial alignment	◕	◑	◔
Computational effort	◔	○	◕
Risk of artificial deformation	◔	◔	◕
Interpretability of result	◑	◔	●
Suitability for large surfaces	◕	◔	◑
Ease of integration in metrology	◑	◔	◕

Legend	
○	Insufficient Performance
◔	Limited Performance
◑	Moderate Performance
◕	Good Performance
●	Strong Performance

The comparison shows that each method offers distinct advantages depending on the type of geometry and deformation involved. CPD provides good robustness to noise but may introduce non-physical deformation, while SDT offers strong non-rigid matching capabilities at the cost of high computational effort. The point-picking and rigid ICP combination performs well when two datasets represent the same structure under comparable loading conditions, although it depends on a reliable initial alignment and cannot capture residual elastic mismatches. Based on the characteristics of the stiffened panel used in this research, the rigid method provides a practical and sufficiently accurate registration strategy.

a) Point-Picking Pre-alignment

The first step of the chosen alignment strategy is a coarse registration based on point-pair picking, as implemented in the **Align (point pairs picking)** tool in CloudCompare. This tool requires the user to manually select at least three pairs of corresponding points on the two shapes (S_{conf}^{meas} and S_{def}^{nom}) [CLOU25]. Once the points are selected in matching order, the algorithm computes a rigid transformation (translation + rotation) that aligns the two geometries, providing an initial estimate of their spatial relationship. This estimate ensures that both surfaces are positioned close enough for an initial guess to be provided to the ICP algorithm for ensuring its convergence.

Coarse alignment is particularly required for ensuring the robustness of ICP, since the algorithm is sensitive to large initial misalignments. By bringing the two surfaces into approximate correspondence, the point-picking step reduces the likelihood of erroneous convergence and improves the stability of the fine alignment process. Although the selection of feature points is performed manually, the number of points required is small and the procedure remains consistent across all measurement configurations.

b) Rigid ICP for Fine Alignment

Following coarse alignment, a rigid ICP algorithm further refines the transformation between S_{conf}^{meas} and S_{def}^{nom}. ICP operates by iteratively associating points on one surface with their nearest neighbours on the other and updating the transformation to minimise the resulting distance. Only rigid transformations, consisting of rotations and translations, are permitted in this approach. This restriction ensures that no artificial deformation is introduced into either dataset during registration. Since the elastic response of the panel is already included in both geometries, allowing non-rigid deformation during alignment would distort the geometry and mask the true form deviations that the workflow seeks to quantify.

The partial rigidity provided by the stringers and frames further supports the use of a rigid ICP algorithm. These stiffening elements maintain the structural form of the panel and prevent significant geometric divergence between the shapes. As a result, rigid ICP is well suited to refining the alignment and achieve a stable and accurate match. The algorithm iterates until the improvement in the alignment metric becomes negligible, at which point the final transformation establishes the common coordinate frame shared by both surfaces, Figure 3-4.

Surface reconstruction and registration establish the geometric relationship between the measured and simulated datasets. The alpha-shape reconstruction converts the scanned point cloud into a continuous and tessellated surface and the combination of manual point-picking and rigid ICP aligns this surface with the simulated deformation without altering its geometry. The resulting registered surfaces form the basis for computing the displacement field associated with manufacturing deviations. Section 3.4 introduces the derivation of this displacement field and its role in approximating the actual GF state shape of the panel.

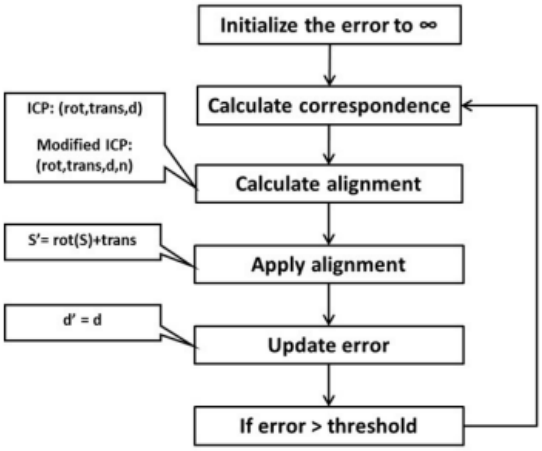


Figure 3-4 Workflow of the ICP algorithm, illustrating the iterative cycle until convergence [HONG14].

3.4 Manufacturing Deviations

Following the registration of the geometries, the measured and deformed shapes are related by a rigid transformation. Any remaining geometric differences between S_{conf}^{meas} and S_{def}^{nom} are therefore attributed to a combination of manufacturing-induced deviations and modelling inaccuracies. These deviations stem from factors such as curing distortions, tooling inaccuracies, fiber placement variations and other process-induced imperfections [SCHL14]. Since both shapes already contain the effect of elastic deformation due to gravity and clamping, the residual differences correspond to the manufacturing deviation field E^{man}, which must be expressed on a continuous, simulation-compatible surface for further analysis.

In the present methodology, manufacturing deviations are defined as the intrinsic geometric differences between the manufactured part and its nominal design. The measured shape, S_{conf}^{meas}, can be expressed as the nominal shape, S_{free}^{nom}, plus the deformation due to loading and clamping, E^{conf}, and the manufacturing deviation, E^{man} [THIÉ17]:

$$S_{conf}^{meas} = S_{free}^{nom} + E^{man} + E^{conf}. \tag{3.1}$$

The objective is to isolate E^{man} so that the S_{free}^{nom} can later be corrected to achieve the true GF state shape of the manufactured part, S_{free}^{man}. To achieve this, the displacement field derived from registration must be reformulated on the nominal surface S_{free}^{nom}. The scanned surface is a dense yet non-uniform point cloud. Although it captures the panel geometry accurately, the vertices do not correspond directly to the nodes of the finite-element mesh. Consequently, the displacement field cannot be directly visualized on the scanned measurement and must be transferred to the mesh of the nominal shape.

The first step of this transfer uses a nearest-neighbor mapping. After registration, the distances between corresponding points on S_{conf}^{meas} and S_{def}^{nom} are defined at the vertices of S_{conf}^{meas}. For each of these vertices, the closest node on S_{free}^{nom} is identified using a k-nearest-neighbor (k-

NN) search, and the corresponding distance value is assigned to that node. This process is conceptually similar to the neighborhood-based approach used by [THIÉ17], where the deviation between a mesh node and its closest neighbors in the measured point cloud is calculated along the normal direction. Mapping the deviations in this manner produces an initial displacement field on S_{free}^{nom}, but the field is incomplete because the number of vertices in S_{conf}^{meas} is smaller than the number of nodes in S_{free}^{nom}. Hence, few nodes remain without assigned deviation values.

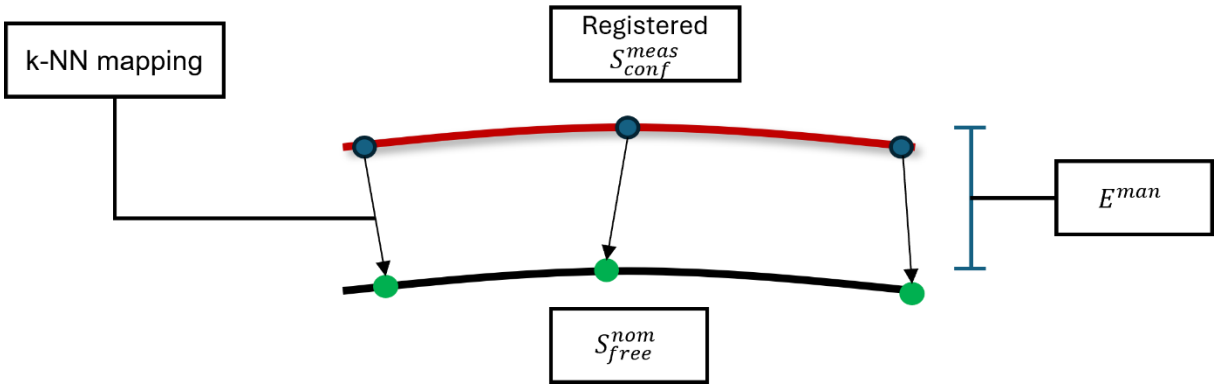


Figure 3-5 Conceptual schematic of the k-nearest-neighbour (k-NN) mapping between registered shape and nominal shape.

To obtain a complete representation, an interpolation step is required. Nodes on S_{free}^{nom} that did not receive a value during the k-NN mapping step are assigned one through inverse distance weighting (IDW). These nodes are provided with interpolated values by combining the distance values of their neighboring nodes, with weights inversely proportional to the Euclidean distance, to produce a smooth and continuous field across the entire surface. Interpolation compensates for local variations in sampling density, measurement noise and occlusions. A similar step is performed during the extrapolation stage in [THIÉ17], where areas of missing data or high digitizing noise are filled to obtain a complete cartography of the measured deviations. The resulting field is defined at each node of S_{free}^{nom} and represents the normal deviation of the measured shape S_{conf}^{meas} relative to the simulated deformed shape, S_{def}^{nom}. This displacement field, E^{man}, is essential for reconstructing the actual gravity-free state geometry, as it isolates the manufacturing deviations from the elastic deformations.

3.4.1 Displacement Field Validation

A validation step is pursued to confirm that the computed displacement field reflects intrinsic properties of the panel rather than the specific loading configuration used during measurement. Since the component is manufactured once, its form deviations in the free form from [ISO10], should remain consistent across all experimental setups. To verify this behavior, measurements were performed under nine different actuator (support) configurations, each producing a distinct loaded shape. For each configuration, the corresponding nominal deformation was simulated, the two surfaces were registered and the displacement field was computed using the same procedure.

The underlying assumption for this validation is that, for a given part, the field E^{man} should remain constant regardless of how it is supported and loaded during measurement. After computing the nine fields, each pair was compared using a root-mean-square (RMS) error metric, averaged over all points of the discretized surface. A low RMS difference indicates that the fields coincide, confirming that the observed deviations are indeed manufacturing-induced and not influenced by the external loads. The consistency of the deviation fields across all configurations validates the mapping and interpolation process and provides confidence that the resulting field can be used to reconstruct the gravity-free shape, S_{free}^{man}. The next section applies this validated deviation field to estimate the true unloaded geometry of the panel.

3.5 Approximation of the GF State Shape

Once the manufacturing deviation field has been determined, the final step in the workflow is to approximate the true gravity-free shape of the manufactured part. The measured configuration S_{conf}^{meas} and the deformed configuration S_{def}^{nom} both include the elastic response to gravity and fixture loads, E^{conf}, and their differences were shown in Section 3.4 to represent intrinsic manufacturing deviations, E^{man}. According to [ISO11] these deviations describe the discrepancy between nominal geometry and real extracted geometry. The deviation field E^{man} must now be applied to S_{free}^{nom}, which corresponds to the configuration of the part in the absence of any external loads, including gravity as defined by [CLAU21]. This differs from the ISO-defined free-state shape, which includes the effect of gravity [ISO10]. Using this definition, the resulting geometry S_{free}^{man} represents the GF state shape of a manufactured component, which cannot be directly measured and must be reconstructed using the nominal geometry and the deviation field.

The relationship between the various shapes can be described as follows. The nominal free shape S_{free}^{nom} is the design intent of the component in an unloaded state. When subjected to the loading condition E^{conf}, this geometry deforms into S_{def}^{nom}, which is obtained using finite-element simulation. The measured configuration, S_{conf}^{meas}, represents the manufactured component under the same loading conditions. In Section 3.4, the difference between S_{conf}^{meas} and S_{def}^{nom} was isolated to form the manufacturing deviation field E^{man}. Applying this field to the nominal free shape yields S_{free}^{man}, an approximation of the actual, unloaded geometry. This step therefore converts the theoretical nominal design into a representation that includes production-related form deviations but excludes any load-induced deflection.

The deviation field must be expressed on the nominal shape before it can be added as a boundary condition. The initial deviation values are initially associated with points on the measured surface, as explained in the previous section, and are therefore transferred to the corresponding regions of S_{free}^{nom}. A barycentric mapping approach is used for this transfer. For each measured point, the closest element on the nominal mesh is identified and the deviation value is distributed to the element's nodes according to barycentric weights. Barycentric coordinates generalize planar coordinates to surfaces and retain desirable properties such as unique reconstruction, edge linearity and smoothness. They are intrinsic, insensitive to

isometric deformations, and can be computed efficiently, which makes them particularly well suited for interpolating fields on complex surfaces [RUST10]. In the present context, barycentric mapping provides a more accurate and physically consistent transfer than the k-nearest-neighbour (k-NN) approach used previously for visualization of E^{man}. The barycentric approach Aligns with the way E^{man} was computed and ensures that the field can be applied reliably as a displacement boundary condition.

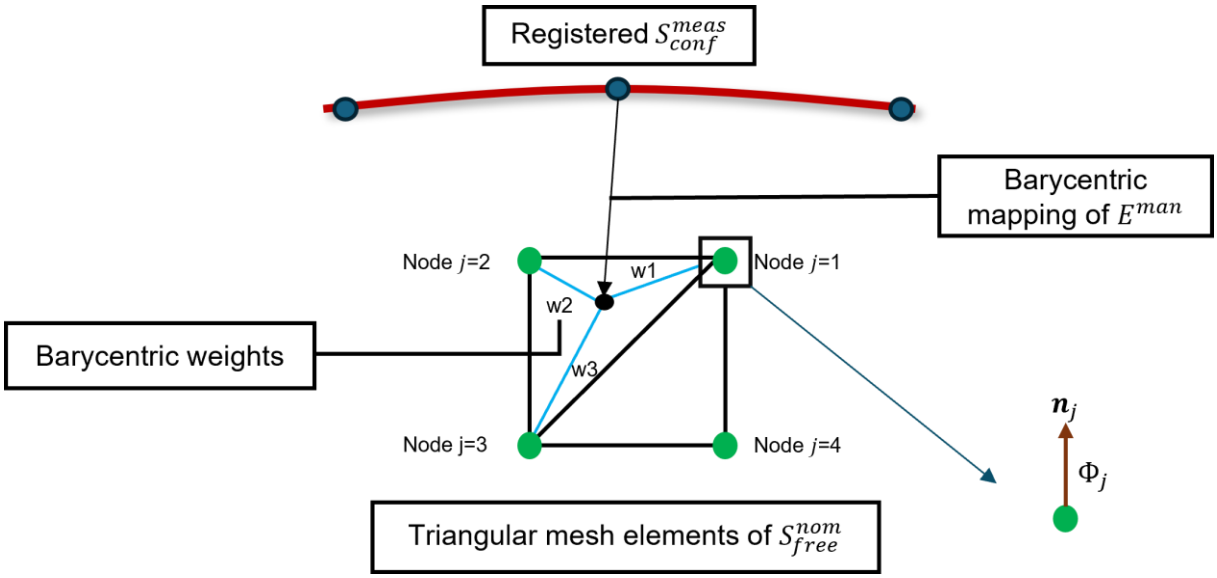


Figure 3-6 Illustration of the barycentric mapping used to transfer the manufacturing deviations from registered shape to the triangular elements of the nominal shape.

After mapping, each node of S_{free}^{nom} carries a deviation vector $\vec{U}_i$ whose components (U_i^x, U_i^y, U_i^z) describe the manufacturing deviation in the local coordinate system. The nominal free shape is then updated by shifting each node by its corresponding deviation components. This produces a new mesh in which every node reflects both the original nominal position and the measured manufacturing deviation. The resulting surface represents the approximated GF state shape of the manufactured panel, S_{free}^{man}. The method conceptually follows the idea proposed by [THIÉ17] for mapping and interpolating deviations, while the overall strategy of computing a gravity-free shape from measurements draws inspiration from approaches used in [CLAU21] and [SABR16].

3.5.1 Workflow Validation

A validation step is performed to verify that the approach yields a consistent approximation. This step confirms that the reconstructed gravity-free shape is consistent with both the measured data and the finite element simulations. Two intermediate shapes are compared: (i) the measured configuration corrected for manufacturing deviations and for gravity, and (ii) the nominal free shape modified solely by the fixture-induced deformation. These shapes are derived specifically for validation and are not themselves gravity-free. The first shape is obtained by subtracting the gravity-induced deformation from S_{conf}^{meas} and then adjusting for the manufacturing deviation field; the second is obtained by simulating S_{free}^{nom} under the fixture loads

alone. If the workflow and assumptions are accurate, the two shapes should be similar. Agreement between the shapes indicates that the deviation mapping and application steps have correctly isolated manufacturing deviations and that the resulting S_{free}^{man} is a reliable approximation of the true gravity-free geometry.

The concepts introduced throughout this chapter establish the methodological foundation for approximating the GF state geometry of the manufactured panel. The chapter first outlined the workflow and its terminology, then described the clamping and scanning conditions that define the measurement environment. The subsequent sections introduced the reconstruction of the scanned surface, the selection of an appropriate registration strategy and the derivation of the manufacturing deviation field. Finally, the procedure for mapping these deviations to the nominal free geometry and reconstructing the corresponding GF state shape was presented. Together, these steps define the complete conceptual pipeline that supports the experimental and numerical investigations undertaken in this work. Further chapters build upon this framework by describing the experimental procedures, simulation setups and implementation aspects required to carry out the methodology in practice and to evaluate the resulting data.

4 Experimentation and Hybrid Pipeline

The following chapter describes the implementation of the methodology developed in Chapter 3. While the previous chapter established the conceptual framework of the workflow, the current chapter and Chapter 5 focuses on the execution of these concepts. The first part outlines the experimental setup and the measurement strategy adopted to capture the loaded configurations of the component. The second part details the computational operations used to process, reconstruct and align the acquired data in preparation for the deviation analysis and GF state shape approximation.

4.1 Experimentation

The experimental phase provides the measured configurations of the panel under controlled loading conditions. The section introduces the measurement environment, the key components of the test bed and the scanning procedure used to acquire the surface data. The discussion is limited to the aspects necessary for understanding how the raw data is acquired before being processed and modelled in the subsequent computational steps.

4.1.1 Experimental Setup

The measurement setup used in this research is based on the experimental infrastructure developed for the investigation of large compliant structures that are non-rigid in nature. The smart jig configuration provides a controlled clamping setup in which the panel can be supported, manipulated and scanned under defined conditions. The aim of this section is to describe the elements of the setup, shown in Figure 4-1, that are relevant to understanding how the measured configuration, S_{conf}^{meas}, was obtained.

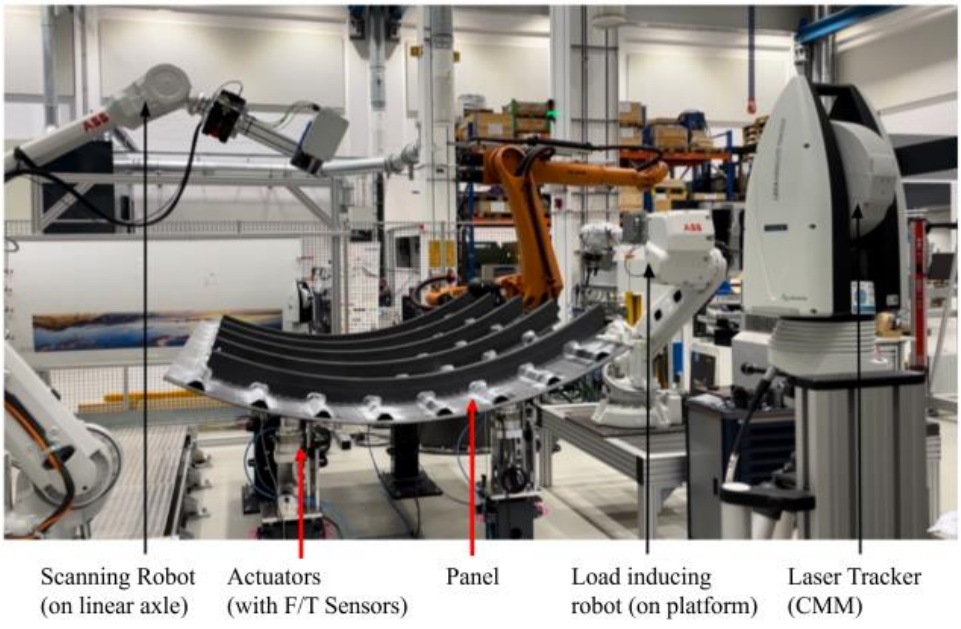


Figure 4-1 Smart jig configuration for capturing the geometry of large compliant thin-walled structures (LCTS) [TRÉH25].

The panel is mounted on a smart jig system equipped with four linear actuators, each fitted with a vacuum suction cup controlled by a pneumatic valve connection, to provide stable support during testing (Figure 4-2). In addition to their clamping function, the actuators can translate vertically along the global z-axis, allowing the panel to be positioned at different heights when required. The actuator body is controlled through mechanical guide pins, and the actuator head is regulated through the connected pressure valves, ensuring precise head movement when the panel is placed on the clamping setup. The panel also carries a set of spherically mounted retroreflectors that serve as reference features for positioning within the global coordinate frame.

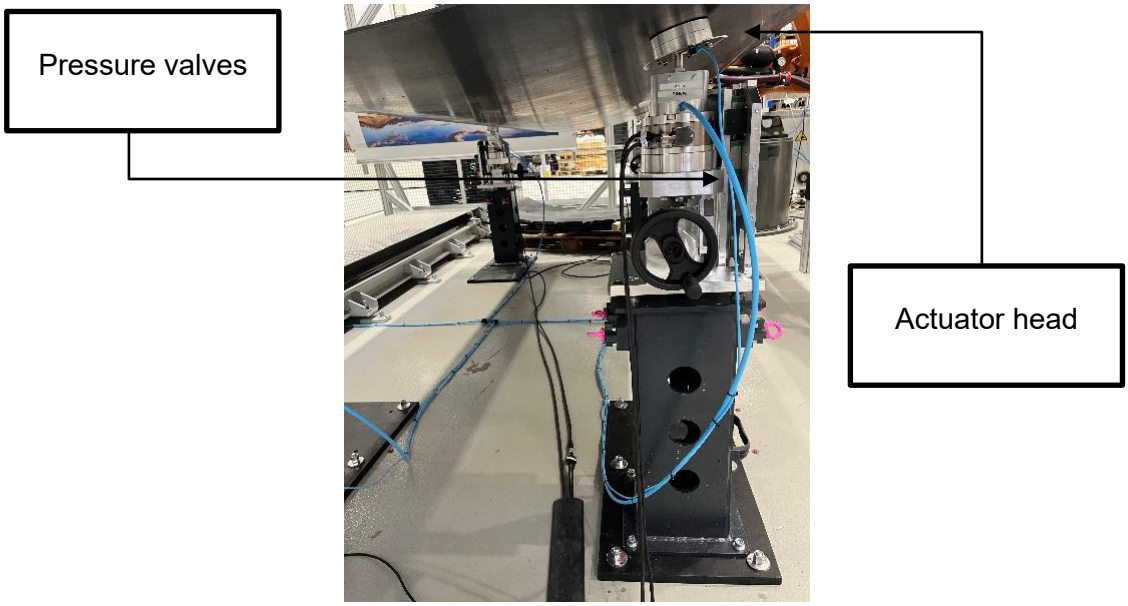


Figure 4-2 Clamping arrangement of the smart jig with one of the actuators connected to the vacuum system through the blue pneumatic pipes.

Surface measurements are acquired using a structured-light scanning system mounted on an industrial robot. A Zivid 2+ M60 scanner is used as the primary sensor and is mounted on an ABB IRB 4600-2.55 robot placed on a linear axis which enables it to reach the complete surface of the panel. The scanner pose is tracked from the Laser Tracker AT 960 MR using a Leica T-Mac target, which ensures that all measurements are recorded in a common coordinate system (CMM) [TRÉH25]. The arrangement of the robot and scanner is shown in Figure 4-3.

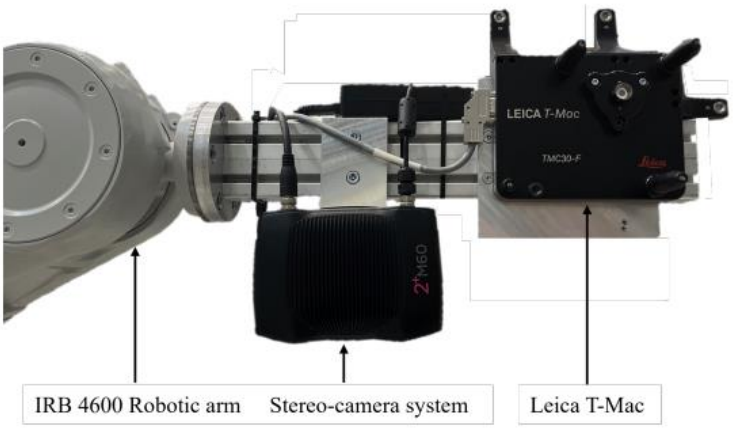


Figure 4-3 Zivid 2+ M60 scanner and a Leica T-Mac mounted on the Scanning Robot's arm [TRÉH25].

To generate the deformed states required for the measurement, a second industrial robot (ABB IRB 4600-2.55) applies localized loads to the panel by means of a tooling ball mounted on a force–torque sensor, Figure 4-4. This configuration allows different loading conditions to be realized in a repeatable manner. All components of the setup, including the scanning system, the load-inducing robot and the actuators, operate within the global coordinate frame defined by the laser tracker, which provides a consistent reference throughout the experiment.

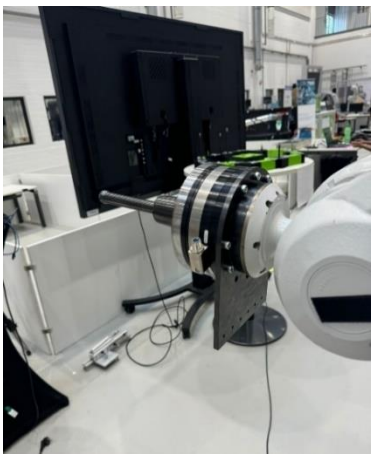


Figure 4-4 End-effector assembly mounted on the Load Inducing Robot's arm.

The combination of these elements comprises of the complete experimental setup and forms the basis for the acquisition of the panel under differing configurations. The next section describes the surface data collection using the Zivid 2+ M60 scanner.

4.1.2 Data Acquisition

The acquisition of the measured panel geometries followed a consistent scanning procedure across all configurations. The scanner is positioned at its recommended focus distance of 600 mm to ensure that the captured data reflected the sensor's optimal depth accuracy. Operating within this range allows the surface geometry to be recorded with a level of detail suitable for the subsequent reconstruction and alignment steps. Multiple scans are recorded for each

configuration. The size of the panel and its curvature require several viewpoints to obtain complete coverage and to reduce occlusions.

To improve measurement reliability, an anti-reflective spray is applied to the panel before scanning. The CFRP surface can produce glare and local intensity variations that affect reconstruction quality, and the spray helps minimise these effects while preserving surface fidelity. The application is kept consistent across all measurement configurations.

Each scan is stored as a point cloud in the point cloud (*.PCD*) format. The individual scans contained overlapping regions to support the merging operations which results in a point cloud representation of the measured shape, S_{conf}^{meas}. The resulting datasets forms the input for the computational modelling steps that follow.

4.1.3 Design of Experiments (DOE)

The panel is measured under nine different actuator configurations to capture its response to controlled clamping variations. These configurations are designed to introduce quantifiable deformation patterns while keeping all other measurement conditions identical. As the panel is manufactured only once, modifying the actuator positions provides multiple loaded states of the same geometry, which later enables the validation of the displacement field derived through the computational workflow.

The selection of actuator settings follows a fractional factorial design prepared using a statistical experimentation tool, Minitab [KLEP91]. The resulting layout offers a structured set of actuator combinations without requiring unnecessary exploration of all possible arrangements due to the large number of possible experimental combinations and the time required for single experiment. The actuator positions and lock states are reported in Table 4-1. The underlying statistical structure forms part of the original DOE formulation but is not required for the methodology described in this thesis.

Each actuator could be either free to move in three translational directions or mechanically constrained. A lock value of 0 indicates that the actuator is pinned in the horizontal plane, allowing motion only along the vertical z direction, while a value of 1 reflects a fully unlocked state. The lock sequence is written in the order of the four actuators, for example 0,0,1,0 means that only actuator 3 was unlocked during that measurement, whereas 1,0,0,0 means that only actuator 1 was unlocked. This control mechanism ensured that the influence of horizontal and vertical constraints could be investigated consistently across all configurations.

The actuator values listed in Table 4-1 represent the upward displacement applied along the positive z-direction. Across the first eight configurations, all four actuators are engaged in the measurement process, each contributing to a distinct deformation pattern depending on its locked state and displacement. Configuration 9 represents the only exception: in this case, actuator 1 does not participate in the loading sequence, which is why no displacement value is reported for that actuator. The remaining three actuators are still adjusted according to the prescribed settings, allowing this configuration to introduce an additional, deliberately asymmetric deformation state. After each actuator arrangement was applied, the system is

allowed to stabilize before scanning. This ensured that all nine datasets are collected under controlled and repeatable conditions.

Table 4-1 Actuator lock states and vertical displacements for the configurations used in the measurements.

Run Order	Locks	Actuator 1 (mm)	Actuator 2 (mm)	Actuator 3 (mm)	Actuator 4 (mm)
1	0,0,1,0	0	0	0	0
2	0,0,1,0	0	15	0	15
3	0,0,1,0	15	0	15	0
4	0,0,1,0	15	15	15	15
5	1,0,0,0	15	0	0	15
6	1,0,0,0	15	15	0	0
7	1,0,0,0	0	0	15	14
8	1,0,0,0	0	10	0	10
9	0,0,0	-	15	5	15

These nine measured configurations form an essential input for the subsequent analysis. Although the deformation induced by each configuration differs, the underlying manufacturing deviations remain constant. The resulting displacement fields obtained from the nine datasets are therefore expected to be consistent with one another. Their comparison in later sections provides a direct means of assessing the robustness of the overall pipeline and supports the validation of the displacement field obtained due to manufacturing deviations.

4.2 Hybrid Pipeline

This section focuses on establishing a consistent and robust pipeline for the computational framework used in this research. The methods outlined in this section deal with the processing and reconstruction of the raw point clouds, the registration of the measured geometries with their corresponding nominal representations and the visualization of the displacement field. These operations convert the scanned measurements into structured datasets and form the basis for the approximation of the GF state shape of the manufactured component presented in Chapter 5.

4.2.1 Data Processing and Surface Reconstruction

The point clouds obtained from the Zivid 2+ M60 scanner serve as the basis for the subsequent computational steps. Each scan is captured at the scanner's recommended focus distance of 600 mm, where the manufacturer specifies a spatial sampling resolution of approximately 0.24 mm and a depth precision of 80µm for repeated measurements. These characteristics determine the achievable point density and noise level and therefore guide the selection of suitable processing parameters. The workflow applied in this section transforms the raw output into a clean and continuous representation of the panel surface. Initial data processing operations are performed in CloudCompare, whereas the surface reconstruction is carried out using the Open3D library [OPEN25a], [OPEN25b], [OPEN23].

a) Data Processing

The raw merged point cloud, Figure 4-7(a), contains approximately 61.6 million points and exhibits local outliers caused by reflections and minor inconsistencies in the acquisition. Statistical Outlier Removal (SOR) tool from CloudCompare is used as the first processing step to suppress these artefacts. A neighborhood size of mean distance of 25 and a standard-deviation threshold of 2.00 are selected. For a spatial resolution of 0.24 mm at the chosen working distance, this neighbourhood size corresponds to a local patch that is sufficiently large to capture meaningful geometric trends while remaining sensitive to the natural curvature of the surface. Applying the filter reduced the dataset to 48.9 million points, removing isolated points around the periphery and regions where local depth noise exceeded the scanner's nominal precision. The SOR configuration and its effect on the cloud are shown in Figure 4-5.

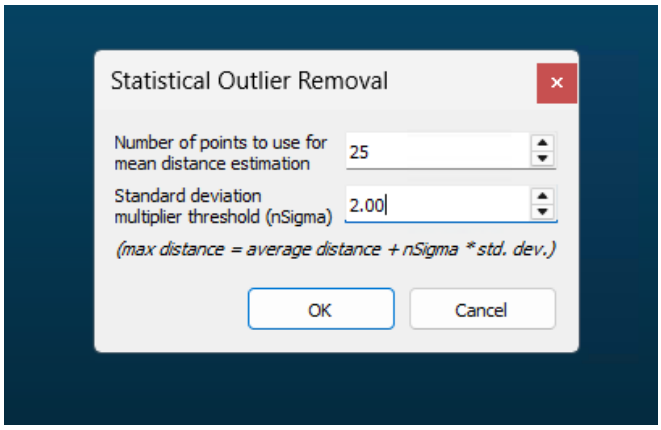


Figure 4-5 Selected distance and deviation parameters.

Following noise reduction, surface normals are generated to support reconstruction and registration. Normals are computed using the quadric local surface model, which approximates the local surface curvature within each neighborhood. A search algorithm with an automatically determined radius of 1.2 mm is used, which is aligned with the spatial sampling characteristics of the scanner. A neighborhood size of k = 12 is chosen for the normal estimation tree, reflecting the density of points expected in a 0.24 mm sampling grid. Normals are oriented consistently using the scanner's viewing direction to avoid flipped or inconsistent orientations, which are common in large datasets. The normal estimation configuration window is shown in Figure 4-6.

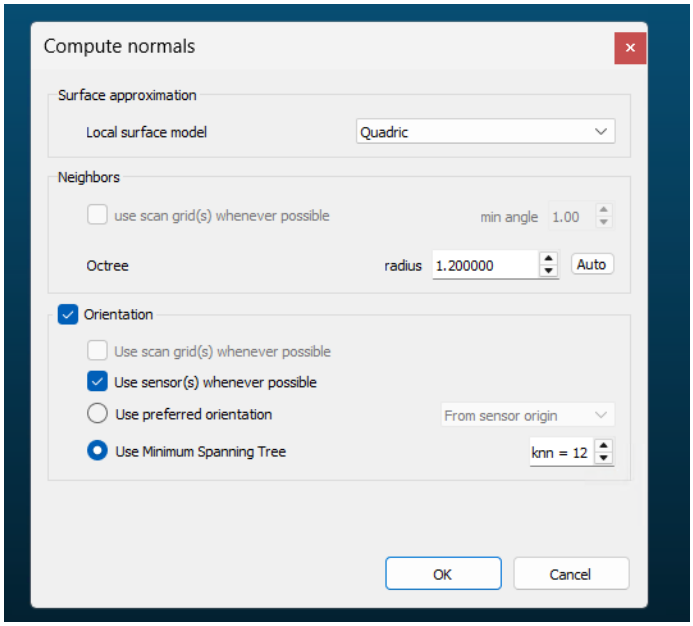
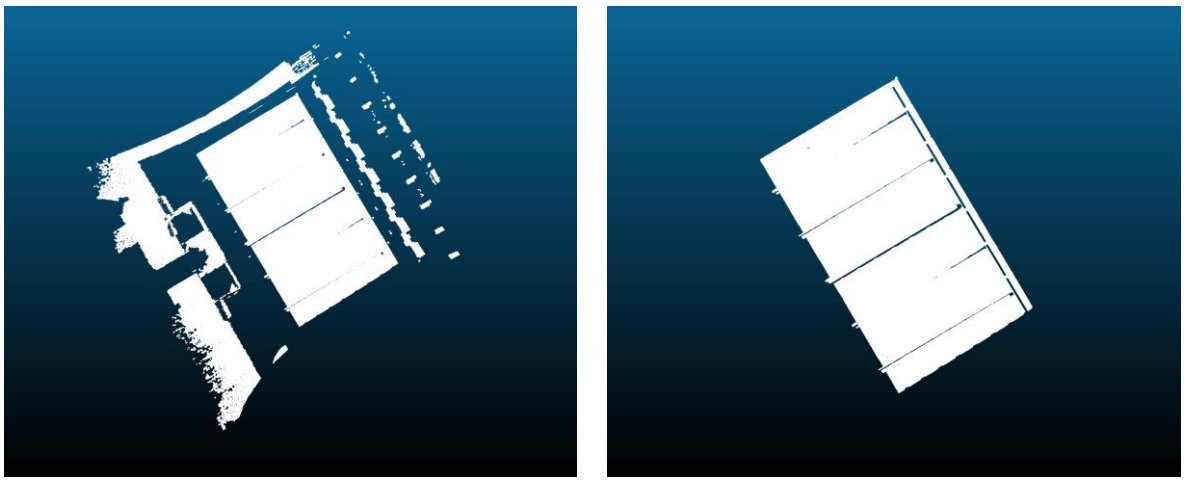
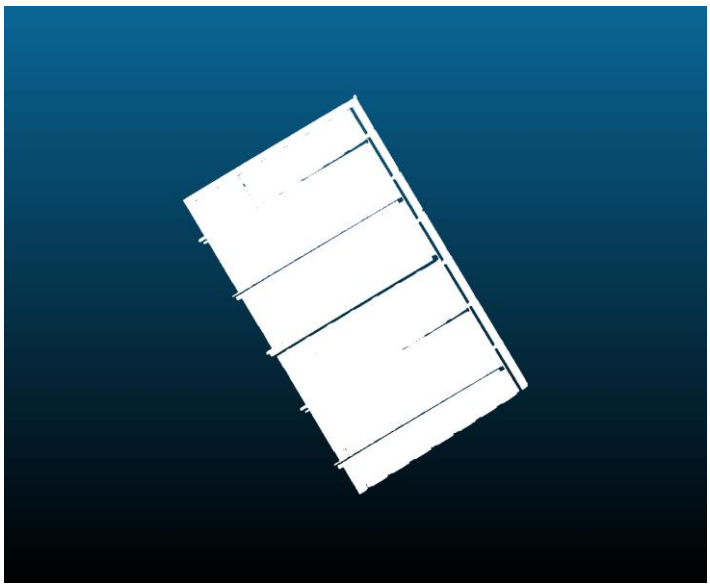


Figure 4-6 Normals were estimated using a quadric surface model with an octree radius of 1.2 mm and k = 12 neighbors and orientation aligned with the scanner direction.

After normals computation, the next step involves sampling the cloud to improve computational efficiency. The dataset is reduced to approximately 26.3 million points, corresponding to a down-sampling ratio of approximately 50%. To accurately represent structural features such as the stiffener interfaces and skin transitions, consideration is taken to maintain increased point density in regions with evident curvature. This step substantially reduces memory requirements and computation time for later reconstruction and registration, without compromising surface fidelity. The unfiltered, filtered and down sampled states of the point cloud are illustrated in Figure 4-7.



(a) (b)



(c)

Figure 4-7 (a) Raw unfiltered point cloud of the panel; (b) filtered point cloud after SOR; (c) down sampled point cloud.

b) Surface Reconstruction

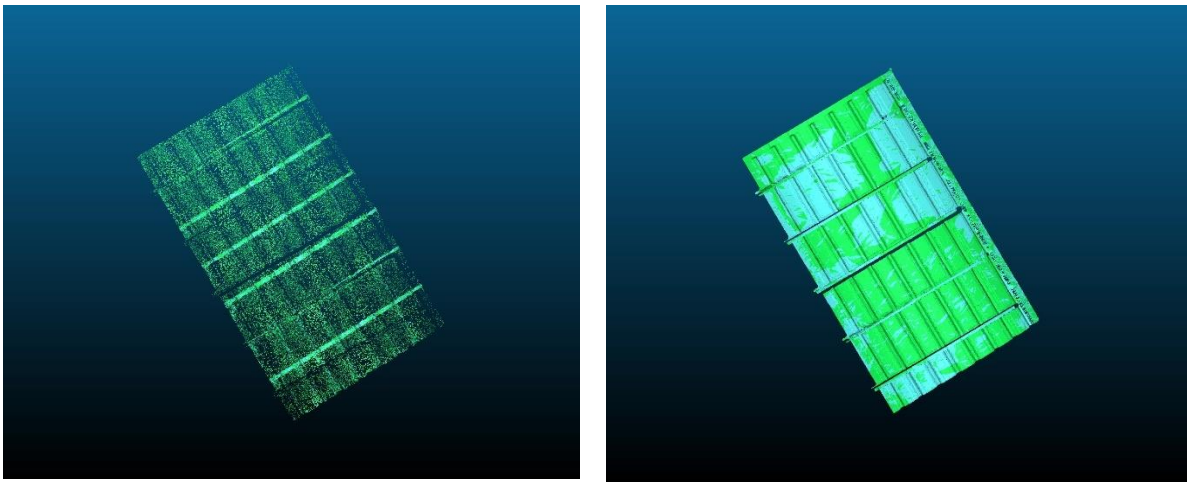
The surface reconstruction step is carried out using the alpha-shape approach because of its accuracy and computational efficiency [EDEL83]. Several values of the alpha parameter are tested to identify a suitable balance between computational efficiency and mesh completeness. A small alpha, such as $\alpha = 1$, generated a mesh rapidly and at low computational cost, but the resulting surface was highly fragmented, with a low triangle density and a significant number of open regions (Figure 4-8(a)). Increasing the alpha value improves mesh continuity. An $\alpha = 20$ reconstruction produces a surface with fewer holes and a higher degree of completeness, but at the expense of considerably longer computation times (Figure 4-8(b)). The value $\alpha = 10$ offers the most favorable compromise: it closes approximately 80–90% of the measurement gaps, preserved the overall curvature of the panel and remains computationally manageable. For these reasons, the $\alpha = 10$ reconstruction is selected for the processing pipeline. The resulting mesh contains approximately 84.3 million vertices and 28.2 million triangles and is illustrated in Figure 4-8(c).

Mesh quality is evaluated to ensure that the reconstructed surface would support reliable geometric registration in Section 4.2.2. The triangle distribution across the mesh is uniform, with no excessive stretching or degenerate elements that could impair alignment. Visual inspection confirms that the reconstruction captures the global curvature of the panel as well as the geometry of the stringer and frame regions. In areas where scanner coverage is low, small residual holes are visible, however these do not affect the downstream registration processes.

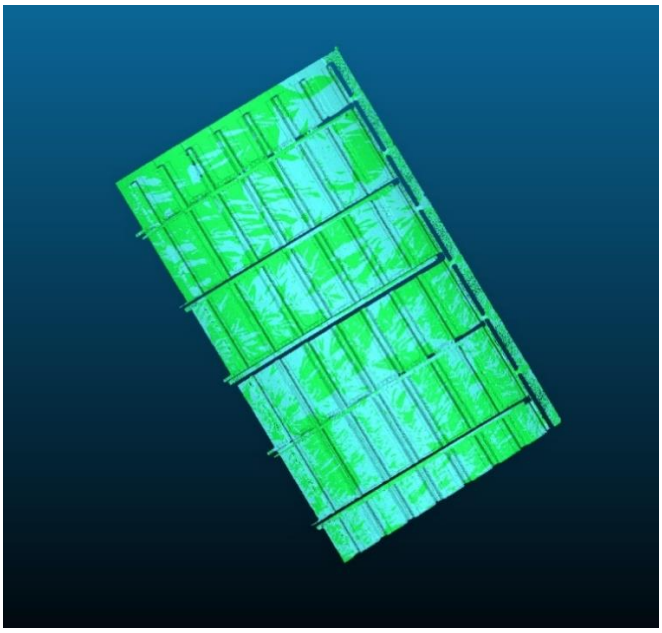
The final output of this processing pipeline is the reconstructed measured surface S_{conf}^{meas}. This surface can now be represented as a set of points:

$$P_i = \begin{bmatrix} x_i \\ y_i \\ z_i \end{bmatrix} \tag{4.1}$$

which collectively describes the shape of the panel in its loaded configuration. This surface provides the basis for the registration and displacement analysis procedures introduced in the next sections.



(a) (b)



(c)

Figure 4-8 Alpha-shape reconstructions of the measured panel: (a) $\alpha = 1$; (b) $\alpha = 20$; (c) $\alpha = 10$, selected for the workflow.

4.2.2 Datasets Registration

The registration step aligns the reconstructed measured surface S_{conf}^{meas} with the simulated deformed nominal surface S_{def}^{nom}. As indicated in Figure 4-9 both geometries are in different coordinate systems, hence this alignment is essential, as both shapes represent the panel under similar loading conditions and must be placed in a common coordinate system before computing the displacement field. The complete registration pipeline is processed with python scripts that interface with the CloudCompare software's functions, ensuring consistent processing across all measurement configurations.

In this context, the deformed surface S_{def}^{nom} represents the simulated configuration of the ideal GF state shape under the same loading conditions described in Section 4.1.1. The undeformed nominal geometry, denoted as S_{free}^{nom} is described as a set of points:

$$\hat{P}_j = \begin{bmatrix} \hat{x}_j \\ \hat{y}_j \\ \hat{z}_j \end{bmatrix} \tag{4.2}$$

which is the CAD-based GF state shape of the component and does not include the influence of external loads or manufacturing deviations. To obtain the deformed nominal configuration used for registration, this nominal shape is subjected to the gravity and clamping conditions of the measurement setup, denoted as E^{conf}, through a finite element simulation. The resulting surface is expressed as S_{def}^{nom}, which is described as a set of points:

$$\acute{P}_k = \begin{bmatrix} \acute{x}_k \\ \acute{y}_k \\ \acute{z}_k \end{bmatrix} \tag{4.3}$$

where each point $\ddot{P}_k$ is obtained by adding E^{conf} into $\dot{P}_j$. Hence:

$$\acute{P}_k = \hat{P}_j + E^{conf}(\hat{P}_j) \tag{4.4}$$

where E^{conf} represents the scalar field which is a function of the gravitational and clamping forces acting on the panel:

$$E^{conf}(\hat{P}_j) = u(F^{grav}(\hat{P}_j), F^{fix}(\hat{P}_j)) \tag{4.5}$$

A two-stage registration approach is adopted to ensure robustness when handling large, compliant geometries with complex curvature. Application of the Iterative Closest Point (ICP) algorithm is insufficient, as the initial positions of the two surfaces differed significantly as shown in Figure 4-9. In such cases, ICP may converge to an incorrect local minimum or fail to register the datasets altogether. Introducing a coarse pre-alignment step based on manually selected corresponding points allows the fine alignment to operate within a constrained search space, increasing stability and accuracy of the resulting registration.

a) Point-Picking

The first stage during the registration step consists of manual point-picking, where corresponding features are selected on both surfaces. These points are chosen on structural features that are present and clearly identifiable on both the nominal and measured geometries, primarily at intersections of frames and stringers, stiffener corners and other well-defined geometric transitions. Selecting points on these manufactured reinforcement features ensures repeatable and unambiguous correspondences.

The distribution of the selected points across the entire panel limited bias and prevents the initial transformation from being dominated by local deviations. The manually picked points are shown in Figure 4-9, illustrating the correspondence pairs on the nominal deformed surface S_{def}^{nom} and the measured surface S_{conf}^{meas}.

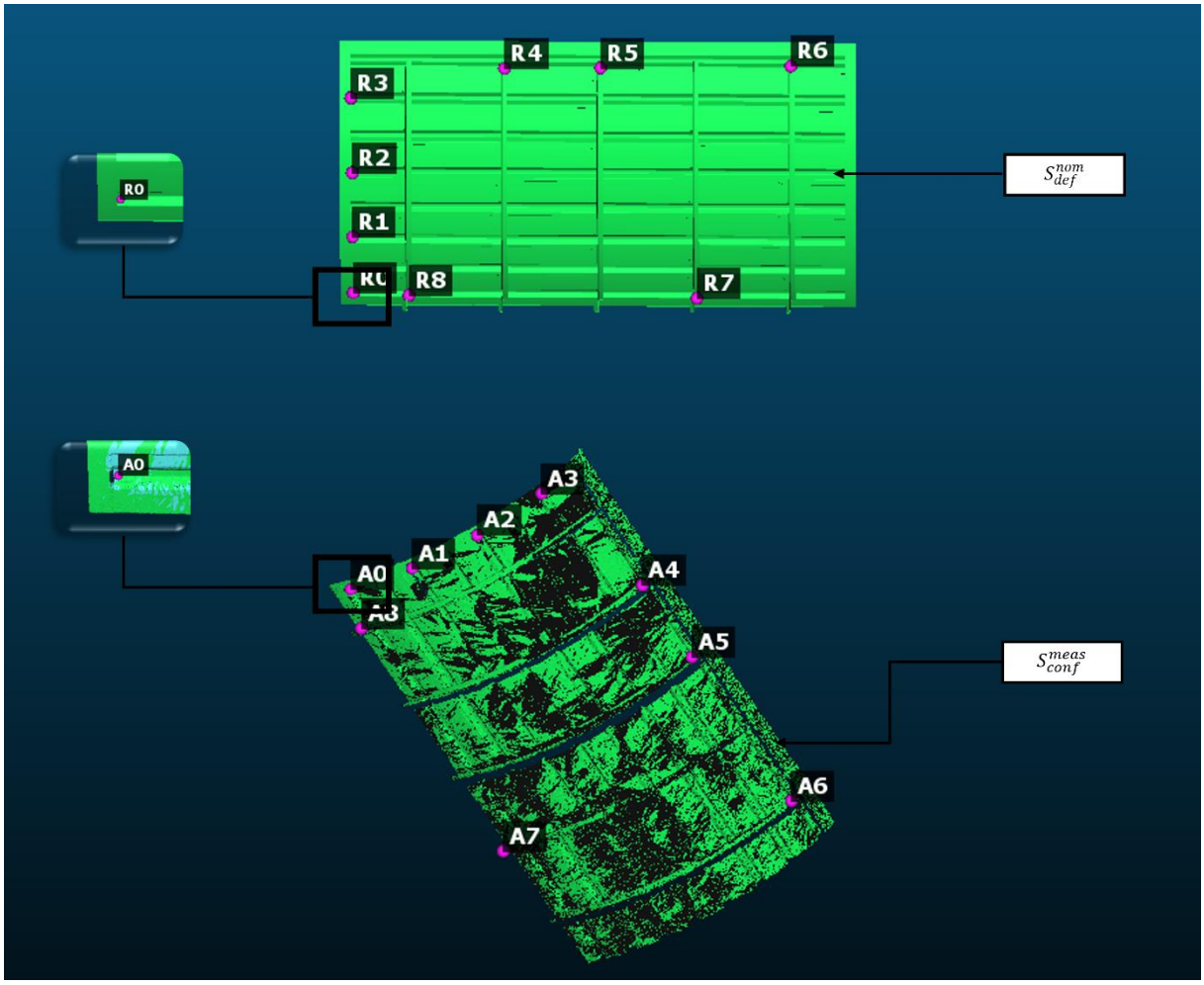


Figure 4-9 Manually selected correspondence points on the measured surface (A-points) and the simulated deformed surface (R-points).

b) Coarse Alignment

Once the common feature pairs are defined, CloudCompare computed an initial rigid transformation matrix. Applying this transformation to the measured surface produced the coarsely aligned dataset, $S_{conf}^{meas\,0}$, which can be described as a set of points:

$$P_i{}^0 = \begin{bmatrix} x_i{}^0 \\ y_i{}^0 \\ z_i{}^0 \end{bmatrix}. \tag{4.6}$$

The coordinate values of the reference and target points used for this computation illustrate how each selected point on the measured surface is paired with its corresponding location on the nominal model. Following this step, the measured surface is approximately aligned with the nominal deformed geometry but still exhibits visible offsets. This intermediate state is shown in Figure 4-10, where the global positioning is correct but local discrepancies remain in form of unaligned coordinate system.

c) Iterative Closest Point (ICP)

The next stage of the pipeline applies the Iterative Closest Point (ICP) algorithm to refine the alignment. ICP iteratively reduces the Euclidean distance between the two surfaces by updating the transformation matrix until convergence. The algorithm is executed through the python interface controlling CloudCompare, ensuring identical settings for all measurement runs. The final registered surface is denoted as $S_{conf,ICP}^{meas}$, which is described as a set of points:

$$P_i{}^{ICP} = \begin{bmatrix} x_i{}^{ICP} \\ y_i{}^{ICP} \\ z_i{}^{ICP} \end{bmatrix}. \tag{4.7}$$

Table 4-2 shows the RMS values between $S_{conf,ICP}^{meas}$ and S_{def}^{nom} for the 9 measurement configurations. Furthermore, the resulting shapes are shown in Figure 4-11, where the two surfaces closely overlap with minimal visible offset. The quality of the registration is assessed using CloudCompare's cloud-to-mesh (C2M) distance tool, yielding a root-mean-square (RMS) deviation of 2.084742 mm. This value represents the average point-to-surface distance from the registered measured mesh to the nominal deformed mesh and is consistent with expected geometric variability for large, compliant structures.

The final registration establishes a common coordinate frame for both S_{def}^{nom} and $S_{conf,ICP}^{meas}$. As both surfaces represent the same physical configuration, one simulated and one measured, the gravitational and clamping-induced effects embedded in both shapes effectively cancel when the surfaces are compared. This alignment therefore provides a reliable basis for computing the displacement field associated with manufacturing deviations, which is presented in the next section.

Table 4-2 Pairwise RMS values between the registered shape and the deformed shape for the measured configurations after ICP registration.

Experiment	RMS/mm
Run 1	2.079
Run 2	2.085
Run 3	2.072
Run 4	1.980
Run 5	2.661
Run 6	2.637
Run 7	2.623
Run 8	2.412
Run 9	2.743

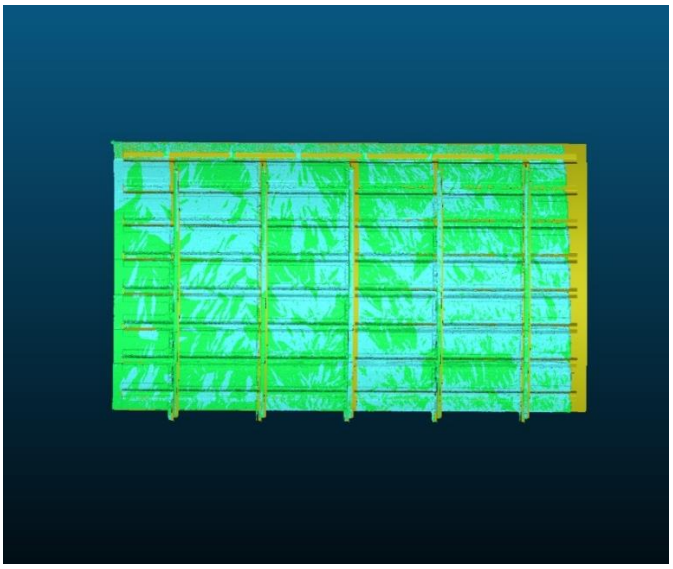


Figure 4-10 Coarse alignment of the measured shape (green) with the deformed shape (yellow), using the transformation matrix obtained from the point-picking correspondences.

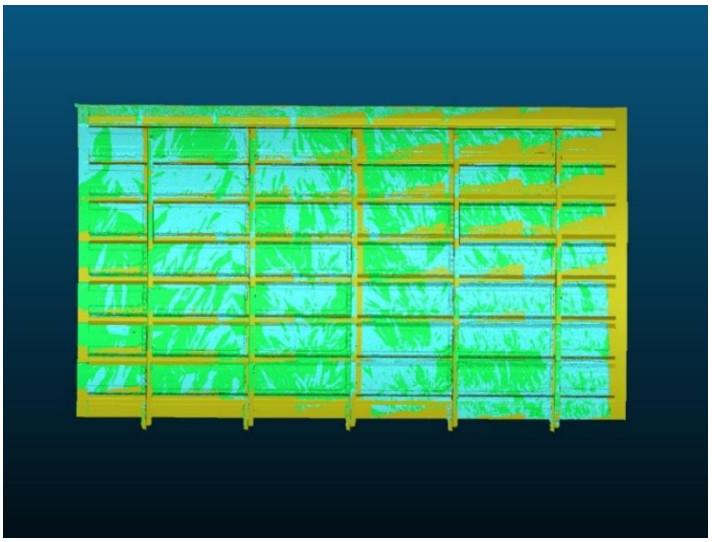


Figure 4-11 Fine alignment obtained after applying the ICP algorithm.

4.2.3 Displacement Field Computation

This section describes the computation, filtering and mapping of the displacement field that is later used for the approximation of the actual GF state shape of a manufactured panel in Chapter 5. Once the measured shape $S_{conf,ICP}^{meas}$ has been registered to the simulated deformed shape S_{def}^{nom}, the next step involves quantifying the deviations that remain between both surfaces. These deviations represent the manufacturing-induced discrepancies in the panel geometry, since the influence of gravity and the clamping configuration is present in both datasets and therefore cancels out after registration.

The displacement values are obtained using the CloudCompare cloud-to-mesh (C2M) tool, which computes the signed point-to-surface distance from each point of the measured mesh to the nearest point on the reference mesh, S_{def}^{nom}, in the direction of surface normals of S_{def}^{nom}. These signed distances form the displacement field, E^{man}. The statistical distribution of these distances is shown in Figure 4-12.

The relationship between both surfaces can be expressed using the mathematical framework established earlier in this chapter. The manufacturing deviation field, E^{man}, is defined as:

$$E^{man} = S_{conf,ICP}^{meas} - S_{def}^{nom}. \tag{4.8}$$

Here, the surface subtraction is interpreted point-wise, and the manufacturing deviation field E^{man} is defined as the scalar distance between corresponding points on the two shapes. Substituting $S_{conf,ICP}^{meas}$ and S_{def}^{nom} from equations 4.8 and 4.3 respectively gives:

$$E^{man} = \left\| P_i^{ICP} - \acute{P}_k \right\| \tag{4.9}$$

The E^{man} at each point of $S_{conf}^{meas\,ICP}$, P_i^{ICP},may also be written as:

$$E^{man}\left(P_i^{ICP}\right) = \Phi_i \tag{4.10}$$

with:

$$\Phi_i = \sqrt{(x_i - \ddot{x}_k)^2 + (y_i - \ddot{y}_k)^2 + (z_i - \ddot{z}_k)^2} \tag{4.11}$$

The scalar value Φ_i represents the Euclidean distance between both surfaces and forms the basis of the displacement field computed through CloudCompare. Φ_i is stored together with the corresponding point coordinates, forming an augmented nodal dataset, however, the manufacturing deviation field itself remains scalar-valued. Evaluating Eq. (4.10) for all $i = 1, \dots, N$ yields the scalar manufacturing deviation field over $S_{conf,ICP}^{meas}$.

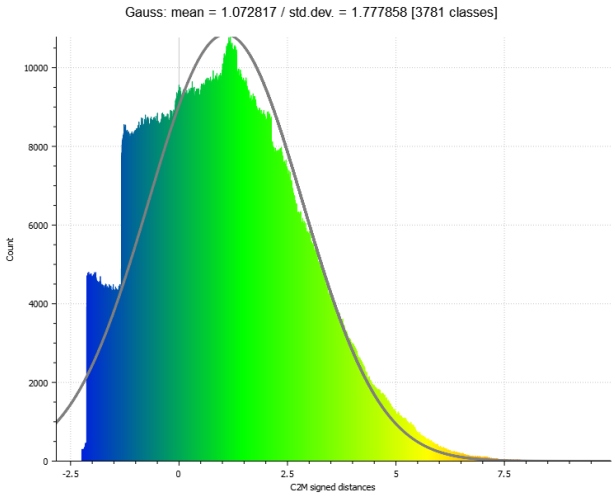


Figure 4-12 Histogram of the signed C2M distances computed between the measured and deformed shapes. The curve indicates the overall spread of the deviations.

The histogram exhibits a smooth, asymmetric profile with a central concentration of values close to zero and a gradual spread on the positive side. Although not perfectly symmetric, the fitted Gaussian curve illustrates the general trend of the displacement distribution and confirms that most of the measured discrepancies lie within a narrow interval. This behavior is typical for manufactured components, in which local form deviations arise from machining tolerances, local fiber misalignments or tooling imperfections [SCHL14].

The spatial variation of the displacement field, E^{man}, is visualized in Figure 4-13. This representation overlays the C2M distances onto the measured surface $S^{meas}_{conf,ICP}$. The coloring highlights regions of larger positive deviations along specific stringers and skin regions, whereas the negative deviations remain limited to isolated areas. The graphical pattern reflects the geometric stiffness distribution of the panel. The stiffened regions, defined by the frames and stringers, constrain the deformation more strongly and therefore exhibit smaller deviations, while the unsupported skin areas display a more noticeable spread. The visualization also confirms the presence of isolated artefacts caused by slight reconstruction noise and local scanning imperfections.

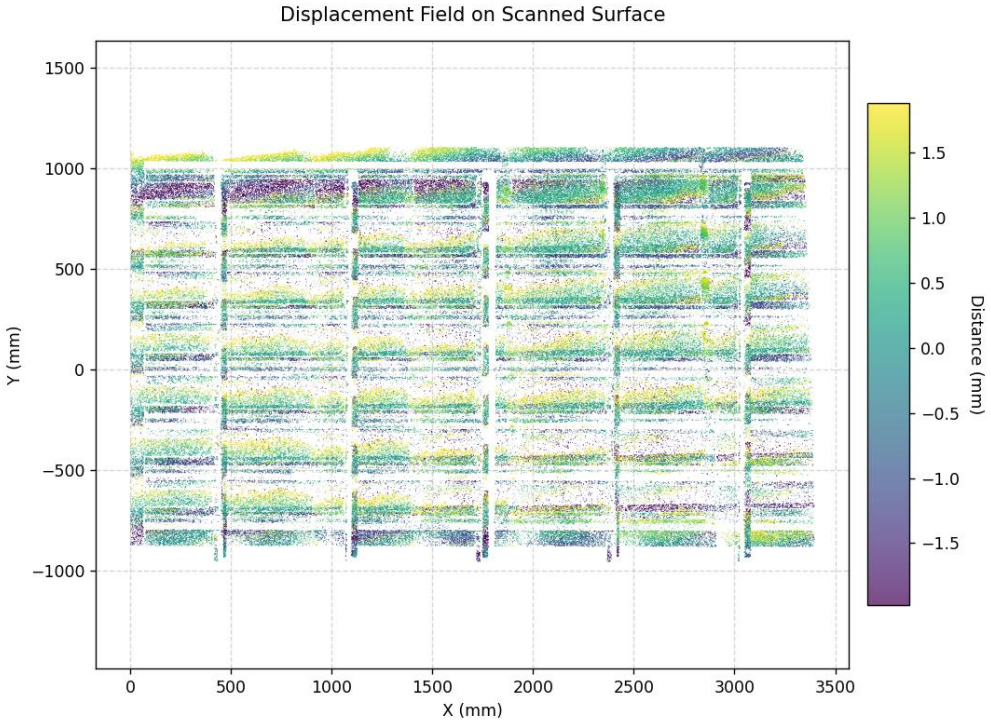


Figure 4-13 Spatial distribution of manufacturing deviations visualized on the nominal shape.

Before the deviation field could be mapped to the nominal surface, it is filtered to remove inconsistent or isolated points. Two filtering procedures are applied: a SOR filter excludes points whose mean neighborhood distance exceeds a threshold derived from the local distribution of distances and a radius outlier removal (ROR) filter that removes points that lack enough neighbors within a specified radius. The combination of both filters ensures that the remaining field captures only spatially coherent deviations while suppressing noise originating from either the scanning process or incomplete surface reconstruction.

To prepare the field for further modelling, the filtered deviations are mapped from $S_{conf,ICP}^{meas}$ onto the nodes of the nominal shape S_{free}^{nom} . This step is necessary because the subsequent finite element reconstruction requires the displacement field to be defined directly on the nominal mesh. A k-nearest neighbor (k-NN) search associates each filtered $S_{conf,ICP}^{meas}$ point with the closest nodes of the nominal surface, as described in Section 3.4. This operation produces a sparse deviation field on S_{free}^{nom} in which only a subset of nodes receives mapped displacement values. The remaining nodes are assigned to displacement values through inverse distance weighting (IDW) interpolation. This method distributes the mapped deviations smoothly across the entire surface by weighing contributions according to the inverse of their spatial distance. The resulting continuous field represents the E^{man} expressed directly on the nominal geometry.

To express the approach in mathematical form E^{man} on $S_{conf,ICP}^{meas}$ is defined as $E^{man}(P_i^{ICP})$ and E^{man} on S_{free}^{nom} as $E^{man}(\acute{P}_k)$. In the k-NN mapping $E^{man}(P_i^{ICP})$ is transferred to $E^{man}(\hat{P}_j)$ using $k = 4$ to have better accuracy. For each $S_{conf,ICP}^{meas}$ vertex P_i^{ICP}, the four closest nominal vertices are identified:

$$N_4\left(P_i^{ICP}\right) = \left\{\hat{P}_{j1}, \hat{P}_{j2}, \hat{P}_{j3}, \hat{P}_{j4}\right\} \tag{4.12}$$

with Euclidean distances:

$$r_{ij} = \left\|P_i^{ICP} - \hat{P}_j\right\|. \tag{4.13}$$

The value Φ_i is then distributed to these four nominal nodes by inverse-distance weighting. If a nominal node, $\hat{P}_j$, receives displacement values from a set of vertices of $S_{conf,ICP}^{meas}$, L_i, a provisional mapped value $\hat{E}^{man}(\hat{P}_j)$ is obtained as:

$$\hat{E}^{man}(\hat{P}_j) = \frac{\sum_{i \in L_j}\left[\frac{1}{r_{ij}}\Phi_i\right]}{\sum_{i \in L_j}\left[\frac{1}{r_{ij}}\right]}\ . \tag{4.14}$$

This step yields a sparse scalar field on the nominal mesh, since some vertices $\hat{P}_j$ do not receive any contribution and remain undefined.

In the next step, the missing values on S_{free}^{nom} are filled by an additional IDW interpolation, again with $k = 4$. Let $\mathcal{M}$ denote the set of nominal vertices that already carry a provisional displacement value $\hat{E}^{man}$. For each vertex $\hat{P}_j$ without a value, the four nearest neighbors in $\mathcal{M}$ are found as:

$$N_4{}^{\mathcal{M}}\left(\hat{P}_j\right) = \left\{\hat{P}_{j_{\ell 1}}, \hat{P}_{j_{\ell 2}}, \hat{P}_{j_{\ell 3}}, \hat{P}_{j_{\ell 4}}\right\} \tag{4.15}$$

with distances:

$$d_{j\ell} = \left\|\hat{P}_j - \hat{P}_{j_\ell}\right\|. \tag{4.16}$$

The final displacement value at $\hat{P}_j$ is then computed as:

$$E^{man}(\hat{P}_j) = \frac{\sum_{\hat{P}_{j_\ell} \in N_4{}^{\mathcal{M}}\left(\hat{P}_j\right)}\left[\frac{1}{d_{j\ell}}\hat{E}^{man}\left(\hat{P}_{j_\ell}\right)\right]}{\sum_{\hat{P}_{j_\ell} \in N_4{}^{\mathcal{M}}\left(\hat{P}_j\right)}\left[\frac{1}{d_{j\ell}}\right]}\ . \tag{4.17}$$

Vertices that already possess a displacement value keep it, hence:

$$E^{man}(\hat{P}_j) = \hat{E}^{man}(\hat{P}_j) \ \text{ for } \hat{P}_j \in \mathcal{M}. \tag{4.18}$$

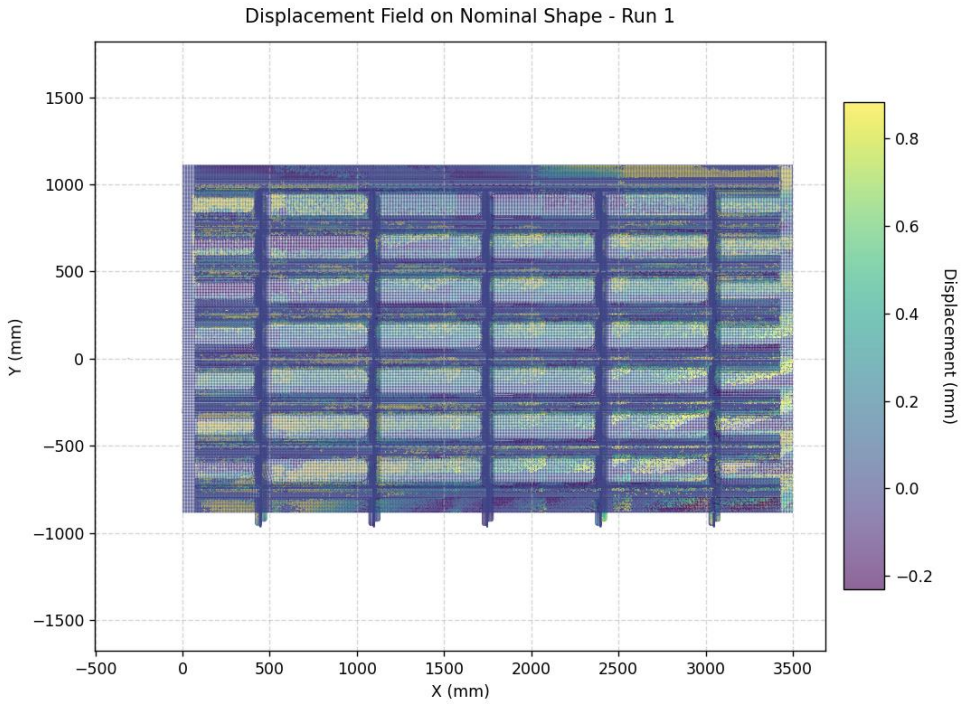


Figure 4-14 Displacement field mapped and interpolated onto the nominal shape.

The mapped displacement field is shown in Figure 4-14. The distribution of deviations follows the same structural pattern observed in the previous representation, Figure 4-13, but is now defined on a continuous and uniform mesh. The smoother appearance reflects the effect of the interpolation step, which reduces local fluctuations while preserving global trends. The mapped E^{man} forms the basis for the GF approximation presented in Chapter 6, where it is applied as a displacement BC to approximate the manufactured panel's unloaded geometry.

5 Numerical Modelling

The numerical model used in this research was based on a nominal CAD representation of the panel that was provided prior to the start of the study. This model reflects the ideal GF geometry of the panel, S_{free}^{nom}, and serves as the basis for generating the simulated deformation under the loading conditions of the measurement setup. The finite element mesh and material definitions are already available in the form of pre-assembled input files, whereas the simulation environment has been modified to fit the boundary and loading conditions that were relevant for the measurements described in Section 4.1.3.

The purpose of this chapter is to describe the finite element methodology (FEM) through which the deformed nominal configuration, S_{def}^{nom}, is obtained. The chapter introduces the nominal geometry and outlines the structure of the provided model before detailing the simulation conditions that were modified for the analysis. As these factors determine the deformation pattern that must be numerically replicated to ensure consistency with the measured data, special attention is paid to the representation of support conditions and the application of gravity. The chapter concludes with the extraction of the simulated deformed shape, which is later used as a reference in the registration and deviation analysis of Chapter 4.

5.1 Construction of the FE Model

The numerical model used in this research was simulated from the nominal CAD geometry of the panel, which was provided as part of the initial project data. This shape represents the ideal GF configuration and forms the basis of the finite element analysis developed for the simulations in this chapter. As the finite element (FE) mesh and material assignments were already included in the input files, the focus of this section is to outline the structure and characteristics of the nominal model. The following subsections describe the geometric definition of the panel and the material properties implemented in the provided model, as these determine the structural response predicted by the numerical analysis.

5.1.1 Nominal CAD Geometry

The nominal panel geometry, S_{free}^{nom}, corresponds to the ideal GF state shape and is shown in Figure 5-1. The model comprises three main structural components: the skin, a set of longitudinal stringers and a series of circumferential frames with local reinforcements. The skin is represented by a shell part that forms the curved outer surface of the fuselage panel. It is modelled as a composite laminate consisting of sixteen plies with an individual thickness of approximately 0.266 mm, resulting in a total skin thickness of about 4.26 mm. Three stringers are defined as separate shell parts: 6 central stringers and two lateral stringers on the left and right. Each stringer uses a composite shell section with an overall thickness of roughly 8 mm. The frames are combined in a single frame part that contains a series of circumferential ribs with different lay-ups assigned. This locally increases the laminate thickness to approximately 15.4 mm.

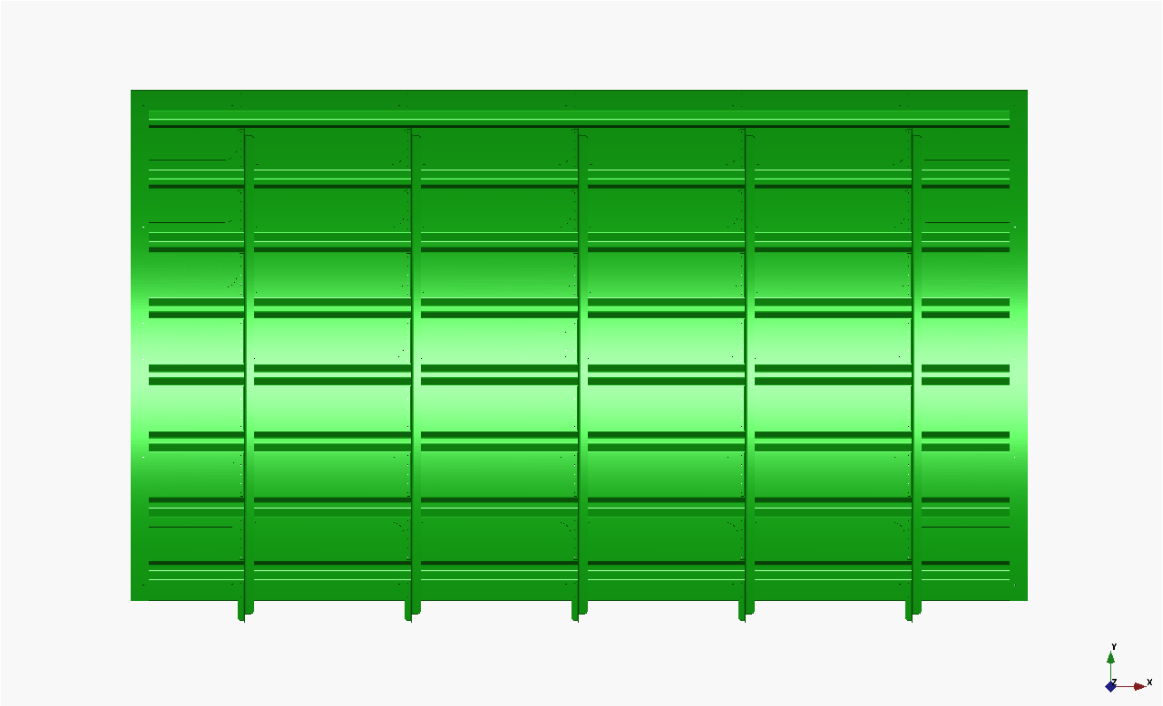


Figure 5-1 Nominal CAD geometry of the panel.

The FE representation follows several assumptions typical for large composite airframe structures [DAVI20]. Small-scale features such as fastener holes, fillets and local chamfers are not included. The stringers and frames are modelled as thin shells with sharp intersections. No door or window cut-outs are present, only the global panel boundaries and the stiffening system are retained. Connections between skin, stringers and frames are realized through interaction definitions rather than fasteners, so that the load transfer between components is idealized while the overall stiffness distribution is preserved. These assumptions reduce complexity while maintaining the behavior needed for the deformation analysis.

The meshed form for all components is contained in the supplied input files and is used without modification. The skin is partitioned along stringer and frame locations so that common nodes exist at the stiffener interfaces. No additional mesh refinement or re-meshing steps are applied for the numerical simulations presented in this chapter. For the subsequent mapping operations presented in Chapter 4, the same nominal geometry is exported as a surface mesh in *.OBJ* format, where the outer surface is represented by triangular elements derived from the original shell model. The OBJ format is an open, non-proprietary geometry file format that represents surfaces using polygonal elements, typically triangles [SIMU15].

The global coordinate system used in the simulations is consistent with the convention adopted for the experimental setup. The x-direction describes the longitudinal axis of the panel along the stringers, with nodal coordinates varying from approximately 75 mm to 3425 mm. The y-direction spans the panel width in the circumferential direction, with coordinates in the range of roughly −900 mm to +1100 mm. The z-direction corresponds to the radial axis normal to the skin and is used as the direction of gravity loading. In the analysis step, gravity is applied in the negative global z-direction. This orientation matches the configuration used during the

measurements in Chapter 4 and ensures that the deformed configuration S_{def}^{nom} is directly comparable to the measured shape S_{conf}^{meas}.

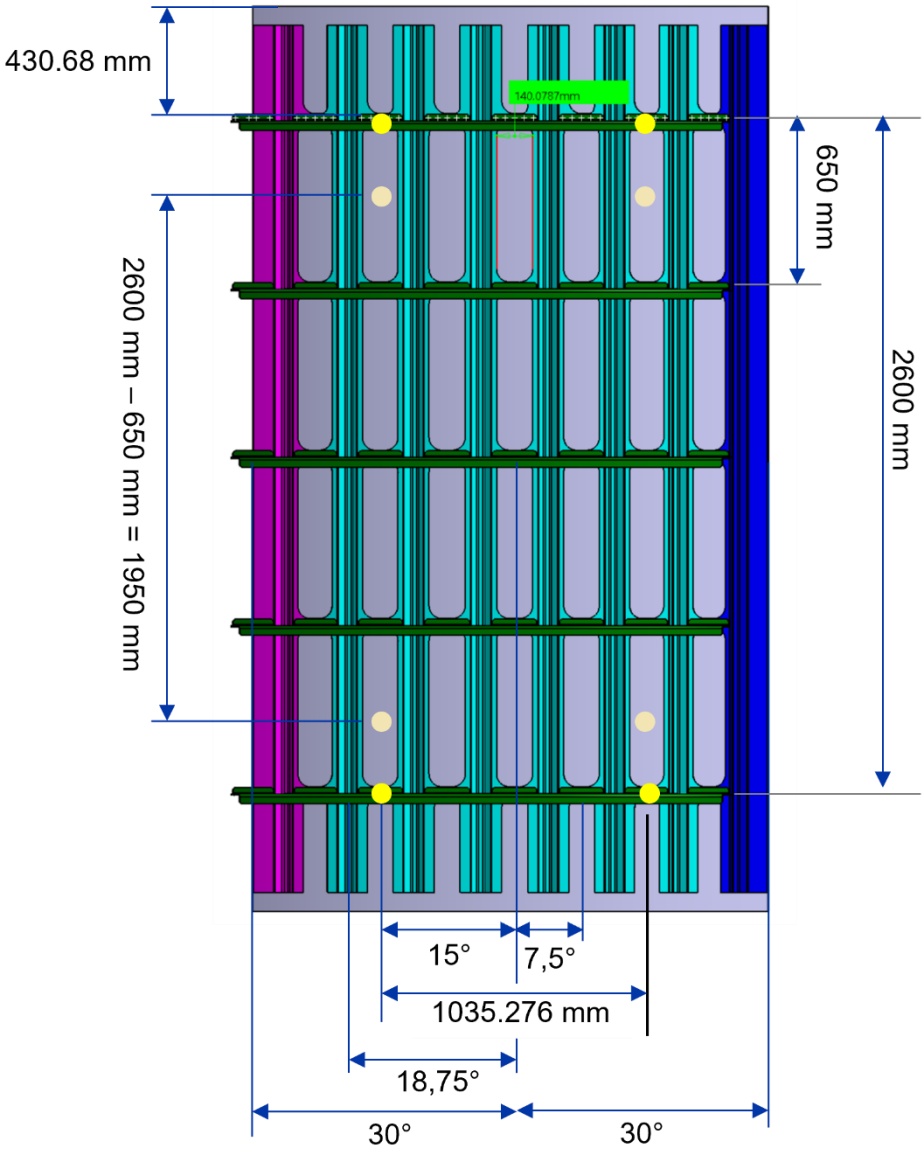


Figure 5-2 Geometric layout of the panel, illustrating the spacing and arrangement of stringers and frames as well as the overall panel dimensions.

5.1.2 Material Definition

The FE model represents the panel as an assembly of composite shell components, each defined through layered orthotropic material properties that reflect the anisotropic behavior of carbon-fiber laminates. All structural parts use the material block STD_CF_UD (Standard Carbon Fiber Unidirectional Lamina), which corresponds to a linear-orthotropic unidirectional carbon-fiber composite. The identical properties therefore govern the stiffness response of the curved skin and the stiffening members, ensuring consistency across the entire structure. A secondary material, Ti-6Al-4V (Alpha-Beta Titanium Alloy), is also defined but it does not contribute to the mechanical behavior of the composite shell system.

The constants associated with STD_CF_UD define a transversely orthotropic elastic response, with a high longitudinal stiffness and lower transverse stiffness as indicated by the values in Table 5-1. The titanium material exhibits a conventional isotropic elastic formulation, with a modulus of approximately $110\ GPa$, a Poisson ratio of about 0.34 and a density of $4420\ \frac{kg}{m^3}$. As titanium appears only in the smart-jig connectors, its mechanical influence on the simulation of the panel is negligible.

Table 5-1 Elastic constants defined for the composite material.

Property	Value
Longitudinal Young's Modulus (E_1)	$135\ GPa$
Transverse Young's Modulus ($E_2 = E_3$)	$10\ GPa$
In-plane Shear Moduli ($G_{12} = G_{13} = G_{23}$)	$5\ GPa$
Poisson's Ratio (v_{12})	0.30
Density (ρ)	$1600\ \frac{kg}{m^3}$

The overall laminate layup is illustrated in Figure 5-3, which provides a representative schematic of the stacking arrangement implemented in the shell sections. Although the figure corresponds to a different section thickness, it captures the organization of plies and the repeated fiber-orientation sequence that characterizes the composite structure of the simulated panel.

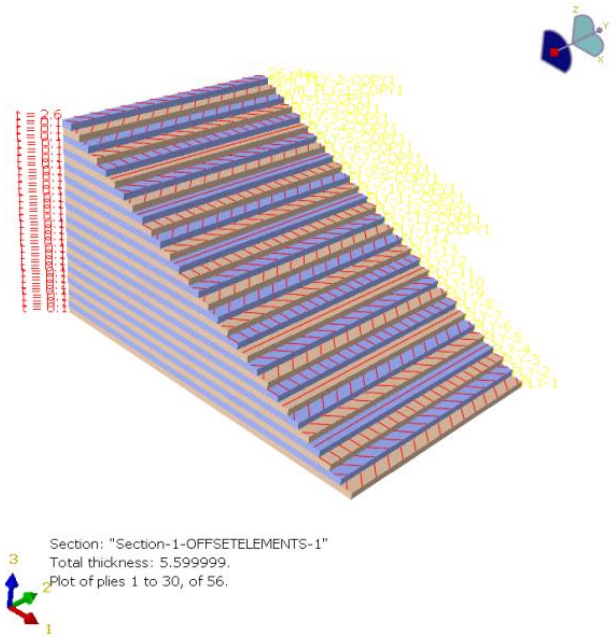


Figure 5-3 Representative schematic of the laminate stacking sequence used in the FE model

5.1.3 Element Type

The FE discretization of the panel is based entirely on continuum-shell elements. The thin-walled composite architecture consists of SC8R and SC6R element formulations. SC8R elements, which are eight-node hexahedral continuum shells, cover most of the skin, stringers and frames. They are joined locally by SC6R triangular wedge elements where the mesh topology requires transitional regions or curved boundaries [SIMU15]. This combination maintains a shell-like kinematic behavior while offering the additional flexibility of a solid-type formulation through the thickness, which is particularly suited to layered composite laminates.

The structural mesh, Figure 5-4, comprises of 817,336 nodes and 176,517 continuum-shell elements, of which 176,201 are SC8R elements and 316 are SC6R elements. A small number of one-dimensional connector elements (CONN3D2) appear at the jig interfaces but do not contribute to the structural stiffness. Across most of the panel, the nodal spacing is approximately 3.8-4 mm in the longitudinal direction. The panel geometry has been partitioned accordingly, allowing coincident nodes at all skin–stiffener interfaces and therefore supporting the tie constraints used in the assembly. The mesh becomes slightly more refined in regions where stringers or frames intersect with the skin, to handle higher stresses, whereas outside these points the elements remain nearly square with uniform spacing.

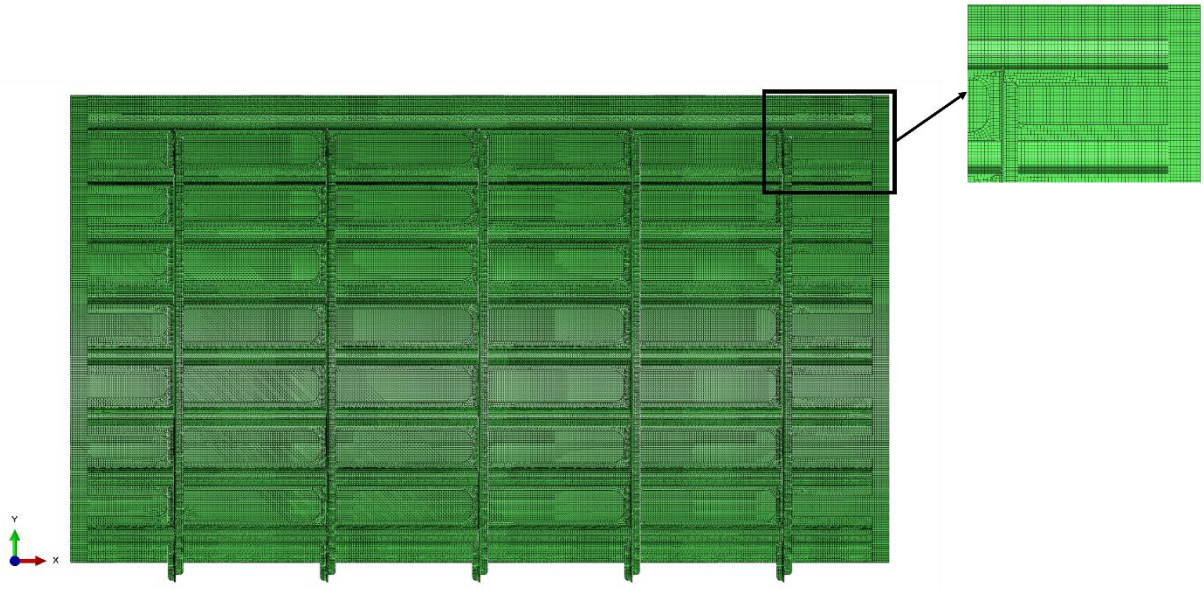


Figure 5-4 Finite element mesh of the nominal panel showing the continuum-shell discretization of the skin, stringers and frames.

The structure of the element discretization in Figure 5-4 suggests that the overall mesh quality is high. The mesh quality refers to how well the mesh elements represent the model geometry and support accurate and stable numerical solutions [DAVI20]. The regular layout of the SC8R continuum-shell elements results in almost square shapes across most regions of the panel, which limits element skewness. In areas of curvature or geometric transition, the use of SC6R wedge elements maintains a smooth representation of the surface. The discretization behaves well under the applied loading conditions as the mesh does not exhibit significant distortions or degraded quality metrics during the simulation and no numerical instabilities were observed,

indicating that the discretization provides a reliable basis for reproducing the panel's structural response.

5.1.4 Assembly Interactions

The connection within the model is to ensure that the skin, stringers and frames deform as an integrated structural system. These interactions determine how loads transfer through the panel and therefore influence the deformation that defines the simulated deformed configuration used in later registration steps. The model hence follows a bonded representation for all structural joints, reflecting the manufacturing assumptions of an integrally stiffened composite panel.

The stiffeners are connected to the skin through displacement-compatible tie constraints. As the mesh is such that nodes coincide along each stiffener base, the ties enforce continuity without requiring intermediate contact formulations. Frames are treated in the same manner as each frame instance along the panel width is connected to the skin at its respective interface. As a result, both stringers and frames act as stiffening substructures and contribute to the global deformation behavior of the panel.

The interaction between frames and stringers is transmitted through the skin rather than by direct connections between them. This design ensures that load redistribution occurs naturally through the skin, maintaining consistency with the intended structural representation. Besides this, the support conditions introduced by the smart jig are represented through connector elements attached to selected frame nodes. These reference points (RPs) are included in the assembly as 1-D elements that link selected frame nodes, represented by yellow markers in Figure 5-5, to reference points representing the smart jig. Their influence is localised to the support regions and does not alter the behavior between the composite components.

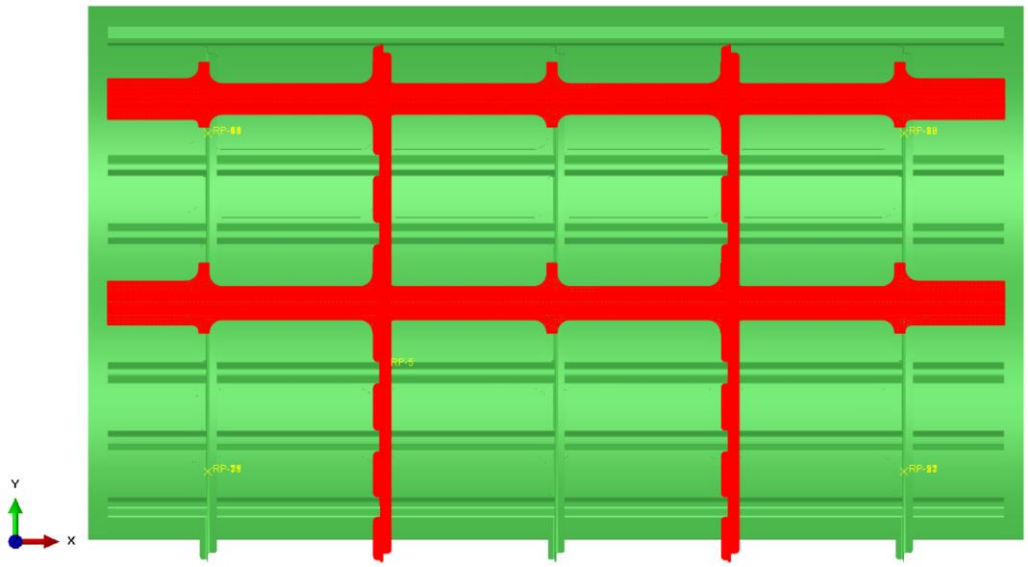


Figure 5-5 Highlighted view of the assembly showing two representative frames (red, vertical) and stringers (red, horizontal) along with the RPs location (yellow markers).

These connections establish an assembly behavior in which the skin provides the primary load path, the stiffeners and frames enhance directional stiffness, and the connectors impose the appropriate BCs associated with the experimental setup. This representation forms the basis for the subsequent simulation.

5.2 Simulation Conditions

The numerical simulation reproduces the loading environment under which the panel was measured in Chapter 4. The material definitions, mesh layout and assembly structure have already been established in Section 5.1 and this section outlines the conditions required to generate the deformed configuration, S_{def}^{nom}. The simulation conditions describe how gravity and support constraints are applied, how the smart-jig connectors are represented in the model and how the analysis is performed to reflect the behavior of the panel under quasi-static loading. These conditions provide a consistent numerical reference for the registration and manufacturing deviation analysis.

5.2.1 Initial Condition

The simulation begins with the undeformed nominal configuration of the panel, which corresponds to the ideal GF state shape S_{free}^{nom}. The skin, stringers and frames were imported as separate part instances and assembled, without additional preprocessing or the introduction of initial strains, displacement fields or temperature effects. This establishes a reference state against which all subsequent deformation induced by boundary and loading conditions can be evaluated.

To initialize the structural system, the analysis starts with a perturbation eigen-frequency step. This step is defined as a linear procedure with geometric nonlinearity disabled. No external mechanical loads are applied at this stage; gravity, actuator-related effects and any connector-mediated actions are excluded from the initial step. The model therefore recognizes only the support conditions already defined within the assembly, allowing the global stiffness characteristics of the jig-mounted panel to be captured without the influence of external loads. The purpose of this step is to establish numerical stability of the supported configuration before the gravity load is introduced.

A single global material orientation is used across all composite components. This orientation system, a rectangular coordinate frame, aligns the local fiber directions with the geometry of the panel. All composite sections reference this same orientation without additional instance-level transformations. As a result, ply angles are interpreted consistently across all parts, with the global orientation defining the reference directions from which composite layup angles are measured.

5.2.2 Boundary Conditions

The boundary conditions (BCs) are defined to reproduce the support behavior of the smart jig described in Chapter 4. In the numerical model the panel is not constrained directly, instead,

all support reactions are applied through the actuator chains, Figure 5-6, that couple the jig to the skin surface. This workflow ensures that the simulated deformation is governed by the same conditions that were present during the measurements. Four actuator chains are included, each associated with one of the actuator heads. The lowest reference point of each chain is fully fixed in all translational and rotational degrees of freedom, representing the rigid foundation of the jig. Above these fixed supports, a series of connector elements transmits motion through a planar, rotational and planar-lock arrangement, following the mechanical sequence of the physical actuator assembly illustrated in Figure 4-2.

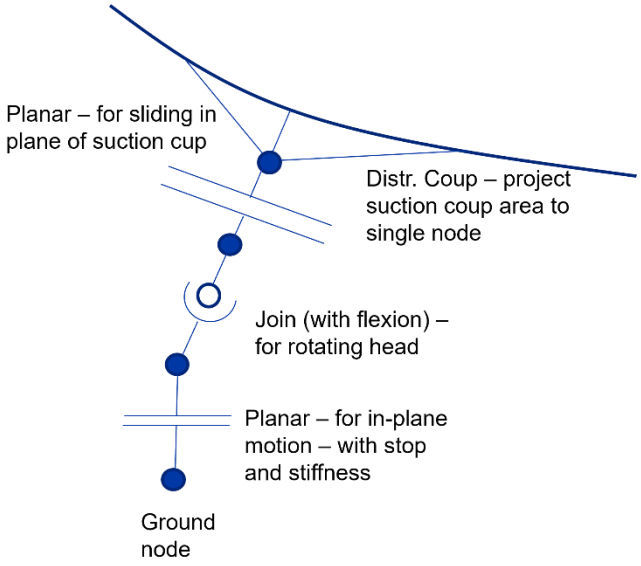


Figure 5-6 Schematic representation of the actuator–panel connector chain used in the numerical model.

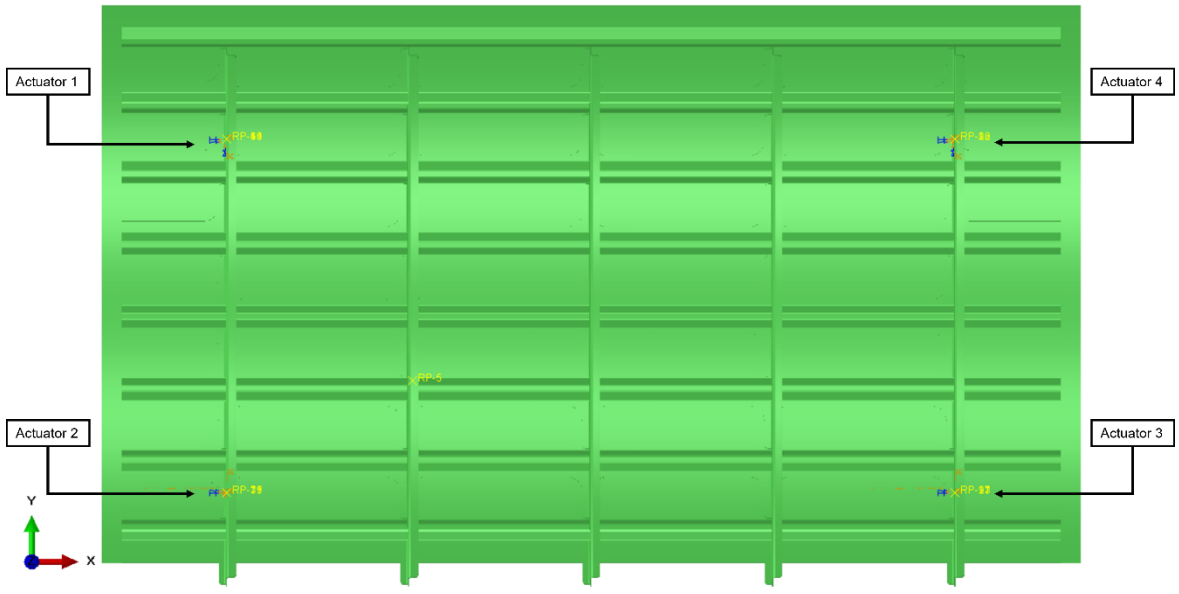
Each actuator chain terminates at a reference point positioned at the center of the suction-cup head. This node is coupled to a circular patch on the outer surface of the skin through a distributed kinematic constraint, which transfers all six degrees of freedom of the connector node uniformly over an influence radius of 20 mm. This coupling model ensures that the actuator does not behave as a point load. Hence, the skin, stringers and frames experience support effects only through the actuator chains as no direct constraints are applied to any structural node of the laminate.

The behavior of each actuator depends on whether its in-plane motion is locked or free, as defined by the experimental configuration. The lock patterns listed in Table 4-1 are implemented by switching the connector behavior in the relevant chain from planar_lock (all in-plane directions restrained) to planar (in-plane sliding allowed). In configurations where an actuator is locked, the suction-cup head constrains the skin in its local plane, as shown in Figure 5-7, whereas an unlocked actuator allows tangential sliding but still restrains motion normal to the surface. In Runs 1–4, only actuator 3 is unlocked, whereas in Runs 5–8 the unlocked actuator is actuator 1. In Run 9 all three actuators are unlocked. These variations give rise to distinct constraint distributions across the panel, which influence the gravity-induced deformation.

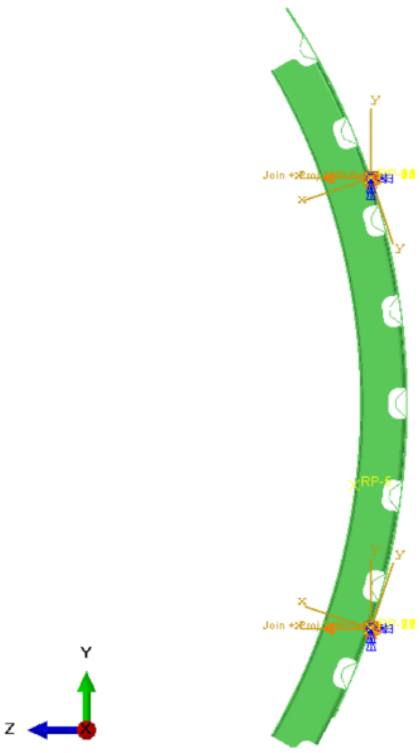
In addition to the locking states, several configurations include actuator displacements that match the measured experimental settings. These displacements act only along the positive global z-direction and are applied at the actuator RPs through displacement BCs in separate quasi-static steps. For example, in Run 2 actuators 2 and 4 are displaced by +15 mm, while the remaining actuators remain at their initial positions. Other runs apply different combinations of upward movements following the values listed in Table 4-1, including the multi-actuator displacement applied in Run 9. As geometric non-linearity is active in all of the displacement steps, the FEM simulation captures the interaction between actuator motion and the changing configuration of the panel under gravity. With the actuator locking states and upward displacements, the BC definition ensures that the FE model reproduces the exact environment of each experimental configuration in a consistent manner.

```
**************************************************
*Connector Behavior, name=planar_lock
*Connector Elasticity, component=2
1e7.,
*Connector Elasticity, component=3
1e7.,
*Connector Elasticity, component=4
1e7.,
**************************************************
```

Figure 5-7 Excerpt illustrating the planar_lock connector behavior. The stiffness values restrict in-plane motion of the actuator head.



(a)



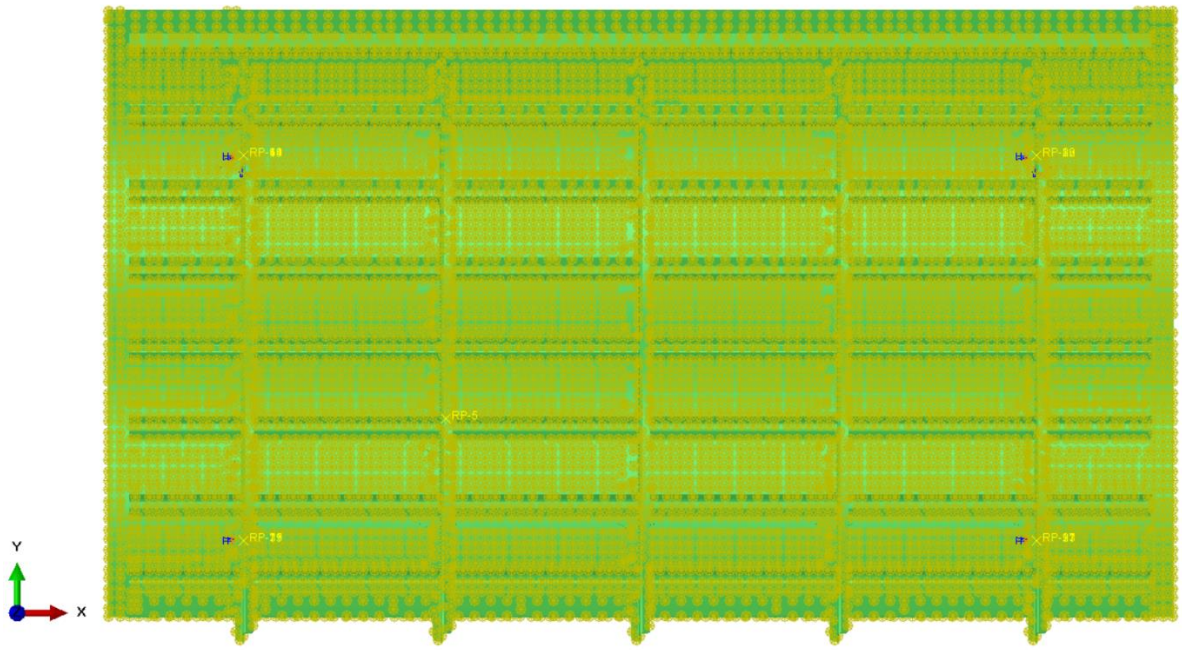
(b)

Figure 5-8 (a) Front (z-axis) view showing the locations of the four actuator attachment regions on the skin surface; (b) side view illustrating the displacement boundary conditions applied to actuators 2 and 4 during Run 2.

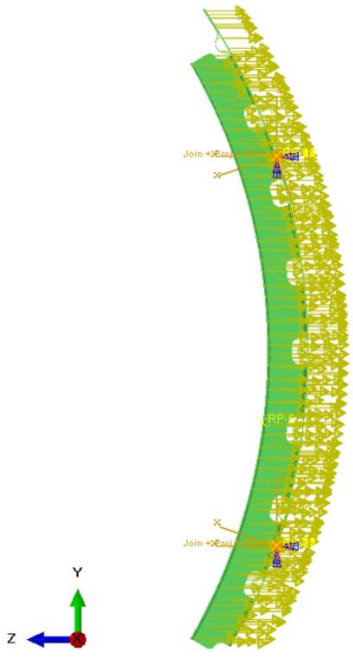
5.2.3 Loading Conditions

The loading applied in the numerical analysis is limited to the mechanical effects that act on the panel during the measurements. This ensured that the deformation predicted by the simulation represents only the influence of self-weight and actuator effects, without introducing additional loads. The loading sequence therefore consists of a gravity step followed by the actuator displacement steps associated with the corresponding experimental configuration.

The gravity step is introduced in a static step, in which the full structural model is subjected to a uniform body force of magnitude 9810 mm/s² acting in the negative global z-direction, shown in Figure 5-9. Furthermore, geometric non-linearity is enabled to account for the curved geometry of the panel. As this step contained no actuator motions, it provides a representation of the gravitational effect on the panel while being supported only by the connector framework. After gravity, the displacements due to actuators are applied in subsequent static steps, each corresponding to the movement prescribed for a specific actuator in the experimental campaign.



(a)



(b)

Figure 5-9 (a) Front view of the gravitational loading on the panel; (b) side view illustrating the uniform body force applied in the negative global z-direction.

As no other loads including pressure, thermal effects or inertia relief are applied in the analysis, the resulting deformation in each simulation arises from the combined effect of gravity and the actuator-induced displacements, consistent with the mechanical conditions reproduced during the measurements.

5.2.4 Solver Setup

The analysis sequence follows the physical progression of the loading stages applied to the panel. The simulation begins with an eigenfrequency extraction, used only to initialise the structural configuration. This step is linear and does not consider geometric nonlinearity. All subsequent steps are formulated as quasi-static analyses with geometric nonlinearity enabled so that the solver accounts for the large-displacement effects associated with gravity loading and actuator-driven deformation during the simulation.

Across these nonlinear steps, the procedure adopts consistent increment settings designed to ensure stable convergence under the combined action of gravity and actuator motion. The step time, initial increment and allowable increment range are chosen to allow automatic reduction to avoid unwanted geometric effects, while still maintaining efficient overall convergence. Therefore, the output focuses on the quantities required for the registration in Chapter 4. Deformed nodal coordinates and displacements are updated at the end of each step to allow the simulated deformed surface to be extracted. Furthermore, element-level stresses and strains are stored to record the mechanical response during the finite element analysis (FEA).

The solver settings follow a streamlined robust configuration starting with a linear initialization step, followed by nonlinear gravity and actuator steps, where each step is solved with consistent incremental control. This provides a stable computational basis for generating the deformed shapes used later in the robust pipeline.

5.3 Integration of Results

The FE simulations described in the previous sections provide the deformed nominal configuration, S_{def}^{nom}, that forms the numerical counterpart to the measured configuration, S_{conf}^{meas}. S_{def}^{nom} represents the response of the ideal GF geometry, S_{free}^{nom}, under gravity and actuator support conditions and is therefore essential for distinguishing manufacturing deviations from deformation caused by the experiment itself. This section aims to outline how the FE results were processed and how the deformed shape was prepared for its role in the robust registration pipeline.

The first stage of the post-processing consists of a visual assessment of the simulated displacement and stress fields to confirm that the model responds accurately to the applied gravity load and actuator conditions. For Run 1, the displacement contour shown in Figure 5-10(a) indicates a smooth deformation of the panel under self-weight, with the largest deformation occurring near the lower central region where the panel is furthest from the actuator supports. The corresponding Von Mises Stress distribution in Figure 5-10(b) indicates that the stresses remain low across most of the skin, with localized concentrations appearing around the intersections of the longitudinal stringers and circumferential frames. These regions coincide with geometric discontinuities and therefore act as natural stiffness transitions.

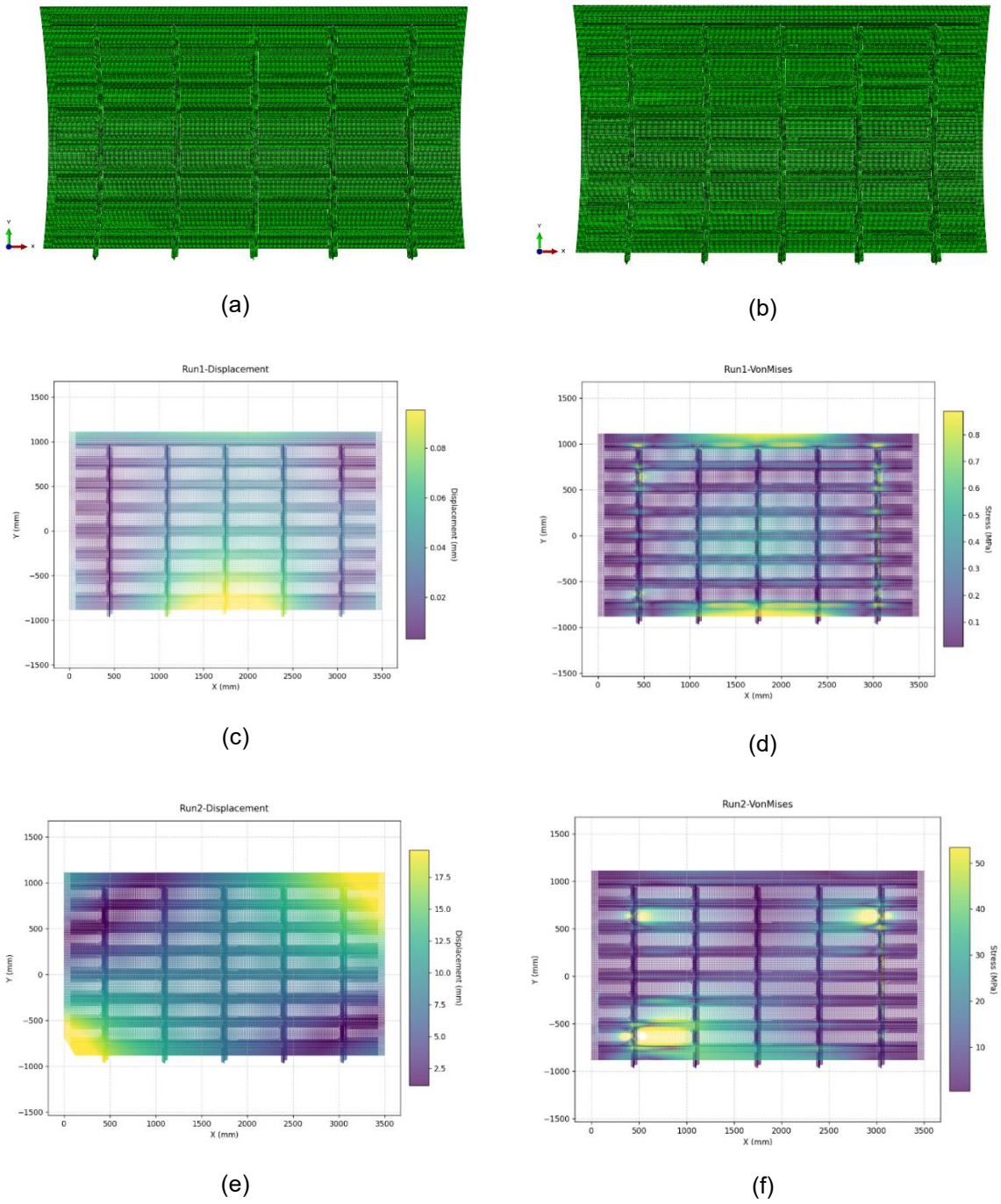


Figure 5-10 (a) Deformed mesh for Run 1; (b) deformed mesh for Run 2; (c) displacement magnitude of deformed shape for Run 1; (d) corresponding von mises stress distribution for Run 1; (e) displacement magnitude of deformed shape for Run 2; (f) von mises stress distribution for Run 2.

The same analysis was repeated for Run 2, where two actuators apply an additional upward displacement. The deformation pattern in Figure 5-10(e) indicates the modified loading state of a central region exhibiting a reduced downward deflection as compared to Run 1, while the edges show the characteristic curvature associated with the cylindrical panel geometry. Whereas the Von Mises Stress contour in Figure 5-10(f) shows a slight redistribution of stresses relative to Run 1, with high stresses near the actuators, especially actuators 2 and 4 that were displaced upwards. This qualitative examination provides confirmation that the FE

model behaves consistently before extracting the deformed geometry for subsequent use in the registration pipeline.

Once the simulation results are validated visually, S_{def}^{nom} is directly exported from ABAQUS as a *.STL* file. The export includes the entire continuum-shell mesh, retaining its full thickness representation. No smoothing, filtering or geometric simplification is applied, and the *STL* is used exactly as exported. This ensures that the deformed geometry reflected the precise nodal positions predicted by the simulation and preserved the structural details defined in the model.

The exported model is then imported directly into the registration pipeline. Within this workflow the exported model is accessed through CloudCompare, where the exported model serves as the reference shape for aligning S_{conf}^{meas}. The deformed geometry provides a reference representation to the experimental data, allowing the registration pipeline to align the measured surface to a representation that already included gravitational and fixture-related deformation. This ensures that any remaining discrepancies between the two shapes are attributable to manufacturing deviations rather than differences in loading, support conditions or global positioning.

The role of the FE output in the overall methodology is therefore central. S_{def}^{nom} is not used in the mapping or interpolation steps that follow later in the computation of the displacement field. Instead, its purpose is to act as a physically consistent reference shape for registration. By aligning the measured surface to a numerically predicted deformation that matches the experimental conditions, the approach isolates the displacement field arising solely from geometric variability. Hence, this distinction is important for approximating the GF state shape of a manufactured component to be discussed in the following Chapter 6.

6 Results and Validation

This chapter presents the workflow and the results obtained from the GF state shape reconstruction and evaluates the performance of the proposed methodology through a series of validation steps. The first section of the chapter introduces the reconstructed GF geometry of a manufactured component derived from the combined registration, deformation simulation and displacement field mapping procedures described in the previous chapters. The approximated S_{free}^{man} is examined both visually and quantitatively to assess its agreement with the ideal GF state model, S_{free}^{nom}. The second section of the chapter focuses on validating the accuracy and reliability of the displacement field along with the methodology. This includes a comparison between displacement fields obtained from the 9 measurements and a detailed examination of the workflow. The chapter concludes with a discussion of the key findings and the limitations that influence the interpretation of the results.

6.1 GF State Shape Approximation

The reconstruction of the GF state shape forms the central outcome of the proposed methodology. It combines the results of the registration procedure, the deformation simulation of the nominal model and the extraction of the displacement field that represents the manufacturing deviations of the measured part. The manufacturing deviations E^{man} is derived in a loaded configuration by comparing the registered measurement S_{conf}^{meas} to the deformed nominal model S_{def}^{nom} where the influence of gravity and fixture loads is already accounted for in the numerical simulation. E^{man} is then mapped to the nominal shape S_{free}^{nom} using the barycentric approach introduced in Chapter 3 and used to displace the nodal coordinates of S_{free}^{nom}. The resulting geometry represents an approximation of the manufactured panel in a GF state that includes only the manufacturing deviations. Section 6.1.1 introduces the mathematical steps required for this reconstruction, while Section 6.1.2 reports the corresponding results.

6.1.1 Mathematical Formulation

The reconstruction of the GF state shape, S_{free}^{man}, begins with the displacement field E^{man}, which represents the manufacturing-induced deviations of the measured panel. As described in Chapter 4, this field is obtained by computing the CloudCompare C2M signed distances between S_{conf}^{meas} and S_{def}^{nom}. For a point ${P_i}^{ICP}$ on the registered measured configuration $S_{conf,ICP}^{meas}$, the displacement field is defined in equation 4.9 as:

$$E^{man}({P_i}^{ICP}) = \left\| {P_i}^{ICP} - \acute{P}_k \right\| \tag{4.9}$$

where $\acute{P}_k$ is the corresponding point on S_{def}^{nom} and $\hat{P}_j$ is the point on a *.obj* file of S_{free}^{nom}. These distances quantify only the manufacturing deviations, since the effects of gravity and fixturing, E^{conf}, are present in both representations and therefore cancel after registration. To

reconstruct the GF shape, this field must be transferred to S_{free}^{nom}, which serves as the reference configuration.

Since the displacement field is defined on $S_{conf,ICP}^{meas}$, a mapping is required to express these displacements on the nodes of S_{free}^{nom}. The mapping follows the same geometric principle as CloudCompare's cloud to mesh distance computation and employs barycentric coordinates on the closest triangle of S_{free}^{nom}.

For each measured point ${P_i}^{ICP}$, the closest point on the nominal mesh surface is first determined. Let this projected point be:

$$Q_i = proj\left({P_i}^{ICP},\ \Delta\left(\hat{P}_{j_1}, \hat{P}_{j_2}, \hat{P}_{j_3}\right)\right) \tag{6.1}$$

where $\Delta\left(\hat{P}_{j_1}, \hat{P}_{j_2}, \hat{P}_{j_3}\right)$ denotes the closest triangular element of S_{free}^{nom}. The point Q_i lies on this triangle, as shown in the schematic in Figure 3-6 and can be expressed in barycentric coordinates as:

$$Q_i = w_{1,i}\,\hat{P}_{j_1} + w_{2,i}\,\hat{P}_{j_2} + w_{3,i}\,\hat{P}_{j_3}, \qquad w_{1,i} + w_{2,i} + w_{3,i} = 1, \qquad w_{k,i} \geq 0. \tag{6.2}$$

These barycentric weights quantify the relative geometric influence of each vertex with respect to the projection of P_i onto the triangle.

To assign the scalar deviation $E^{man}(P_i)$ to the nominal mesh, the vertex with the largest barycentric weight is selected:

$$k = arg_{l\in\{1,2,3\}}\, max\ w_{l,i}. \tag{6.3}$$

The full displacement value is attributed to the corresponding node $\hat{P}_{j_k}$. As multiple measured points may map to the same vertex, only the displacement of the largest absolute magnitude is retained to preserve the most representative contribution:

$$\Phi_{j_k} = \begin{cases} E^{man}(P_i), & if\ |E^{man}(P_i)| > \left|\Phi_{j_k}\right| \\ \Phi_{j_k}, & otherwise \end{cases} \tag{6.4}$$

Here, Φ_{j_k} denotes the scalar deviation value currently assigned to the nominal vertex $\hat{P}_{j_k}$. Although an averaging of vertex values using the barycentric coefficients would be possible, this approach was deliberately avoided, as it introduces additional smoothing of the displacement field. Vertices receiving no contributions through this procedure are assigned a deviation value of zero. This produces a sparse scalar displacement field defined on the nodes of S_{free}^{nom} and consistent with the geometric placement of E^{man}.

To convert this scalar field into a vector field, the scalar deviation at each node is multiplied by the corresponding nodal normal vector. Let $\mathbf{n}_j$ denote the unit normal at node $\dot{P}_j$, computed as

the normalized sum of the normals of all faces adjacent to that node. The manufacturing deviation is expressed as a vector:

$$\boldsymbol{u}_j{}^{man} = \Phi_j \, \boldsymbol{n}_j \tag{6.5}$$

which specifies how the node must shift along the local normal direction to incorporate the manufacturing-induced deviation.

The S_{free}^{man}, which is described as a set of points:

$$\hat{P}_j{}^{GF} = \begin{bmatrix} \hat{x}_j{}^{GF} \\ \hat{y}_j{}^{GF} \\ \hat{z}_j{}^{GF} \end{bmatrix} \tag{6.6}$$

is obtained by adding the displacement vectors calculated in equation 6.5 to S_{free}^{nom} coordinates:

$$\hat{P}_j{}^{GF} = \hat{P}_j + \boldsymbol{u}_j{}^{man} \tag{6.7}$$

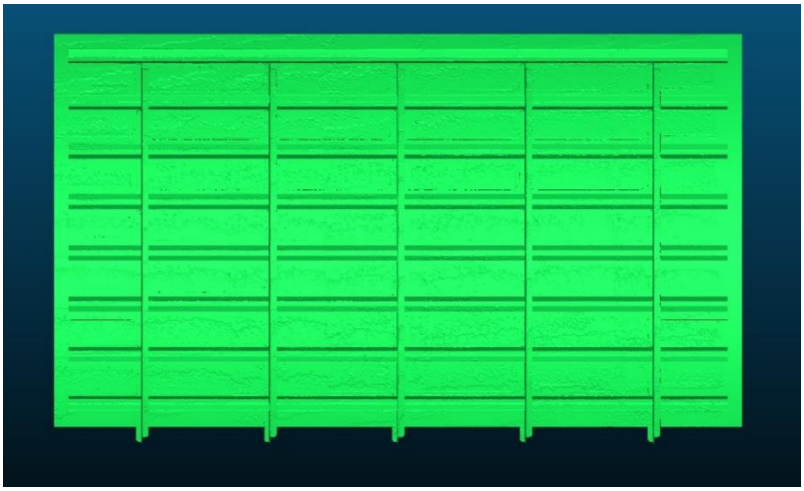
The resulting surface S_{free}^{man} represents the manufactured panel in an unloaded configuration and contains only the intrinsic manufacturing deviations. This reconstructed geometry forms the basis for the visual and quantitative comparisons presented in the following section.

6.1.2 GF State of the Manufactured Panel

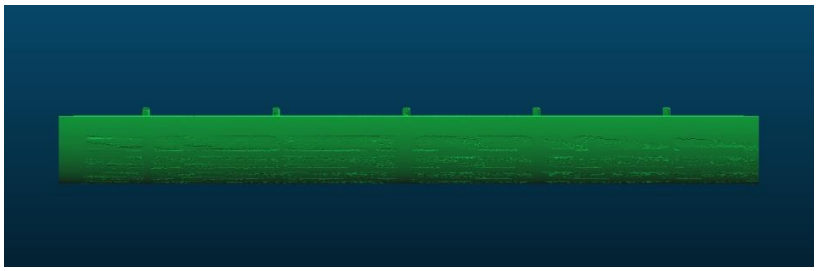
The GF state of the manufactured panel, S_{free}^{man}, obtained from the mapping procedure in Section 6.1.1 is illustrated in Figure 6-1(a) and Figure 6-1(b). The front view of the reconstructed surface shows the complete stiffened panel with all frames and stringers clearly represented, while the side view confirms that the global curvature introduced by gravity and clamping in the measured configurations has been effectively removed. The panel appears almost planar along the longitudinal direction, with only small residual waviness that can be attributed to manufacturing deviations. The meshed representation in Figure 6-1(c) highlights the underlying discretization of the S_{free}^{nom} and demonstrates that the mapped displacement field has been applied consistently across the entire surface.

To quantify the difference between S_{free}^{man} and S_{free}^{nom}, the GF shape of Run 1 was compared to S_{free}^{nom} using CloudCompare's cloud to mesh distance tool. The resulting distance field is shown as a heatmap in Figure 6-2. The color scale ranges approximately from -1.5 mm to +1.5 mm and indicates that most deviations lie within a narrow band around zero. Regions along the stiffeners and frames exhibit relatively small deviations, which reflects the higher local stiffness of these features and their close agreement with the nominal design. Slightly larger deviations occur in the unsupported skin regions, where the laminate is more compliant and therefore more sensitive to manufacturing variations. Localized patches of increased distance near the panel edges and around cut-outs suggest small alignment imperfections or reconstruction

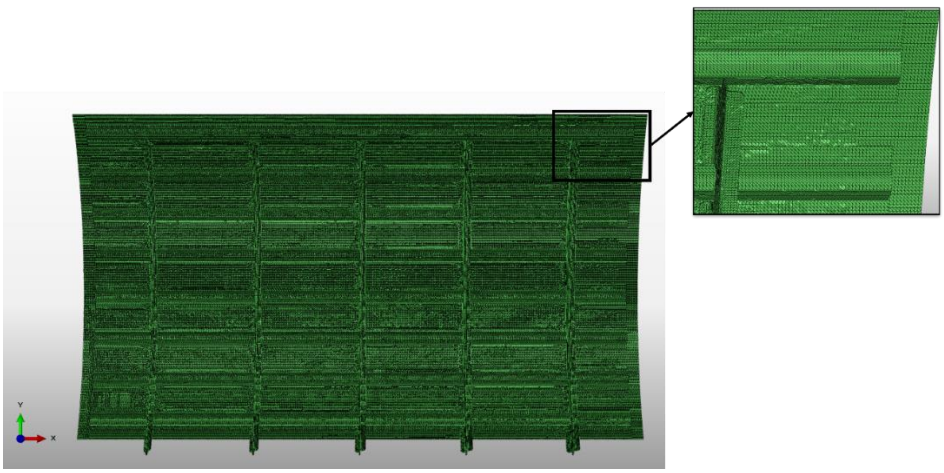
artefacts. The 95th percentile of the distance distribution for Run 1 is approximately 1.49 mm, which indicates that only a small fraction of the surface deviates more than this value from the nominal free-state shape, while most of the panel remains within tighter bounds.



(a)



(b)



(c)

Figure 6-1 (a) Front view of the approximated shape; (b) side view illustrating the effect of deviations; (c) mesh representation of the approximated shape.

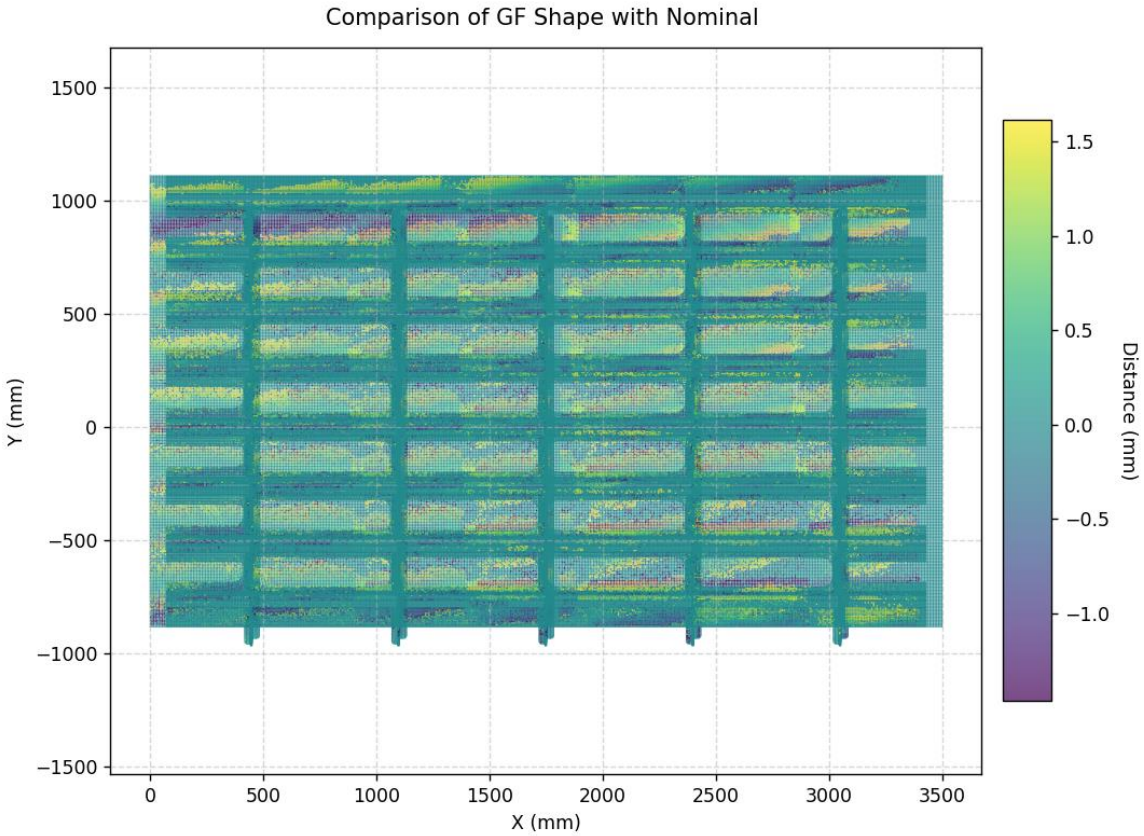
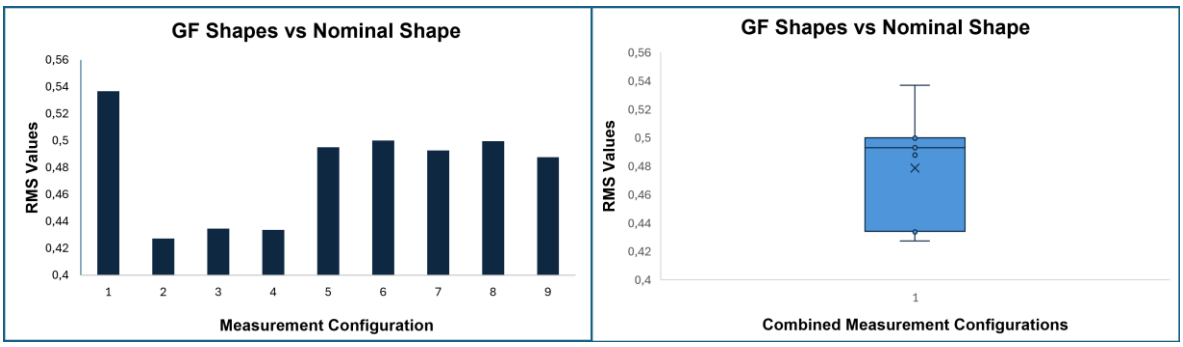
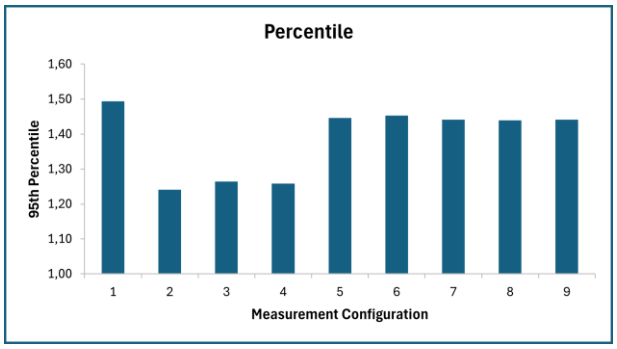


Figure 6-2 Comparison plot between GF state shape for Run 1 and the nominal shape.

The consistency of the GF reconstructed shape across all nine measurement configurations is summarized in the plot in Figure 6-3(a). The graph shows the root-mean-square (RMS) distance between each reconstructed GF shape and the nominal shape. The RMS values range between approximately 0.43 mm and 0.54 mm. Runs 2 to 4 exhibit the lowest RMS values, around 0.43 to 0.44 mm, while Run 1 and Runs 5 to 9 show slightly higher values close to 0.50 mm. A similar trend is observed in the 95th percentile values, which lie between about 1.24 mm and 1.49 mm for all runs, Figure 6-3(b). The limited spread of both RMS and percentile metrics indicates that S_{free}^{man} is independent of the specific actuator configuration used during measurements. In other words, the proposed workflow produces a stable approximation of the manufactured panel's GF state shape even though the loaded configurations differ significantly.



(a)



(b)

Figure 6-3 (a) RMS distances between configurations and nominal shape (left) with a box-and-whisker plot to illustrate the distribution (right) (b) corresponding 95th-percentile deviation values.

Overall, the visualization and distance statistics presented in this section show that the reconstructed S_{free}^{man} remains close to S_{free}^{nom}, with deviations of the order of a few tenths of a millimeter in RMS and below approximately 1.5 mm for most of the shape. The spatial patterns of the deviations are consistent with the structural layout of the panel and reflect plausible manufacturing imperfections rather than artefacts of the reconstruction process. These observations provide an initial indication of the reliability of the proposed method and form the basis for the validation of the displacement field and the workflow presented in Section 6.2.

6.2 Validation

The reliability of the reconstructed GF state shape S_{free}^{man} and the computational workflow depends on the accuracy of the displacement field and the consistency of the method across multiple measurement configurations. This section presents two validation steps designed to assess these aspects. First, the displacement field used to approximate the GF state is examined by comparing the fields obtained from all available measurements. This ensures that the extracted deviation field is consistent and repeatable. The second part of the section focuses on validating the complete workflow by comparing S_{conf}^{meas} corrected for E^{man} and the gravitational effect, with S_{free}^{nom} plus the effect of the smart jig. Visual and numerical comparisons are used to verify that the method produces stable results. These validation steps establish

confidence in the overall methodology and support the interpretation of the reconstructed shapes.

6.2.1 Displacement Field Validation

The accuracy and repeatability of the manufacturing displacement field play a central role in the GF state shape reconstruction. As the reconstructed S_{free}^{man} in Section 6.1 relies on the mapped displacement field E^{man}, verification is performed to ensure that the field remains consistent across the available measurement configurations. The purpose of this subsection is to assess the similarity of the displacement fields derived from each run and to determine whether the observed differences remain within the expected range of measurement uncertainty.

The $E^{man}(P_i)$ of each run was first mapped and interpolated onto the nodes of S_{free}^{nom}, as described in Section 4.2.3. Each run therefore produces a nodal displacement field $E^{man}(\hat{P}_j)$, defined on the same set of nominal coordinates $\hat{P}_j$ where according to equation 4.10:

$$E^{man}(\hat{P}_j) = \Phi_j. \tag{6.8}$$

To evaluate the consistency of the results, the fields of all measurement configurations were compared pairwise. For two run configurations r and s, the point-wise difference at a node j is:

$$\Delta_{jrs} = \left|\Phi_{jr} - \Phi_{js}\right| \tag{6.9}$$

The mean difference across the N nodes of the nominal mesh is:

$$\mu = \frac{1}{N}\sum_{j=1}^{N} \Delta_{jrs} \tag{6.10}$$

and the root-mean-square difference between the two displacement fields is:

$$RMS_{rs} = \sqrt{\frac{1}{N}\sum_{j=1}^{N} \left(\Delta_{jrs}\right)^2}. \tag{6.11}$$

A validation threshold of 0.12 mm was chosen to determine whether the differences between two displacement fields lie within the expected measurement uncertainty. This value is derived from the scanner precision of 0.08mm, which represents the uncertainty of a single scan. When two independent measurements are compared, the uncertainties accumulate in a root-sum-square sense, resulting in a combined uncertainty of:

$$\sigma_{\text{combined}} = \sqrt{0.08^2 + 0.08^2} \approx 0.113\,\text{mm}. \tag{6.12}$$

This value was rounded to 0.12 mm for use as the validation criterion. It represents a lower-bound estimate, as it does not account for additional sources of uncertainty such as robot positioning accuracy and end-effector calibration errors. Differences below this threshold can be attributed to measurement noise, while larger values indicate genuine discrepancies between displacement fields or accumulated modelling and registration errors. Hence:

$$max(RMS_{rs}) \leq 0.12\,\text{mm}. \tag{6.13}$$

The RMS differences between all pairs of runs are summarized in Table 6-1. The RMS values range from approximately 0.18 mm to 0.28 mm. The smallest difference occurs between configurations 9 and 5, with an RMS of 0.183 mm. Although these values exceed the 0.12 mm threshold, they lie within a close range, which suggests that the displacement fields produced by the nine measurement configurations are broadly consistent. The overall pattern indicates that the mapping and interpolation steps produce stable fields across different loading configurations, with variations that remain small relative to the magnitude of the panel's manufacturing deviations.

Table 6-1 Pairwise RMS differences between the mapped displacement fields of the nine configurations.

	Run 1	Run 2	Run 3	Run 4	Run 5	Run 6	Run 7	Run 8	Run 9
Run 1	-	0,24178	0,24687	0,23858	0,27989	0,28006	0,27977	0,2815	0,27595
Run 2	0,24178	-	0,22875	0,19773	0,25051	0,25085	0,25044	0,24987	0,24117
Run 3	0,24687	0,22875	-	0,2009	0,25578	0,2545	0,25493	0,25653	0,24662
Run 4	0,23858	0,19773	0,2009	-	0,24704	0,24659	0,24612	0,2488	0,23594
Run 5	0,2799	0,25051	0,25578	0,24704	-	0,19626	0,19623	0,22101	0,18301
Run 6	0,28006	0,25085	0,2545	0,24659	0,19625	-	0,20626	0,22153	0,18987
Run 7	0,27977	0,25044	0,25492	0,24612	0,19624	0,20626	-	0,21848	0,19172
Run 8	0,28151	0,24988	0,25653	0,24881	0,22101	0,22153	0,21847	-	0,21488
Run 9	0,27595	0,24118	0,24662	0,23594	0,18301	0,18987	0,19172	0,21488	-

The pair of configurations exhibiting the smallest RMS difference (Run 9 and Run 5) was further examined by analyzing the spatial distribution of their point-wise differences. Figure 6-4 shows the heatmap of the displacement field difference between these two runs. The deviations are predominantly below 0.5 mm and form spatially coherent patterns. The reinforced regions exhibit the lowest differences whereas slightly larger discrepancies appear in the unsupported skin areas and near the panel edges. The histogram of the differences, shown alongside the heatmap, follows a right-skewed log-normal distribution with a mean of approximately 0.14 mm, which aligns with the $max(RMS_{rs})$ and reinforces the conclusion that the displacement fields remain close across runs.

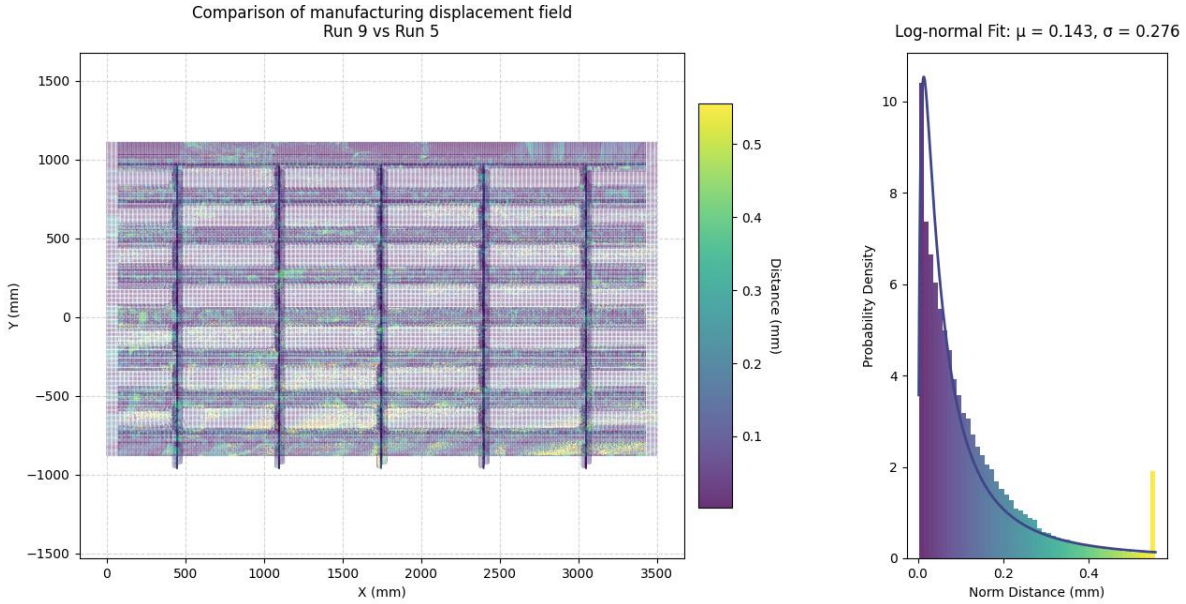


Figure 6-4 Comparison of the fields obtained from Run 9 and Run 5. The heatmap (right) shows the spatial distribution, while the histogram (left) illustrates the overall distribution fitted with a log-normal curve.

Overall, the validation results indicate that the displacement fields are consistent across different measurement configurations, although the RMS values exceed the maximum threshold. The deviations remain small, spatially structured and physically plausible in relation to the panel's geometry. These results support the reliability of E^{man} used in the approximated S_{free}^{man} and motivate the broader workflow validation discussed in Section 6.2.2.

6.2.2 Workflow Validation

The second part of the validation focuses on analyzing the accuracy and internal consistency of the workflow. While Section 6.2.1 emphasized the repeatability of the manufacturing displacement field, the current section evaluates whether the combined procedure of gravitational correction, deviation removal and deformation simulation produces results that agree with the physical behavior of the panel under clamping-only loading. The principle of the validation is that if the gravity and manufacturing deviations are removed from the measured configuration $S_{conf,ICP}^{meas}$, the resulting corrected surface should match the nominal model deformed solely by the fixture forces of the corresponding measurement configuration, r.

To implement this comparison, two independent analyses were carried out as shown in Figure 6-5. In the first analysis, a FE simulation of the nominal shape, S_{free}^{nom}, subjected only to gravity was performed. The resulting deformed configuration, denoted as $S^{nom+grav}$, was processed with CloudCompare to extract the gravitational field, E^{grav}. E^{grav} is initially expressed on the nominal mesh and then mapped onto the registered measured configuration and projected along the vertices surface normals, hence removing gravity-induced deformation from $S_{conf,ICP}^{meas}$. As the manufacturing deviation displacement field E^{man} is already defined on

$S_{conf,ICP}^{meas}$, no additional mapping step was required for its removal. Subtracting both fields produces a GF and deviation-free representation of the measured shape, $S^{corrected}$.

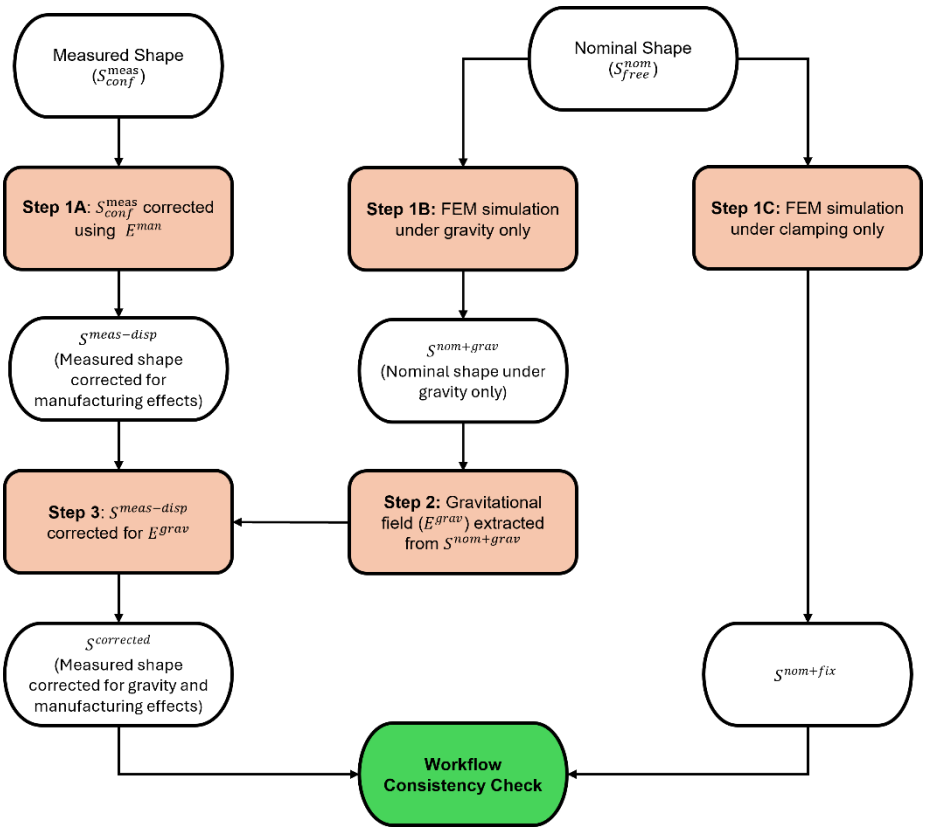


Figure 6-5 Proposed methodology for the workflow validation.

For each run r, the starting point is the registered measured configuration $S_{conf,ICP}^{meas}$ defined as a set of points ${P_{i,r}}^{ICP}$ having N vertices. At each vertex the manufacturing deviation field is defined as a scalar quantity ${E_r}^{man}({P_i}^{ICP})$, obtained from the CloudCompare C2M computation described in Chapter 4. The corresponding vertex normal is denoted by $\boldsymbol{n}_i{}^r$. The manufacturing deviation at vertex i is converted into a displacement vector:

$$\boldsymbol{u}_{i,r}^{man} = {E_r}^{man}({P_i}^{ICP})\ \boldsymbol{n}_i{}^r. \tag{6.14}$$

Removing this displacement from the measured configuration yields an intermediate surface, $S^{meas-disp}$ that is free of manufacturing deviations but still affected by gravity and clamping:

$${S^{meas-disp}}_r = {S_{conf}^{meas}}^{ICP}_r - \boldsymbol{u}_{i,r}^{man}. \tag{6.15}$$

This surface represents how the ideal panel would deform under the combined influence of gravity and fixture loads for run r.

The gravitational deformation is obtained through a FE simulation of the nominal panel subjected only to gravity. The resulting configuration $S^{nom+grav}$ is post-processed in

CloudCompare to obtain a scalar gravitational distance field $E^{grav}(\hat{P}_j) = \psi_j$ defined on the S_{free}^{nom} mesh nodes j. At each of these nodes, the gravitational displacement vector is:

$$\boldsymbol{u}_j{}^{g} = \psi_j \, \boldsymbol{n}_j{}^{g}, \tag{6.16}$$

where $\boldsymbol{n}_j{}^{g}$ is the corresponding nodal normal from the FE mesh. The normals $\boldsymbol{n}_i{}^{r}$ and $\boldsymbol{n}_j{}^{g}$ originate from different surface discretization's and are therefore not assumed to be identical. E^{grav} is then transferred to the intermediate surface $S^{meas-disp}{}_r$ using the k-NN mapping and IDW interpolation, as described in Chapter 4. The result is a mapped scalar field $E_r{}^{grav}(\check{P}_i)$ defined on the vertices of $S^{meas-disp}{}_r$. The mapped gravitational displacement at vertex i is:

$$\boldsymbol{u}_{i,r}{}^{g} = E_r{}^{grav}(\check{P}_i)\, \check{\boldsymbol{n}}_i{}^{r}. \tag{6.17}$$

Here, $\check{\boldsymbol{n}}_i{}^{r}$ denotes the unit surface normal vector at vertex $\check{P}_i$ of the deviation corrected measured configuration $S^{meas-disp}{}_r$, computed from the local mesh geometry. Subtracting $\boldsymbol{u}_{i,r}{}^{g}$ from $S^{meas-disp}{}_r$ generates the corrected configuration:

$$S^{corrected}{}_r = S^{meas-disp}{}_r - \boldsymbol{u}_{i,r}{}^{g}. \tag{6.18}$$

which in theory should approximate the shape of S_{free}^{nom} subjected only to the clamping forces of the specific measurement configuration.

In parallel, a second FE simulation is performed for each measurement configuration, in which S_{free}^{nom} is loaded by the actuator forces while gravity is deactivated. The corresponding deformed configuration is denoted by $S^{nom+fix}{}_r$, defined as a set of points:

$$P_{i,r}{}^{nom+fix} = \begin{bmatrix} x_{i,r}{}^{nom+fix} \\ y_{i,r}{}^{nom+fix} \\ z_{i,r}{}^{nom+fix} \end{bmatrix}. \tag{6.19}$$

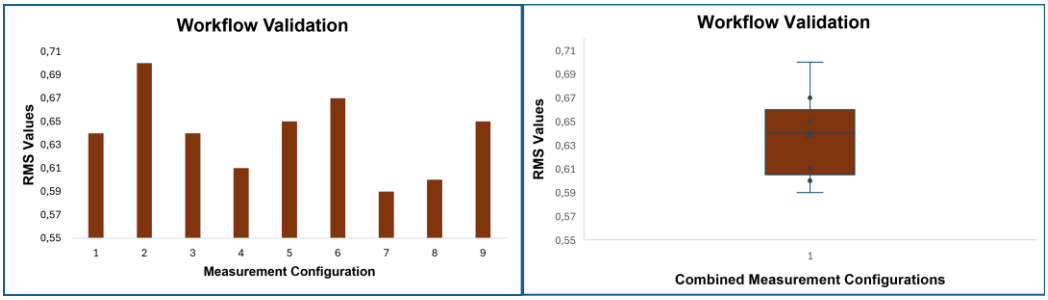
Ideally, the corrected measured configuration $S^{corrected}{}_r$ and the clamping-only simulation $S^{nom+fix}{}_r$ would coincide. In practice, their difference reflects the combined influence of measurement noise, registration errors, mapping approximations and modelling assumptions in the finite element analysis. The point-wise deviation between both shapes for configuration r is expressed as:

$$\delta_{i,r} = S^{corrected}{}_r - S^{nom+fix}{}_r. \tag{6.20}$$

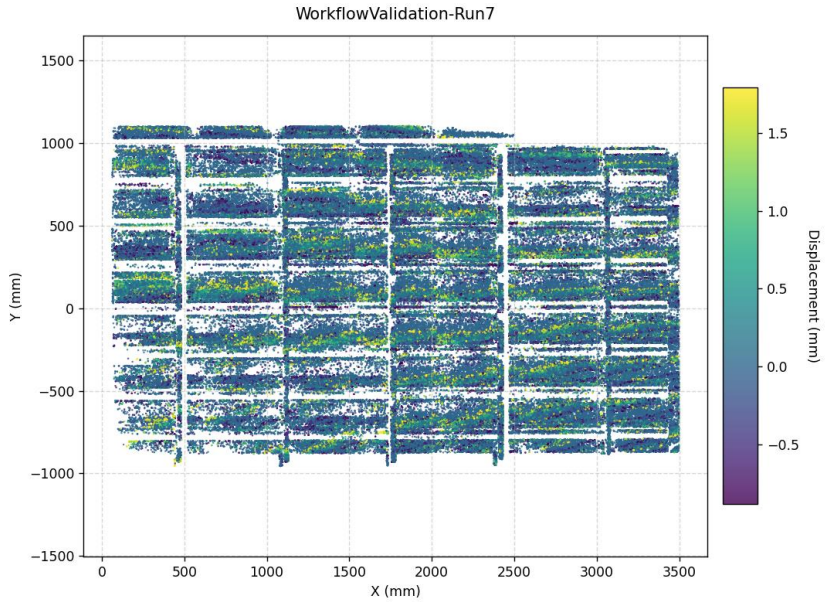
Hence, the overall agreement between the corrected measured shape and the clamping-only prediction is quantified by the root-mean-square (RMS) distance:

$$RMS_r = \sqrt{\frac{1}{N}\sum_{i=1}^{N} {\delta_{i,r}}^2}. \tag{6.21}$$

The RMS values for all nine measurement configurations are summarised in Figure 6-6(a). They fall within a relatively narrow range between approximately 0.59 mm and 0.70 mm, indicating that the workflow behaves consistently across different actuator displacements. Run 7 exhibits the lowest RMS and was therefore selected for a more detailed visual comparison. The corresponding distance heatmap, shown in Figure 6-6(b), indicates that the residual deviations between ${S^{corrected}}_r$ and ${S^{nom+fix}}_r$ remain within about $\pm$1.5 mm over most of the surface. Stiffened regions show relatively small deviations, whereas the more flexible skin areas exhibit larger differences. These trends are consistent with the structural behavior of the panel and suggest that the remaining discrepancies can be allotted to the combined uncertainties rather than fundamental flaws in the workflow.



(a)



(b)

Figure 6-6 (a) RMS distances between the corrected measured surfaces and the clamping-only FEM predictions (left) with a box-and-whisker plot (right); (b) spatial distance field for configuration 7, showing the distances between the corrected measured shape and the corresponding clamping-only FEM prediction.

Overall, the validation methodology demonstrates that the corrected measured configurations approximate the clamping-only FE predictions with moderate but systematic discrepancies as discussed in Section 6.4. The magnitude and spatial distribution of these deviations are physically reasonable and remain stable across varying measurement configurations. The potential consequences of these findings, together with the sources of residual error, are discussed further in Section 6.4.

6.3 Discussion of Key Findings

The results presented in the previous sections provide an assessment of the proposed methodology for the approximation of the GF state shape of a manufactured component. This section analyses the results obtained from the GF state shape approximation, the displacement field and the workflow validation. The objective of this section is to interpret the observed results and identify the factors that influence the quality and reliability of the results.

The first set of results concerns the GF shape reconstructed in Section 6.1. The RMS distances between the GF shapes, S_{free}^{man}, and the nominal shape, S_{free}^{nom}, lies between approximately $0.43mm$ and $0.54mm$ for all 9 measurement configurations. The average RMS is relatively smaller than the manufacturing deviation level of approximately $2.08mm$ reported in Chapter 4. Several factors contribute to this discrepancy:

- The comparison measures the distance between two *free-state shapes* which dampens certain deformation modes.
- Components of the manufacturing deviation that are tangential to the surface may cancel out after the mapping.
- The normalization of displacements along local surface normals tends to reduce their magnitude once transferred to the nominal mesh.
- Barycentric mapping introduces a degree of smoothing, which can reduce sharp local variations present in the original deviation field.

On the other hand, it should be acknowledged that some of the discrepancies may arise from workflow-related imperfections such as registration inaccuracies, mapping artefacts or modelling simplifications. Overall, the results remain physically interpretable, but the magnitude of deviations indicates that further refinement of the mapping procedure may be beneficial.

The displacement field validation in Section 6.2.1 provides further insight into the repeatability of E^{man}. The RMS differences between E^{man} from different measurement configurations fall within $0.18mm$ to $0.28mm$, which is higher than the threshold of $0.12mm$ derived from the scanner precision. This threshold represents a lower bound, as it does not account for additional sources of uncertainty such as the calibration accuracy of the robot end-effector, which was not repeated between measurements. Hence, the variability between the configurations cannot be explained by the measurement noise alone. Several factors contribute to these elevated values:

- Small oscillations of the scanner mounting system, which is subject to ambient vibrations during acquisition, may introduce uncertainty.
- Each E^{man} results from a sequence of operations; stitching, registration, distance computation and mapping that introduces uncertainty beyond the nominal scanner precision.
- The transfer of displacement fields onto the nominal mesh may smooth or amplify deviations depending on local point density and surface geometry.

Although these RMS values exceed the threshold, the spatial patterns remain smooth and structurally coherent rather than random. The extracted displacement field, E^{man}, therefore provides a reliable representation of the manufactured component, however, the results also highlight the presence of cumulative uncertainties.

The workflow validation in Section 6.2.2 adds another verification step to the analysis. The RMS comparison between the corrected measured configurations, ${S^{corrected}}_r$, and the clamping-only FE prediction, ${S^{nom+fix}}_r$, ranges between approximately $0.59mm$ and $0.70mm$. The main observations include:

- Stiffeners and frame intersections exhibit smaller deviations, reflecting the close agreement between simulation and measurement in the reinforced areas.
- Skin regions show larger deviations, consistent with the region's higher compliance and sensitivity to minor changes.
- Global patterns indicate that the residual differences are not random and reflect the underlying structural behavior of the panel.

These results indicate that the workflow captures the main deformation characteristics of the panel, even though few discrepancies remain.

When viewed together, the results from the actual GF state shape reconstruction, the displacement field and the workflow validation constitute a consistent illustration:

- The reconstruction of S_{free}^{man} is consistent across runs.
- E^{man} displays stable behavior with moderate but predictable variation.
- ${S^{corrected}}_r$ approximates ${S^{nom+fix}}_r$ with structured and interpretable residuals.

Each validation step supports the robustness of the methodology and their combined interpretation suggests a workflow that is stable and physically meaningful.

Despite the strengths of the approach, several uncertainties are evident. The stitching of raw point clouds without dedicated markers contributes significantly. As the panel exceeds the scanner's field of view, multiple scans must be combined and the stitched model provided for this work showed small misalignments, particularly in the out-of-plane, z, direction. These misalignments are visible as horizontal and vertical variations in the scalar distance fields and influence the results. Other factors that may affect the results include errors during rigid ICP registration, surface reconstruction and sampling of the point cloud data (PCD), assumptions during the mapping and interpolation steps, simplifications in FEM and variations in surface normals. These limitations are examined in more detail in Section 6.4.

In conclusion, the outcomes indicate that the proposed methodology performs reliably across multiple measurement configurations and successfully captures the structural behavior of the manufactured panel. On the other hand, the analysis also highlights areas of improvement, including point cloud processing, registration accuracy and E^{man} transfer. These observations form the basis for the uncertainty and limitation assessment presented in the following section.

6.4 Limitations and Uncertainty Analysis

The results obtained in this chapter indicate that the proposed workflow produces consistent displacement fields and a plausible approximation of the GF state shape of a manufactured component. However, several limitations and uncertainties influence the accuracy of these results. These limitations arise from experimental data, numerical modelling and the interaction between the two domains. The main uncertainty sources affecting the workflow are summarized in Figure 6-7, while the following paragraphs describe each category in detail.

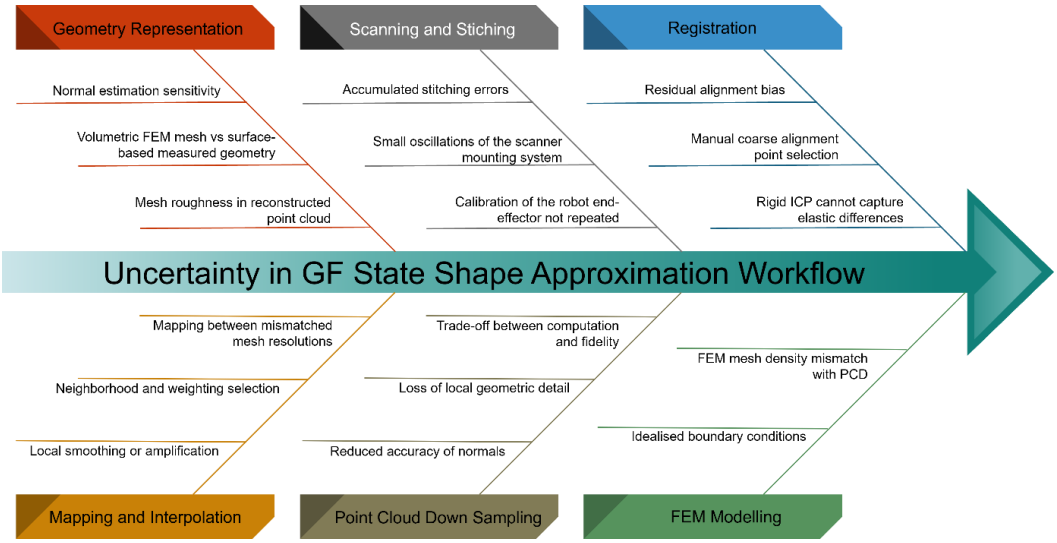


Figure 6-7 Ishikawa diagram summarizing the principal sources of uncertainty in the approximation of the GF state shape workflow [SMAR25], [BILU24] .

The primary set of uncertainties arises from the geometric representation of the surfaces. The most significant uncertainties include:

- The deformed shape S_{def}^{nom} is volumetric, while the measured shape S_{conf}^{meas} is a surface scan. This mismatch complicates point-to-surface projection and introduces sensitivity to how distances are computed.
- Surface normals were not available in the OBJ file of S_{free}^{nom}, requiring them to be recomputed during the approximation of S_{free}^{man}. Errors in the estimated normals influence mapping accuracy and the decomposition of displacement components.
- The meshing of the reconstructed point cloud is not fully smooth and irregularities affect CloudCompare's distance values and local normal estimation.

The stitching of the raw scans represents another source of uncertainty. Due to the size of the panel, multiple partial scans were combined into a single point cloud. The stitched model

showed small misalignments, particularly along the out-of-plane direction. These misalignments appear as horizontal and vertical distortions in the scalar distance fields and extend into the manufacturing deviation field, the GF state shape approximation and the workflow validation.

Further uncertainties originate from the registration process. The following aspects influence the fine alignment between the surfaces:

- Fine alignment relies on rigid ICP, which cannot accommodate local elastic differences between the measured and simulated shapes.
- Coarse alignment depends on manual point selection, introducing operator-dependent variations.
- Residual registration errors accumulate and influence the displacement field, the approximated shape and the comparison against numerical predictions.

Moreover, the mapping and interpolation of fields also introduce certain limitations. Both the gravitational, E^{grav}, and manufacturing deviation, E^{man}, displacement fields must be transferred between surfaces with different mesh densities. The following factors contribute to mapping uncertainty:

- Dependence on neighbourhood size and weighting functions.
- Differences in point density between nodes of S_{def}^{nom} and vertices of S_{conf}^{meas}.
- Local smoothing or amplification introduced by interpolation.

Another cause of the uncertainty is the down-sampling of the point cloud. Although it is necessary for practical computation times, it reduces the local resolution of the surface, influencing normal estimation, cloud-to-mesh distance computation accuracy and the distribution of mapped fields. The resulting loss of detail can affect the results.

Several limitations also arise from the FEM model. These include:

- Idealised actuator displacement BCs in the FEM simulations that does not fully reproduce physical clamping and loading conditions.
- Differences in mesh resolution between the FEM model and the point cloud, which influence subsequent mapping and comparison steps.

When combined, these uncertainties help explain the variations observed in the results. While each factor makes a minor contribution, their cumulative effect increases throughout the workflow, from scanning and stitching to registration, mapping, and numerical comparison. This accumulation results in RMS values that are higher than the thresholds and the local deviations present in some reconstructed regions. Despite these limitations, the reconstructed shapes and displacement fields demonstrate persistent patterns that are consistent with the structural characteristics of the panel. The identified uncertainties provide recommendations for potential refinement of the workflow and context for evaluating the results presented in this chapter. These insights form the basis for the conclusion and recommendations presented in the following chapter.

7 Conclusion and Future Work

7.1 Conclusion

The aim of this thesis was to approximate the GF state shape of a large, compliant thin-walled structure (LCTS) using a combined experimental, numerical and automated workflow. This objective supports the broader goal of enabling adaptive tooling processes, where accurate knowledge of the GF geometry is important for virtual clamping and offers an alternative to traditional shimming during assembly. The workflow developed in this research integrates non-contact measurement, shapes registration, FEM simulation and displacement field mapping to approximate the GF state shape of a manufactured component from loaded measurement configurations.

The results indicate that the proposed methodology can reconstruct a plausible GF state shape that reflects the main structural characteristics of the manufactured component. The manufacturing displacement fields obtained from multiple measurement runs remain consistent, with RMS differences ranging between approximately 0.18 mm and 0.28 mm, and exhibit coherent deformation patterns, indicating that the deviations can be captured reliably. The reconstructed shapes also display smooth and interpretable features, and while the RMS distances to the nominal GF shape lie in the range of approximately 0.45 mm to 0.55 mm, their behavior is explained by the uncertainties identified in Chapter 6. These observations confirm that the approach provides a meaningful approximation of the panel's GF state form.

The validation procedures further demonstrate that the approximated GF state shapes and corrected measurement configurations preserve the global behavior predicted by the numerical model. While some deviations arise from measurement, stitching, mapping and registration errors, the major deformation trends remain intact. This indicates that the methodology is robust enough to serve as a basis for virtual clamping applications.

Overall, the work presented in this research establishes that the GF state shape approximation from constrained measurements is a viable approach for supporting data-driven assembly operations. The insights gained here provide a foundation for refining the workflow and for integrating GF state shape information into adaptive tooling, where it has the potential to reduce reliance on shimming and improve the efficiency of future assembly processes.

7.2 Future Work

The workflow developed in this thesis provides a promising foundation for GF state shape approximation, however several opportunities exist to refine the workflow. A first direction concerns improvements in the measurement pipeline. The use of enhanced stitching algorithms would help reduce geometric misalignment and improve the accuracy of the manufacturing deviations displacement field. Further benefits may be obtained by adopting more advanced registration strategies, such as non-rigid or hybrid alignment methods used in [SABR16], which could reduce residual bias that cannot be captured by rigid ICP.

Further improvements in the mapping and interpolation procedures also require consideration. More sophisticated approaches for transferring scalar fields between shapes as provided by [THIÉ17], together with improved surface normal estimation and point cloud meshing, may help preserve local geometric features and reduce smoothing effects. Adjusting the point cloud down-sampling strategy or exploring adaptive sampling could further balance computational efficiency with geometric accuracy [SABR17].

Another possible approach involves a more thorough physical integration of the displacement field into the numerical modelling. In the current workflow, the manufacturing deviations are introduced through a mathematical coordinate adjustment, which applies displacements to individual nodes without taking into consideration how the surrounding structure would physically respond. Incorporating the manufacturing displacement field as a displacement boundary condition in a FE analysis would be a helpful extension [ESPO22]. This would allow the panel to deform according to its actual stiffness distribution, providing a more realistic representation of how the manufactured component loosens into its GF state. This approach could improve the accuracy of the GF state shape reconstruction and streamline the workflow.

Besides these, expanding the methodology to different panel geometries or larger structural assemblies would also provide valuable insight into the scalability and robustness of the approach. Overall, these improvements would strengthen the methodology introduced in this research and support its transition from an investigative workflow to a practical tool for assembly operations.

7.3 Outlook

The methods and results presented in this research provide a foundation for integrating the GF state shape of a manufactured component into adaptive assembly strategies. As manufacturing moves toward increasingly digital and model-based processes, the ability to approximate the GF geometry from constrained measurements holds considerable importance. A reliable GF shape enables virtual clamping, which supports automated fixture adjustment and reduces reliance on manual shimming during assembly. Integrating the reconstructed and approximated shape within a digital workflow could contribute to the development of adaptive tooling systems.

There is significant opportunity to extend this workflow using various approaches. Improvements in scanning technology, more accurate manufacturing deviations mapping and more advanced registration algorithms will support the generation of GF shapes with greater accuracy and robustness. Similarly, integrating physics-based modelling of the manufacturing deviation field, by applying it as a BC in FE analysis, could enable more predictive and physics-based refinements of the approximated shape. As these developments progress, GF shape estimation can become an important part of intelligent assembly operations to enhance structural quality.

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